MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

STANDARD SERIAL INPUT, DIGITAL, KEY CODE EQJ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic standard serial input (SSI) module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.293 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.4.1 Cooling requirements. The module shall be cooled with 27.6°C maximum temperature ambient air at a minimum of 12 ft/s, or with an alternate temperature velocity which will produce the equivalent cooling condition.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EQJ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A. The cross sectional area of the entire fin shall not exceed 57 percent of the area designated in MIL-STD-1389.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/215-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ± 0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

3.10 Functional Tests. All modules shall be subjected to functional tests which conform to the requirements specified in 3.10.1 through 3.10.3.

3.10.1 Test patterns. This specification includes a set of unique input patterns (table VIII) capable of detecting all nonredundant failures in the module logic networks. These test patterns are generated by assuming "stuck at H state" and "stuck at L state" in each of the basic logic functions and applying the necessary input patterns to detect these failures. A "stuck at H state" indicates a failure resulting in a steady high level output. A "stuck at L state" indicates a failure resulting in a steady low level output. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the

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NOTES:
1. V_{CC} = 5.0 V ±10%
2. ENA-R - Enable receiver
3. RDY-R - Ready
4. RT - Reset timing
5. P.U. - Pull up
6. CLOB - Clock OB
7. DEOW - Data end-of-word
8. CLBI - Clock B
9. D33 - Data bit 33
10. GND - Ground
11. CLAI - Clock A
12. $\overline{BCC-1}$ - Bit counter clock - 1
13. g - Word length line
14. $\overline{RCT}$ - Bit counter reset
15. DTA 0-R - Data 0
16. $\overline{SOW-R}$ - Start of word
17. f - Word length line
18. c - Word length line
19. $\overline{BCC}$ - Bit counter clock
20. b - Word length line
21. $\overline{LRC-R}$ - Lower rank clock R
22. ENA-I - Enable (transmission)
23. URC-R - Upper rank clock
24. STB-CR - Strobe
25. $\overline{DTA-A}$ - Data accepted
26. SDI - Serial data input
27. CLOA - Clock OA
28. OVR-I - Override
29. D32 - Data bit 32
30. $\overline{MCLR-R}$-Master clear
31. LOC - Load counter
32. EOC - End of count
33. e - Word length line
34. OSC-R - Oscillator, receiver
35. DTOR - Data OR'd
36. DTA 1-R - Data 1
37. SEL-BC - Select clock
38. d - Word length line
39. a - Word length line
40. CLA - Clock A gated

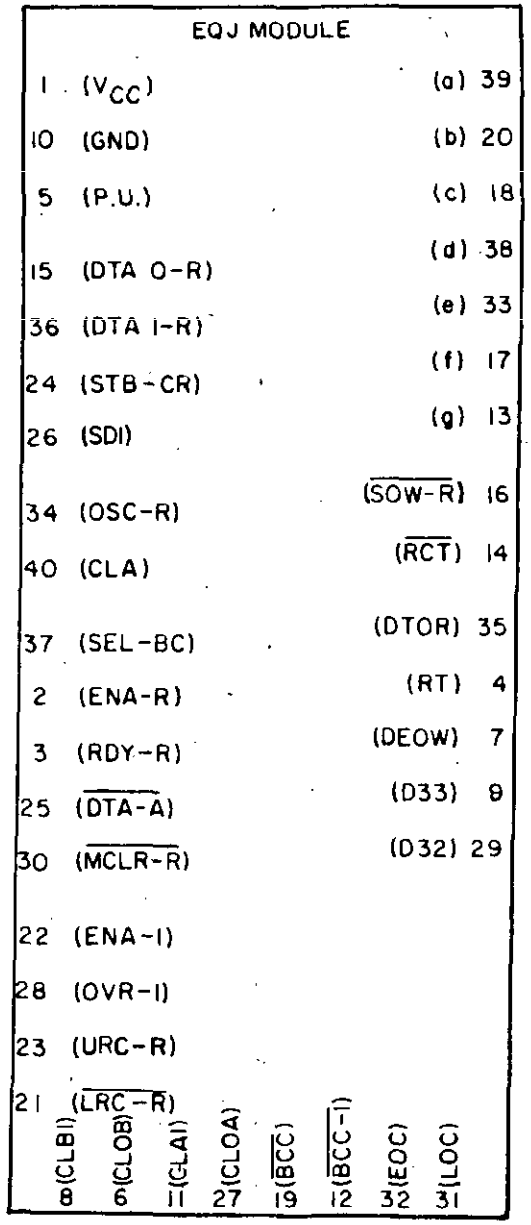


FIGURE 1. EQJ module function diagram, standard serial input.

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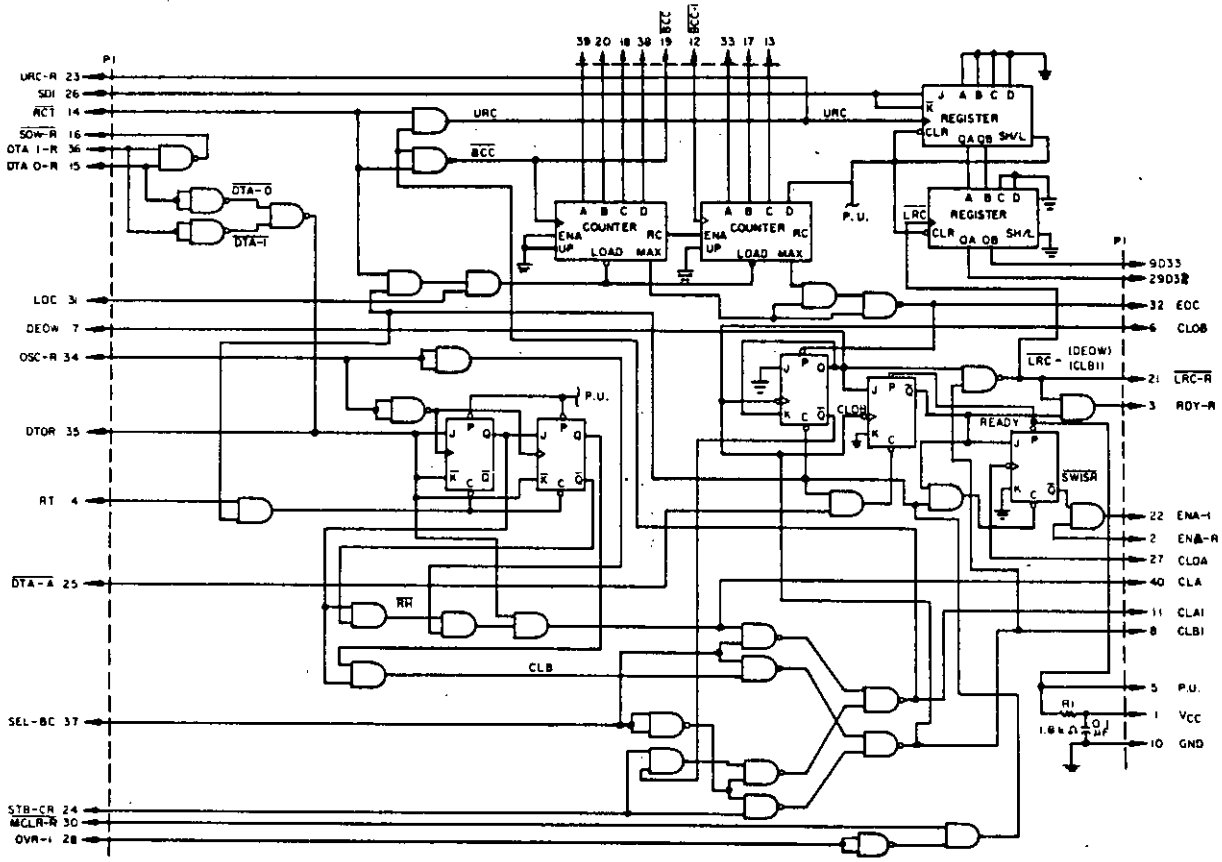


FIGURE 1. EQJ module function diagram, standard serial input - Continued.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B14	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	mV
C1-C8,C10-C15,C19	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
C9,C17, C18,C23	Input breakdown current	V	S	0	2	0	2	0	2	0	2	mA
C16,C20,C21	Input breakdown current	V	S	0	3	0	3	0	3	0	3	mA
C22	Input breakdown current	V	S	0	4	0	4	0	4	0	4	mA
D1-D12	Output short circuit current	VI	X	-20	-55	-20	-55	-20	-55	-20	-55	mA
D13,D14	Output short circuit current	VI	X	-6	-130	-6	-130	-6	-130	-6	-130	mA
E1-E17,E19-E21,E23	Input clamp diode voltage	VII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E18,E22	Input clamp diode voltage	VII	X	-0.1	-1.5	-0.1	-1.5	-0.1	-1.5	-0.1	-1.5	V
F0-F112	Functional	VIII	X	Test vectors must produce output vectors as specified in test table.								
G1-G14	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	mV
H1-H12	Output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H13,H14	Output voltage, high level	X	X	2.5	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J13	Input current, high level	XI	X	0	40	0	40	0	40	0	40	µA
J14,J16	Input current, high level	XI	X	0	80	0	80	0	80	0	80	µA
J15	Input current, high level	XI	X	-2250	80	-2250	80	-2250	80	-2250	80	µA
J17-J19	Input current, high level	XI	X	0	120	0	120	0	120	0	120	µA
J20	Input current, high level	XI	X	-4500	160	-4500	160	-4500	160	-4500	160	µA
J21,J22	Input current, high level	XI	X	0	160	0	160	0	160	0	160	µA

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J23	Input current, high level	XI	X	0	40	0	40	0	40	0	40	µA
K1-K13	Input current, low level	XII	X	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	mA
K14,K15, K17-K19	Input current, low level	XII	X	-0.2	-3.46	-0.2	-3.46	-0.2	-3.46	-0.2	-3.46	mA
K16	Input current, low level	XII	X	-0.2	-3.2	-0.2	-3.2	-0.2	-3.2	-0.2	-3.2	mA
K20,K21	Input current, low level	XII	X	-0.3	-5.2	-0.3	-5.2	-0.3	-5.2	-0.3	-5.2	mA
K22	Input current, low level	XII	X	-0.4	-6.4	-0.4	-6.4	-0.4	-6.4	-0.4	-6.4	mA
K23	Input current, low level	XII	X	-0.1	-0.78	-0.1	-0.78	-0.1	-0.78	-0.1	-0.78	mA
L	Power supply current	XIII	X	135	520	135	520	135	520	135	520	mA
M1-M12	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	4	30	4	30	4	35	4	35	ns
M13,M14	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	5	25	5	25	5	25	5	25	ns
M15-M26	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	4	50	4	50	4	55	4	55	ns
M27,M28	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	6	35	6	35	6	35	6	35	ns
M27,M28	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	6	35	6	35	6	35	6	35	ns
N1-N12	Transition time (t_{THL}), C_L = 50 pF	XV	S	4	13	4	13	4	15	4	15	ns
N13,N14	Transition time (t_{THL}), C_L = 50 pF	XV	S	1	10	1	10	1	10	1	10	ns
N15-N26	Transition time (t_{THL}), C_L = 150 pF	XV	Q	4	20	4	20	4	23	4	23	ns
N27,N28	Transition time (t_{THL}), C_L = 150 pF	XV	Q	2	15	2	15	2	15	2	15	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P1	Propagation delay time (t_{PLH}), C_L = 50 pF	XVI	S	4	26	4	26	4	26	4	26	ns
P2	Propagation delay time (t_{PLH}), C_L = 150 pF	XVI	Q	4	32	4	32	4	32	4	32	ns
Q1	Propagation delay time (t_{PHL}), C_L = 50 pF	XVI	S	4	19	4	19	4	19	4	19	ns
Q2	Propagation delay time (t_{PHL}), C_L = 150 pF	XVI	Q	4	25	4	25	4	25	4	25	ns
R1	Propagation delay time (t_{PLH}), C_L = 50 pF	XVI	S	8	45	8	45	8	45	8	45	ns
R2	Propagation delay time (t_{PLH}), C_L = 150 pF	XVI	Q	8	57	8	57	8	57	8	57	ns
S1	Propagation delay time (t_{PHL}), C_L = 50 pF	XVI	S	8	45	8	45	8	45	8	45	ns
S2	Propagation delay time (t_{PHL}), C_L = 150 pF	XVI	S	8	57	8	57	8	57	8	57	ns
T1	Propagation delay time, end-of-count (measuring output pulse width), C_L = 50 pF	XVII	S	35	135	35	135	35	135	35	135	ns
T2	Propagation delay time, end-of-count (measuring output pulse width), C_L = 150 pF	XVII	Q	40	142	40	142	40	142	40	142	ns
T3	EOC Spike	XVII	Q	0	3	0	3	0	3	0	3	ns
U1	Propagation delay time (t_{PHL}), minimum pulse width, C_L = 50 pF	XVIII	S	20	85	20	85	20	85	20	85	ns
U2	Propagation delay time (t_{PHL}), minimum pulse width, C_L = 150 pF	XVIII	Q	25	92	25	92	25	92	25	92	ns
V1	Propagation delay time (t_{PLH}), C_L = 50 pF	XIX	S	20	130	20	130	20	130	20	130	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
V2	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	15	45	15	45	15	45	15	45	ns
V3	Propagation delay time (t_{PLH}), C_L = 150 pF	XIX	Q	25	137	25	137	25	137	25	137	ns
V4	Propagation delay time (t_{PHL}), C_L = 150 pF	XIX	Q	20	52	20	52	20	52	20	52	ns
W1	Propagation delay time (t_{PLH}), C_L = 50 pF	XX	S	6	42	6	42	6	42	6	42	ns
W2	Minimum pulse width, C_L = 50 pF	XX	S	90	110	90	110	90	110	90	110	ns
W3	Propagation delay time (t_{PLH}), C_L = 50 pF	XX	S	14	91	14	91	14	91	14	91	ns
W4	Minimum pulse width, C_L = 50 pF	XX	S	70	110	70	110	70	110	70	110	ns
W5	Propagation delay time (t_{PHL}), C_L = 50 pF	XX	S	10	79	10	79	10	79	10	79	ns
W6	Minimum pulse width, C_L = 50 pF	XX	S	82	135	82	135	82	135	82	135	ns
W7	Propagation delay time (t_{PLH}), C_L = 150 pF	XX	Q	11	49	11	49	11	49	11	49	ns
W8	Minimum pulse width, C_L = 150 pF	XX	Q	95	117	95	117	95	117	95	117	ns
W9	Propagation delay time (t_{PLH}), C_L = 150 pF	XX	Q	19	98	19	98	19	98	19	98	ns
W10	Minimum pulse width, C_L = 150 pF	XX	Q	77	117	77	117	77	117	77	117	ns
W11	Propagation delay time (t_{PHL}), C_L = 150 pF	XX	Q	15	86	15	86	15	86	15	86	ns
W12	Minimum pulse width, C_L = 150 pF	XX	Q	87	142	87	142	87	142	87	142	ns
X1	Propagation delay time (t_{PLH}), C_L = 50 pF	XXI	S	25	110	25	110	25	110	25	110	ns
X2	Propagation delay time (t_{PHL}), C_L = 50 pF	XXI	S	16	77	16	77	16	77	16	77	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
X3	Pulse width, C_L = 50 pF	XXI	S	65	100	65	100	65	100	65	100	ns
X4	Propagation delay time (t_{PLH}), C_L = 150 pF	XXI	Q	30	117	30	117	30	117	30	117	ns
X5	Propagation delay time (t_{PHL}), C_L = 150 pF	XXI	Q	20	84	20	84	20	84	20	84	ns
X6	Pulse width, C_L = 150 pF	XXI	Q	60	107	60	107	60	107	60	107	ns
Y1	Maximum operating frequency check, C_L = 50 pF	XXII	S	850	1150	850	1150	850	1150	850	1150	ns
Y2	Maximum operating frequency check, C_L = 50 pF	XXII	S	850	1150	850	1150	850	1150	850	1150	ns
Z1	Maximum operating frequency check, C_L = 50 pF	XXIII	S	850	1150	850	1150	850	1150	850	1150	ns
Z2	Pulse width, C_L = 50 pF	XXIII	X	51	107	51	107	51	107	51	107	ns
Z3	Maximum operating frequency check, C_L = 50 pF	XXIII	S	850	1150	850	1150	850	1150	850	1150	ns
Z4	Pulse width, C_L = 50 pF	XXIII	S	220	390	220	390	220	390	220	390	ns
Z5	Pulse width, C_L = 50 pF	XXIII	S	220	390	220	390	220	390	220	390	ns
AA1-AA14	Output crosstalk (low level), C_L = 50 pF	XXIV	Q	0	400	0	400	0	400	0	400	mV
BB1-BB14	Output crosstalk (high level), C_L = 50 pF	XXV	Q	0	500	0	500	0	500	0	500	mV
CC	Power supply transient amplitude	XXVI	Q	0	400	0	400	0	400	0	400	mV
DD	Power supply filter capacitance	XXVII	S	0.07	0.13	0.07	0.13	0.07	0.13	0.07	0.13	µF
EE1-EE13	Input capacitance	XXVIII	Q	2	15	2	15	2	15	2	15	pF
EE14-EE16	Input capacitance	XXVIII	Q	2	25	2	25	2	25	2	25	pF
EE17	Input capacitance	XXVIII	Q	2	30	2	30	2	30	2	30	pF

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
EE18,EE20	Input capacitance	XXVIII	Q	2	35	2	35	2	35	2	35	pF
EE19	Input capacitance	XXVIII	Q	3	35	3	35	3	35	3	35	pF
EE21-EE23	Input capacitance	XXVIII	Q	2	45	2	45	2	45	2	45	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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test patterns submitted do not meet these critieria, the additional tests required shall be added to table I. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at L state" or "stuck at H state", the vendor shall be required to submit to the procuring activity and Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) the proposed new set of test patterns for approval.

3.10.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level - Input voltage shall be 0.8 V ±1 percent.

High level - Input voltage shall be 2.0 V ±1 percent.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows (per figure 19 output loads):

Low level - Output voltage shall be 0 V minimum, 0.4 V maximum

High level - Output voltage shall be 2.4 V minimum, V_{CC} maximum

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through EE shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through EE in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through DD shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 6.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 8.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 9. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 6. Following the life test and while still at the maximum class temperature, electrical tests D through DD shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through EE shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/215-1 supersedes part number M28787/215. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Supply voltage		5 V dc ± 10 percent
Maximum power dissipation		2.86 W maximum (module exceeds 2.5 W per fin of MIL-STD-1389)
Input voltage		
Low level, SDI		-0.5 V minimum, 0.7 V maximum
Low level, all other inputs		-0.5 V minimum, 0.8 V maximum
High level		2 V minimum, V_{CC} maximum
Input current		
SDI		
Low level	(V_{IL} = 0.0 V)	-0.78 mA maximum
	(V_{IL} = 0.4 V)	-0.72 mA maximum
High level	(V_{IH} = 2.7 V)	40 µA maximum
	(V_{IH} = 5.5 V)	2 mA maximum
ENA-R, RT, $\overline{\text{DTA-A}}$, $\overline{\text{MCLR-R}}$, Word length a,b,c,d,e,f,g		
Low level	(V_{IL} = 0.0 V)	-1.73 mA maximum
	(V_{IL} = 0.4 V)	-1.6 mA maximum
High level	(V_{IH} = 2.4 V)	40 µA maximum
	(V_{IH} = 5.5 V)	1 mA maximum
STB-CR, OVR-I, DTA 0-R, DTA 1-R, OSC-R		
Low level	(V_{IL} = 0.0 V)	-3.46 mA maximum
	(V_{IL} = 0.4 V)	-3.2 mA maximum
STB-CR, OVR-I		
High level	(V_{IH} = 2.4 V)	80 µA maximum
	(V_{IH} = 5.5 V)	2 mA maximum
$\overline{\text{RCT}}$, DTA 0-R, DTA 1-R		
High level	(V_{IH} = 2.4 V)	120 µA maximum
	(V_{IH} = 5.5 V)	3 mA maximum
$\overline{\text{RCT}}$, SEL-BC		
Low level	(V_{IL} = 0.0 V)	-5.2 mA maximum
	(V_{IL} = 0.4 V)	-4.8 mA maximum

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SEL-BC, OSC-R		
High level	(V_{IH} = 2.4 V)	160 µA maximum
	(V_{IH} = 5.5 V)	4 mA maximum
Input voltage (test inputs)		
Low level		-0.5 V minimum, 0.8 V maximum
High level		2 V minimum, V_{CC} maximum
Input current (test inputs)		
LOC, $\overline{BCC\text{-}I}$		
Low level	(V_{IL} = 0.0 V)	-1.73 mA maximum
	(V_{IL} = 0.4 V)	-1.6 mA maximum
High level	(V_{IH} = 2.4 V)	40 µA maximum
	(V_{IH} = 5.5 V)	1 mA maximum
CLOA		
Low level	(V_{IL} = 0.0 V)	-3.46 mA maximum
	(V_{IL} = 0.4 V)	-3.2 mA maximum
High level	(V_{IH} = 2.4 V)	80 µA maximum
	(V_{IH} = 5.5 V)	1 mA maximum
CLOB		
Low level	(V_{IL} = 0.0 V)	-6.92 mA maximum
	(V_{IL} = 0.4 V)	-6.4 mA maximum
High level	(V_{IH} = 2.4 V)	160 µA maximum
	(V_{IH} = 5.5 V)	4 mA maximum
Output drive current maximum capability, D32, D33		
Low level	(V_{OL} = 0.4 V)	4 mA
High level	(V_{OH} = 2.5 V)	-400 µA
Output voltage		
D32, D33		
Low level	(I_{OL} = 4 mA)	0.4 mA maximum
High level	(I_{OH} = 400 µA)	2.5 V minimum
$\overline{SOW\text{-}R}$		
Low level	(I_{OL} = 16 mA)	0.4 V maximum
High level	(I_{OH} = 400 µA)	2.4 V minimum
RDY-R, ENA-I		
Low level	(I_{OL} = 16 mA)	0.4 V maximum
High level	(I_{OH} = 800 µA)	2.4 V minimum
DTOR		
Low level	(I_{OL} = 9.6 mA)	0.4 V maximum
High level	(I_{OH} = -240 µA)	2.4 V minimum
DEOW		
Low level	(I_{OL} = 11.2 mA)	0.4 V maximum
High level	(I_{OH} = -280 µA)	2.4 V minimum
CLAI		
Low level	(I_{OL} = 12.8 mA)	0.4 V maximum
High level	(I_{OH} = -320 µA)	2.4 V minimum
$\overline{LRC\text{-}R}$		
Low level	(I_{OL} = 14.0 mA)	0.4 V maximum
High level	(I_{OH} = -340 µA)	2.4 V minimum
$\overline{BCC}$, CLBI, EOC		
Low level	(I_{OL} = 14.4 mA)	0.4 V maximum
High level	(I_{OH} = -360 µA)	2.4 V minimum
CLA		
Low level	(I_{OL} = 14.4 mA)	0.4 V maximum
High level	(I_{OH} = -760 µA)	2.4 V minimum

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Output voltage - Continued

URC-R	
Low level (I_{OL} = 15.5 mA)	0.4 V maximum
High level (I_{OH} = -780 µA)	2.4 V minimum
Maximum operating frequency	
D32, D33	
C_L = 50 pF	2 MHz minimum
CLAI, CLBI	
C_L = 50 pF	2 MHz minimum
Pulse width (t_{pW})	
CLAI	71 ns minimum, 93 ns maximum
CLBI	220 ns minimum, 390 ns maximum

* 6.5 Module function. This module contains the logic necessary to facilitate control of serial data reception. When used in conjunction with other modules, including registers, line drivers, line receivers, line terminators and oscillators, the control module is capable of processing asynchronous or synchronous serial transmission at rates up to 2 MHz.

6.5.1 Transmission types. There are four serial transmission types possible when the SSO and SSI modules are used. These types include asynchronous or synchronous transmission, each of which can be either controlled or uncontrolled. The asynchronous transmission differs from the synchronous type in that the former accomplishes message transfer without the necessity of a synchronizing clock line from the transmitter to the receiver. The control of either transmission type is accomplished by the use of discrete control lines connected between the appropriate subsystems.

6.5.2 Input to output relationship. Each SSI module has the input to output relationships of figures 2, 3, and 4. The length of the data word is programmed in accordance with table II.

6.5.3 Input and output definitions. The interface to the SSI module consists of several types of general interfaces. These include user controls, clocks and strobes, data paths, test point jumpers, and serial line interfaces. Test point jumper inputs and outputs are for test purposes only.

6.5.3.1 User controls. The user controls consist of $\overline{\text{MCLR-R}}$, SEL-BC, ENA-R, DTA-A, RDY-R, DEOW and word length.

$\overline{\text{MCLR-R}}$: MASTER CLEAR, RECEIVER is an active low input pulse which is set active by the user to clear all internal SSI control logic. Random input data occurring after $\overline{\text{MCLR-R}}$ will be accepted by the SSI thus shifting the SSI from the "cleared" state.

SEL-BC: SELECT-CLOCK is an active high input level which is used to select the source of the internal SSI clock. If SEL-BC is a logic high, the SSI internal clock is sourced from OSC-R, if logic low, the internal clock is sourced from STB-CR.

ENA-R: ENABLE RECEIVER is an active high input level which is used to generate ENA-I output signal.

$\overline{\text{DTA-A}}$: DATA ACCEPTED is an active low input signal level which is activated by the user electronics, after the data which is present in holding registers has been strobed into the user electronics. RDY-R will be deactivated as a result of the activity of $\overline{\text{DTA-A}}$.

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RDY-R — READY is an active high output level and indicates, when active, that a stable word exists in the SSI holding register and is ready to be transferred into the user electronics. RDY-R will be cleared by OVR-I, $\overline{\text{MCLR-R}}$, or $\overline{\text{DTA-A}}$.

DEOW: — DATA END OF WORD is an active high output level that indicates that the last bit of the pin programmed word has been clocked into the SSI shift register.

Word length: — These seven inputs are pin programmed to identify the number of bits to be contained in each serial word that will be transmitted across the serial interface. The range of serial word lengths range from 1 to 127 bits in length. The usual word length in the fire control system is 34 bits and will be programmed: A=1, B=0, C=1, D=1, E=1, F=0, and G=1. Other word lengths are shown in table II. Start-of-Word (SOW) and End-of-Word (EOW) bits are not counted by the SSI bit counter.

6.5.3.2 Clocks and strobes. The clocks and strobes in the SSI include OSC-R, CLA, URC-R, and $\overline{\text{LRC-R}}$.

OSC-R: — OSCILLATOR RECEIVER is an active high input pulse train which when selected by SEL-BC will control the internal timing of the SSI module. For all asynchronous transmission modes or anytime STB-CR is not used, an OSC-R signal is required. Refer to mode notes for OSC-R criteria.

CLA: — CLOCK A is an active high output pulse train that has the same frequency as the input serial data changes on the DTA-1 or DTA-0 lines and will change coincident with OSC-R changes. CLA is active in the asynchronous mode or filtered synchronous mode.

URC-R: — UPPER RANK CLOCK-RECEIVER is an active high output pulse train which is used to clock each bit from the serial interface into the SSI shift register. The clock is generated either by CLA or STB-CR depending upon the level of SEL-BC.

$\overline{\text{LRC-R}}$: — LOWER RANK CLOCK-RECEIVER is an active low signal generated by the SSI to load a full serial word from the shift register into the holding register.

6.5.3.3 Data paths. The data paths in the SSI, neglecting the serial interface, include D33, D32, and SDI.

D33: — DATA BIT 33 is the MSB of the parallel word available as an output to the user, assuming a 34-bit word. If the word is shorter than 34 bits, this bit location will be unused.

D32: — DATA BIT 32 is the next bit in succession from the MSB assuming a 34-bit word. If the serial word is shorter than 33 bits, this bit location will be unused.

SDI: — SERIAL DATA INPUT is the serial line into the SSI from external register modules. This line feeds the registers whose outputs are D33 and D32.

6.5.3.4 Test point jumpers. The test point jumpers in the SSI include CLBI-CLOB, CLAI-CLOA, $\overline{\text{BCC}}$-$\overline{\text{BCC-I}}$, EOC-LOC, DTOR-RT, SOW-RCT, and P.U. The first term of each dashed pair is the output. Test point jumper inputs and outputs are for test purposes only.

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$\overline{SOW-R}$: The Start-of-Word output is generated by ANDing DTA 1-R and DTA 0-R. In the broadcast mode $\overline{SOW-R}$ is tied to the counter reset ($\overline{RCT}$), and will reset the bit counter during SOW time to assure sender-receiver synchronization for each word.

$\overline{RCT}$: Reset Counter is an active low input used to reset the SSI bit counter. In the broadcast mode, $\overline{RCT}$ is tied to $\overline{SOW-R}$ in order that the bit counter can be reset during Start-Of-Word time.

P.U.: PULL UP is a static high logic level output signal which is available to the user for other modules if the appropriate conditions exist. There is a capability for 30-unit loads when the user is in the high state. A unit load is equal to 40 µA.

6.5.3.5 Serial line interfaces. The serial line interface includes ENA-I, OVR-I, DTA 1-R, DTA 0-R, and STB-CR.

ENA-I: ENABLE is an active high output level which in general is used to control the word transmission frequency into the SSI. When ENA-I is logic high, the SSI registers are in the position to receive data. ENA-I is active if ENA-R is active and the SSI:

a. Ready signal (RDY-R) is inactive, or
b. Has a current word in its holding register (RDY-R is a high level) and the next word has not started to enter the SSI.

If the first bit of the next word has arrived, and the conditions in b. exist, the ENA-I becomes inactive. Transmission of that word will continue until the bit count has expired.

OVR-I: OVERRIDE is an active high input pulse which clears the SSI control logic in preparation for serial word acceptance. OVR-I or $\overline{MCLR-R}$ will clear the RDY-R signal.

DTA 1-R: DATA 1 is an active high serial data input line. The information contained on this line is a TRUE representation of the data transferred from the sender. This line becomes active only when the logic highs are transferred. The data can either be RZ or NRZ coded which is dependent upon the serial digital transmission type. RZ coding is generally used in the asynchronous transmission mode.

DTA 0-R: DATA 0 is an active high serial data input line. The information contained on this line is a FALSE representation of the data transferred from the sender. The line becomes active only when logic lows are transferred. The data can either be RZ or NRZ coded which is dependent upon the serial digital transmission type. RZ coding is generally used in the asynchronous transmission mode. Both DTA 1-R and DRA 0-R are active during the first bit time of each word (SOW) when asynchronous transmission is used.

STB-CR: STROBE is an active high input pulse train which is sourced at the serial interface. In general, STB-CR is used instead of the OSC-R to generate internal SSI timing when synchronous transmission is used. If STB-CR were used, SEL-BC must be at a logic low level.

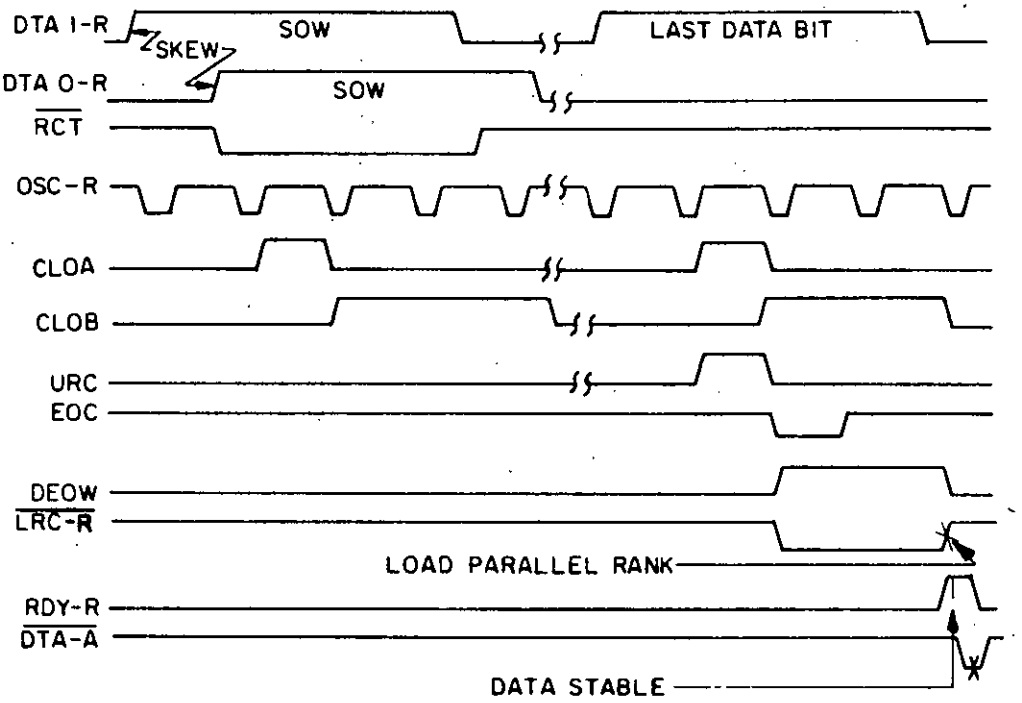
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6.5.4 Pin nomenclature. Module pin assignments and the function description are as follows:

Function	Pin no.	Description
ENA-R	2	Enable receiver
RDY-R	3	Ready
RT	4	Reset, timing
P.U.	5	Pullup (internal)
1/ CLOB	6	Clock OB
DEOW	7	Data end-of-word
1/ CLBI	8	Clock B
D32	29	Data bits
D33	9	Data bits
1/ CLAI	11	Clock A
1/ $\overline{\text{BBC-1}}$	12	Bit counter clock-1
a, b, ---- g	39,20,18,38,33,17,13	Word length lines
$\overline{\text{RCT}}$	14	Bit counter reset
DTA 0-R	15	Data 0
$\overline{\text{SOW-R}}$	16	Start-of-word
1/ $\overline{\text{BCC}}$	19	Bit counter clock
$\overline{\text{LRC-R}}$	21	Lower rank clock
ENA-I	22	Enable (transmission)
URC-R	23	Upper rank clock
STB-CR	24	Strobe
$\overline{\text{DTA-A}}$	25	Data accepted
SDI	26	Serial data input
1/ CLOA	27	Clock OA
OVR-I	28	Override
$\overline{\text{MCLR-R}}$	30	Master clear receiver
1/ LOC	31	Load counter
1/ EOC	32	End of count
OSC-R	34	Oscillator, receiver
DTOR	35	Data OR'd
DTA 1-R	36	Data 1
SEL-BC	37	Select Clock
CLA	40	Clock A

1/ These pins are jumpers provided for testing purposes only. It is not intended they be used for any other purpose.

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NOTES:
1. Maximum data rate = 1 MHz.
2. Minimum positive data lobe = 470 ns.
3. Minimum negative data lobe = 250 ns.
4. User clock frequency = See note 8.
5. Maximum skew = 60 ns/200 ns (see note 9).
6. In the broadcast mode, internal SSI timing is generated as indicated.
7. User clock must comply with criteria defined in note 8.

FIGURE 2. Broadcast (mode timing).

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NOTES:
8. User clock (OSC-R).

The following criteria shall apply to the user clock:



a. Positive lobe shall be a minimum of 60 ns at 2 V.
b. Negative lobe shall be a minimum of 40 ns at 0.8 V.

c. The minimum clock frequency shall be such that two (2) complete clock periods occur in the time that remains when 240 ns are subtracted from the positive data lobe.

Example:

Data lobe width 500 ns - 240 = $\frac{260}{2}$

Therefore, clock period = $\frac{260}{2}$ = 130 ns (7.66 MHz).

From the example it can be seen that the 7.66 MHz HVA output will not permit reliable operation for a positive data lobe whose width is < 500 ns.

d. The maximum clock frequency (OSC-R) will be limited by the criteria set forth in a and b (10 MHz).

9. With skew in excess of 60 ns between data lines (data 1 MHz data rate) a URC and $\overline{\text{BCC}}$ pulse may be generated during SOW time. This will not affect the internal operation of the SSI because the counter load lines are brought low later during SOW time. If these signals are used externally, however, they will not relate correctly to the actual data bits received.

A maximum skew of 200 ns between data lines shall apply when URC and $\overline{\text{BCC}}$ are not used external to the SSI and the Serial to Parallel Converter (EQF) (MIL-M-28787/214).

This skew tolerance in this case relates to negative data lobe width and will increase as lobe width increases.

10. Minimum pulse width shall be 60 ns for the following inputs:

OVR-I	Override
$\overline{\text{MCLR-R}}$	Master clear receiver
$\overline{\text{DTA-A}}$	Data accepted

11. RDY-R - Data in the Holding Register will be stable a maximum of 50 ns after RDY-R becomes active.

FIGURE 2. Broadcast (mode timing) - Continued.

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12. Broadcast Mode Sync Clearing Technique for Data Lobes > 460 ns and < 500 ns.

The clearing technique depicted below will permit reliable EOJ operation for data lobes greater than 460 ns and less than 500 ns using a 7.66 MHz User Clock (OSC-R).

This methods uses the first negative going clock edge after DTOR falls to reset the sync circuit. This will provide additional counter cycle time under worst case sync generation conditions. The figure below shows the waveform relationships in this mode. It can be seen that this method will delay the fall of the CL08 clock until approximately 525 ns have elapsed from the leading edge of DTOR, thereby providing adequate time for the bit counter to cycle. This additional time permits reliable operation with DATA 0 and DATA 1 lobes as low as 460 ns.

It must be noted that this method does reduce skew tolerance by a maximum of one User Clock (OSC-R) period. This should not cause any problems under normal circumstances since the negative lobe widths are usually equal to positive lobe widths and allow more than adequate time for skew accommodation.

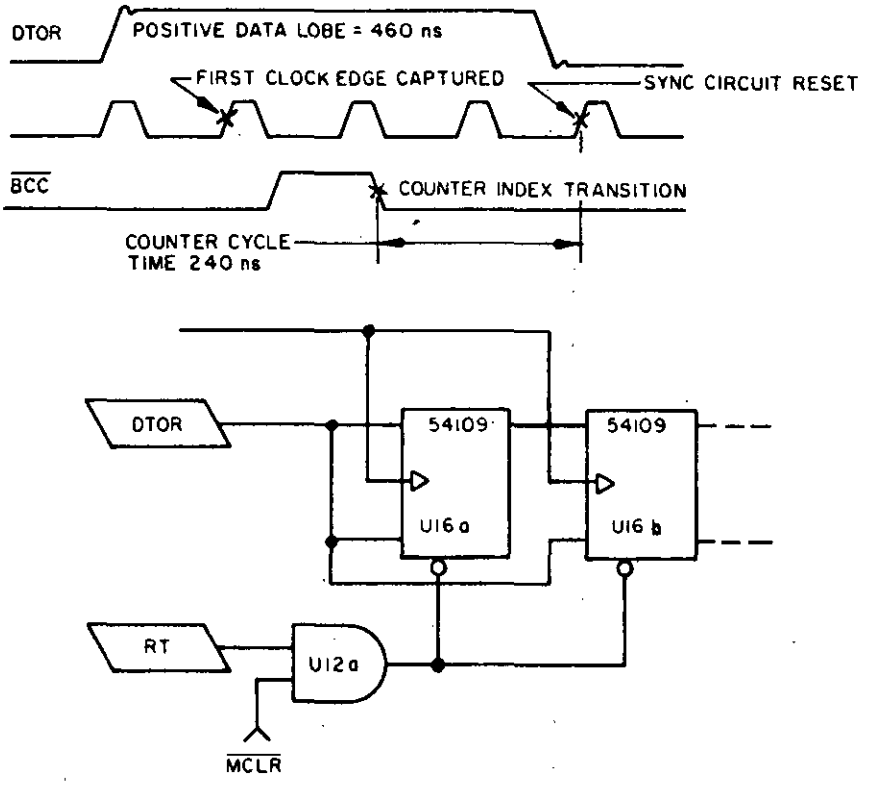
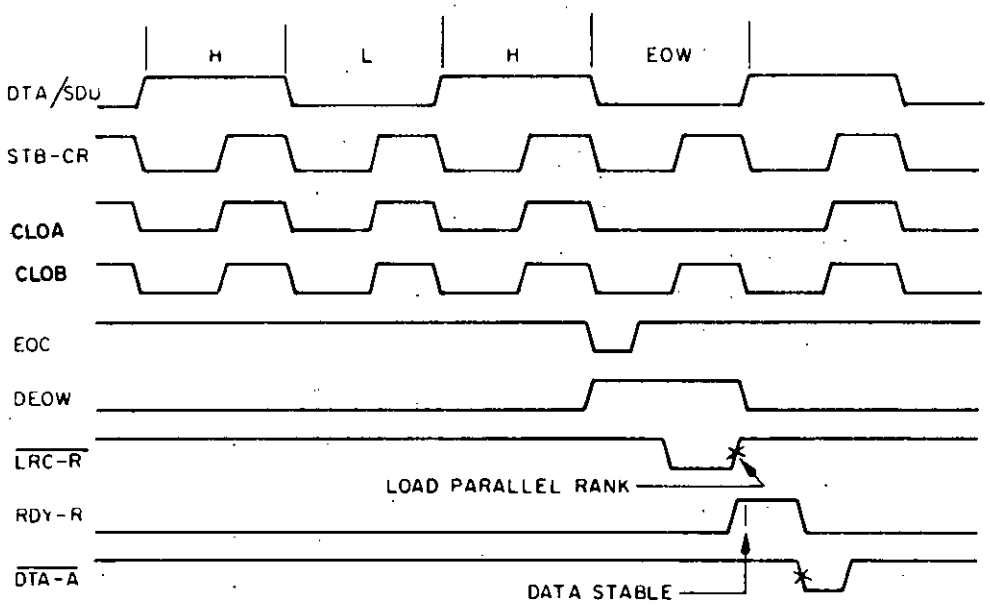


FIGURE 2. Broadcast (mode timing) - Continued.

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NOTES:
1. Maximum data rate = 2 MHz.
2. Minimum STB-CR positive lobe = 190 ns.
3. Maximum skew DTA STB at 2 MHz = 90 ns (see note 5).
4. In control serial L, internal SSI timing is generated as indicated.

FIGURE 3. Control serial "L".

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NOTES:

5. At the maximum data rate, skew tolerance is limited by the positive lobe width of STB-CR and the internal delay of the SSI in generating the URC signal. At lower transmission rates, the skew tolerance can be determined by subtracting 100 ns from the width of the positive lob of STB-CR.

   Example:

   Data rate = 1 MHz
   STB-CR Positive Lobe Width = 400 ns

   Therefore, 400 ns - 100 ns = 300 ns skew tolerance.

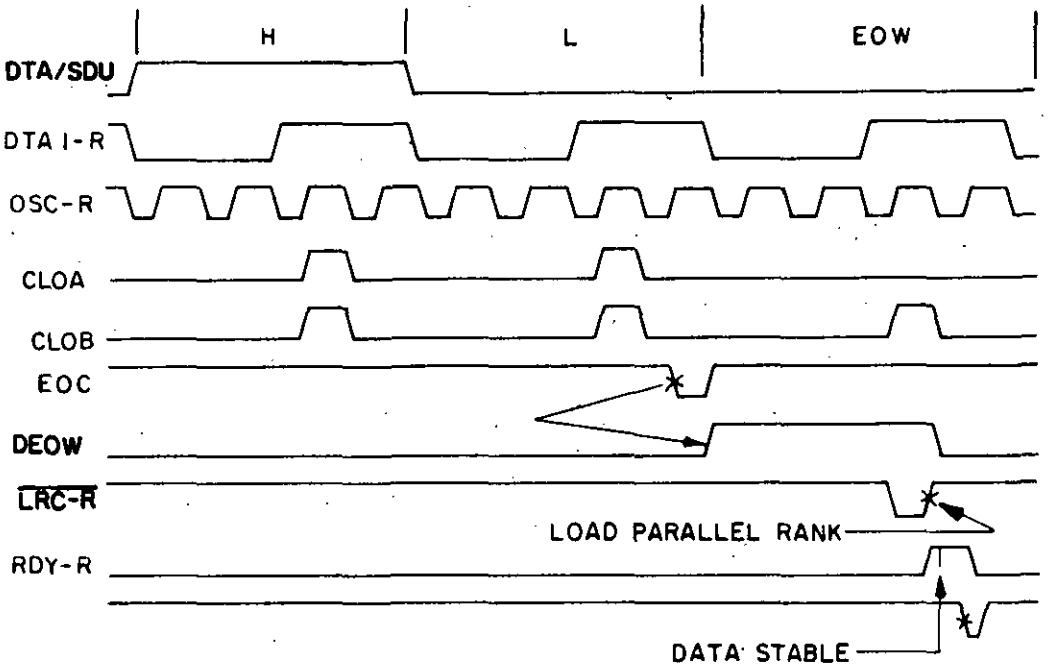
6. Minimum pulse width shall be 60 ns for the following inputs:

OVR-I	Override
$\overline{\text{MCLR-R}}$	Master Clear
$\overline{\text{DTA-A}}$	Data Accepted

7. RDY-R - Data in the Holding Register will be stable a maximum of 50 ns after RDY-R becomes active.

FIGURE 3. Control serial "L" - Continued.

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NOTES:
1. Maximum data rate = 1 MHz.
2. Minimum strobe lobe at DTA 1-R input = 470 ns.
3. User clock frequency = See note 7.
4. Maximum skew = See note 8.
5. In Control Serial H, internal SSI timing is generated as indicated.
6. User clock must comply with criteria defined in note 7.

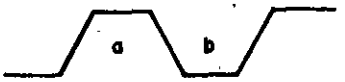
FIGURE 4. Control serial "H".

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NOTES:

7. User clock - (OSC-R)

The following criteria shall apply to the user clock.



a. Positive lobe shall be a minimum of 60 ns at 2 V.
b. Negative lobe shall be a minimum of 40 ns at 0.8 V.
c. The minimum clock frequency shall be such that two (2) complete clock periods occur in the time that remains when the following considerations are subtracted from the positive lobe of the strobe appearing at the DTA 1-R input of the SSI.

(a). Required skew tolerance between the data at the SDI and the strobe input to the SSI DTA 1-R input.
(b). 100 ns to provide for delay and clock criteria internal to the SSI and SPC.

Figure 4 covers a range of usage parameters.

Example:

Assume the following:

Data rate = 1 MHz
Positive STB Lobe = 470 ns
Required skew tolerance = 100 ns + 100 ns SSI factor (b) = 200 ns

$$470 \text{ ns} - 200 \text{ ns} = \frac{270 \text{ ns}}{2} \text{ (clock periods)}$$

Therefore, clock period = $\frac{270}{2}$ = 135 ns (7.41 MHz).

8. Maximum skew - Skew tolerance of the SSI in control serial "H" is a function of the following:

a. The positive lobe width of the strobe input applied to the SSI DTA 1-R terminal.
b. The frequency of the user clock (OSC-R).
c. A fixed 100 ns factor related to the internal delay of the SSI and the clocking requirement of the Serial-Parallel Converter (SPC). Skew tolerance can be determined by subtracting two (2) user clock (OSC-R) periods and the 100 ns factor from the positive lobe width of the strobe input applied to the SSI DTA 1-R terminal.

Figure 4 covers a range of usage parameters.

9. Minimum pulse width shall be 60 ns for the following inputs:

OVR-I	Override
$\overline{\text{MCLR-R}}$	Master Clear
$\overline{\text{DTA-A}}$	Data Accepted

10. RDY-R - Data in the Holding Register will be stable a maximum of 50 ns after RDY-R becomes active.

FIGURE 4. Control serial "H" - Continued.

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11. Digital filtering in Control Serial H

The digital filtering inherent in the Control Serial H mechanization provides noise immunity on the Strobe line from the Serial Output Element. This prevents generation of false sync (CLOA) pulses due to switching transients or line reflections for the duration of time that the user clock (OSC-R) is low. Therefore, the Strobe line has the user clock (OSC-R) low time to become stable. Inasmuch as the strobe is active during the second half of a bit time, it can be assumed the data line will already have become stable.

A typical situation might be:

User clock (OSC-R) = 1.0 MHz (50% duty-cycle).

Therefore, masking would exist for one half (1/2) the period of 1.0 MHz, or a minimum of 500 ns.

It can be seen, therefore, that lower frequency clock rates yield a longer masking interval. Clock criteria are covered above.

FIGURE 4. Control serial "H" - Continued.

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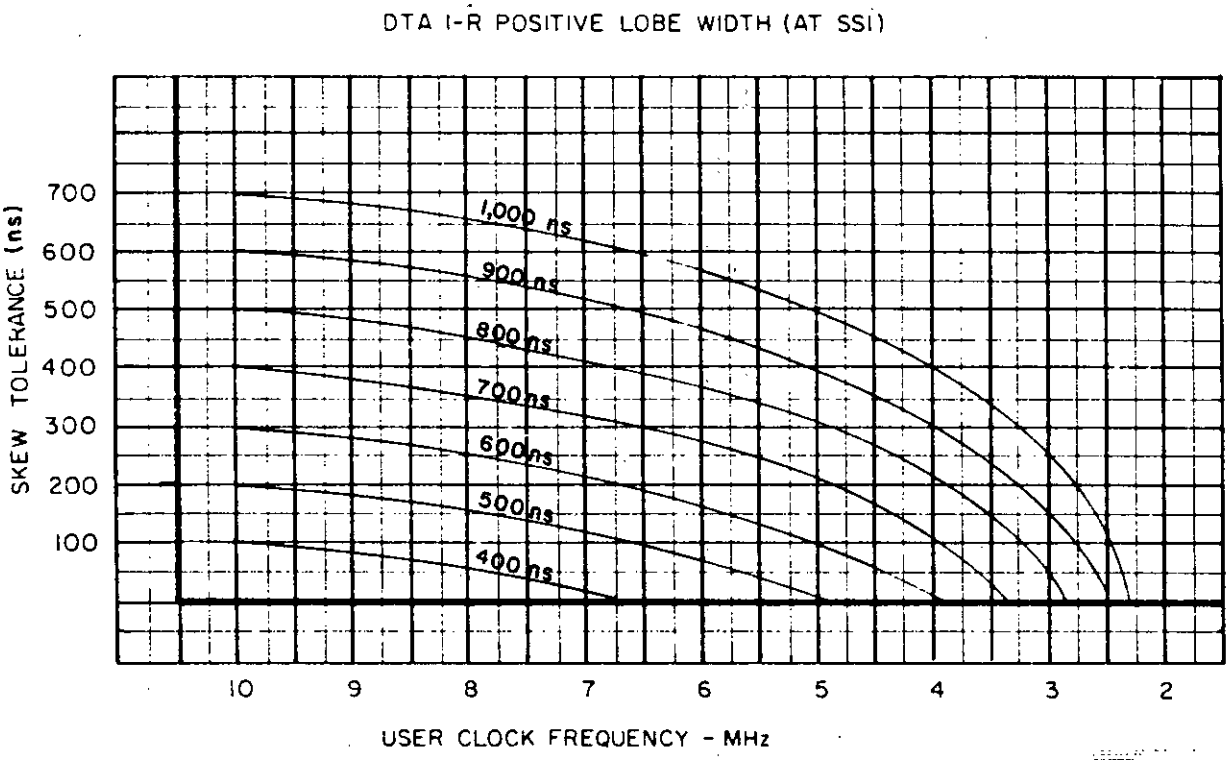


FIGURE 5. Control serial "H" interface variables.

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Table II. Counter pin programming (SSI). 1/ 2/

Mode - broadcast, control L and control H							
39	20	18	38	33	17	13	Word length (bits)
H	H	H	H	H	H	H	0
L	H	H	H	H	H	H	1
H	L	H	H	H	H	H	2
L	L	H	H	H	H	H	3
H	H	L	H	H	H	H	4
L	H	L	H	H	H	H	5
H	L	L	H	H	H	H	6
L	L	L	H	H	H	H	7
H	H	H	L	H	H	H	8
L	H	H	L	H	H	H	9
H	L	H	L	H	H	H	10
L	L	H	L	H	H	H	11
H	H	L	L	H	H	H	12
L	H	L	L	H	H	H	13
H	L	L	L	H	H	H	14
L	L	L	L	H	H	H	15
H	H	H	H	L	H	H	16
L	H	H	H	L	H	H	17
H	L	H	H	L	H	H	18
L	L	H	H	L	H	H	19
H	H	L	H	L	H	H	20
L	H	L	H	L	H	H	21
H	L	L	H	L	H	H	22
L	L	L	H	L	H	H	23
H	H	H	L	L	H	H	24
L	H	H	L	L	H	H	25
H	L	H	L	L	H	H	26
L	L	H	L	L	H	H	27
H	H	L	L	L	H	H	28
L	H	L	L	L	H	H	29
H	L	L	L	L	H	H	30
L	L	L	L	L	H	H	31
H	H	H	H	H	L	H	32
L	H	H	H	H	L	H	33
H	L	H	H	H	L	H	34
L	L	H	H	H	L	H	35
H	H	L	H	H	L	H	36
L	H	L	H	H	L	H	37
H	L	L	H	H	L	H	38
L	L	L	H	H	L	H	39
H	H	H	L	H	L	H	40
L	H	H	L	H	L	H	41
H	L	H	L	H	L	H	42
L	L	H	L	H	L	H	43
H	H	L	L	H	L	H	44
L	H	L	L	H	L	H	45
H	L	L	L	H	L	H	46
L	L	L	L	H	L	H	47
H	H	H	H	L	L	H	48
L	H	H	H	L	L	H	49
H	L	H	H	L	L	H	50
L	L	H	H	L	L	H	51
H	H	L	H	L	L	H	52
L	H	L	H	L	L	H	53
H	L	L	H	L	L	H	54
L	L	L	H	L	L	H	55
H	H	H	L	L	L	H	56

See footnotes at end of table.

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Table II. Counter pin programming (SSI) - Continued. 1/ 2/

Mode - broadcast, control L and control H							
39	20	18	38	33	17	13	Word length (bits)
L	H	H	L	L	L	H	57
H	L	H	L	L	L	H	58
L	L	H	L	L	L	H	59
H	H	L	L	L	L	H	60
L	H	L	L	L	L	H	61
H	L	L	L	L	L	H	62
L	L	L	L	L	L	H	63
H	H	H	H	H	H	L	64
L	H	H	H	H	H	L	65
H	L	H	H	H	H	L	66
L	L	H	H	H	H	L	67
H	H	L	H	H	H	L	68
L	H	L	H	H	H	L	69
H	L	L	H	H	H	L	70
L	L	L	H	H	H	L	71
H	H	H	L	H	H	L	72
L	H	H	L	H	H	L	73
H	L	H	L	H	H	L	74
L	L	H	L	H	H	L	75
H	H	L	L	H	H	L	76
L	H	L	L	H	H	L	77
H	L	L	L	H	H	L	78
L	L	L	L	H	H	L	79
H	H	H	H	L	H	L	80
L	H	H	H	L	H	L	81
H	L	H	H	L	H	L	82
L	L	H	H	L	H	L	83
H	H	L	H	L	H	L	84
L	H	L	H	L	H	L	85
H	L	L	H	L	H	L	86
L	L	L	H	L	H	L	87
H	H	H	L	L	H	L	88
L	H	H	L	L	H	L	89
H	L	H	L	L	H	L	90
L	L	H	L	L	H	L	91
H	H	L	L	L	H	L	92
L	H	L	L	L	H	L	93
H	L	L	L	L	H	L	94
L	L	L	L	L	H	L	95
H	H	H	H	H	L	L	96
L	H	H	H	H	L	L	97
H	L	H	H	H	L	L	98
L	L	H	H	H	L	L	99
H	H	L	H	H	L	L	100
L	H	L	H	H	L	L	101
H	L	L	H	H	L	L	102
L	L	L	H	H	L	L	103
H	H	H	L	H	L	L	104
L	H	H	L	H	L	L	105
H	L	H	L	H	L	L	106
L	L	H	L	H	L	L	107
H	H	L	L	H	L	L	108
L	H	L	L	H	L	L	109
H	L	L	L	H	L	L	110
L	L	L	L	H	L	L	111
H	H	H	H	L	L	L	112
L	H	H	H	L	L	L	113
H	L	H	H	L	L	L	114

See footnotes at end of table

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Table II. Counter pin programming (SSI) - Continued. 1/ 2/

Mode - broadcast, control L and control H							
39	20	18	38	33	17	13	Word length (bits)
L	L	H	H	L	L	L	115
H	H	L	H	L	L	L	116
L	H	L	H	L	L	L	117
H	L	L	H	L	L	L	118
L	L	L	H	L	L	L	119
H	H	H	L	L	L	L	120
L	H	H	L	L	L	L	121
H	L	H	L	L	L	L	122
L	L	H	L	L	L	L	123
H	H	L	L	L	L	L	124
L	H	L	L	L	L	L	125
H	L	L	L	L	L	L	126
L	L	L	L	L	L	L	127

1/ L - Module pin grounded.
2/ H - Module pin tied to pullup resistor.

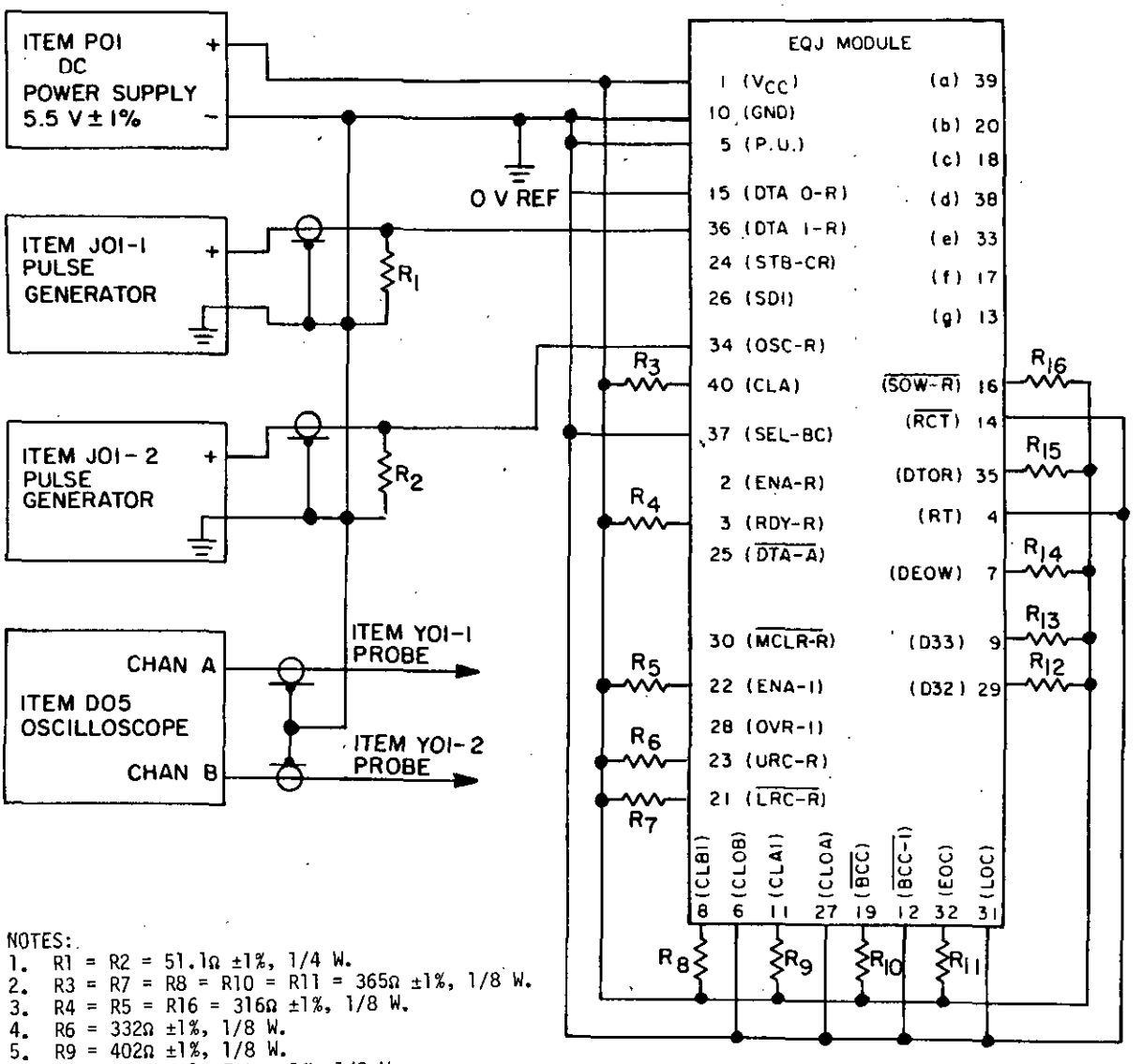
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N606)

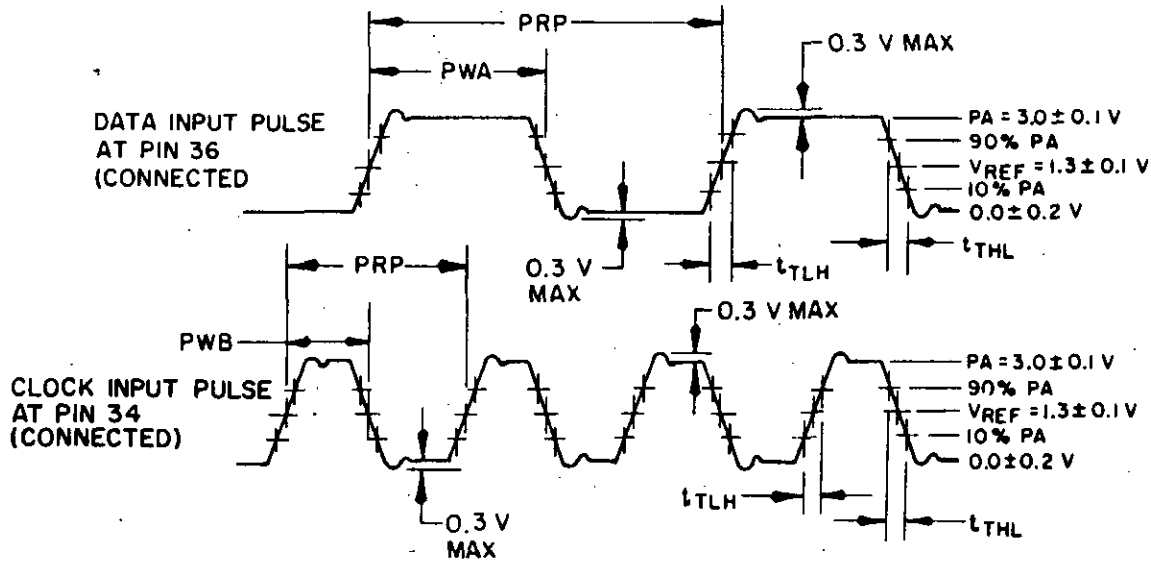
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. R3 = R7 = R8 = R10 = R11 = 365Ω ±1%, 1/8 W.
3. R4 = R5 = R16 = 316Ω ±1%, 1/8 W.
4. R6 = 332Ω ±1%, 1/8 W.
5. R9 = 402Ω ±1%, 1/8 W.
6. R12 = R13 = 1.27 kΩ ±1%, 1/8 W.
7. R14 = 464Ω ±1%, 1/8 W.
8. R15 = 536Ω ±1%, 1/8 W.
9. See figure 7 for input pulse parameters.

* FIGURE 6. Operating temperature and life test circuit.

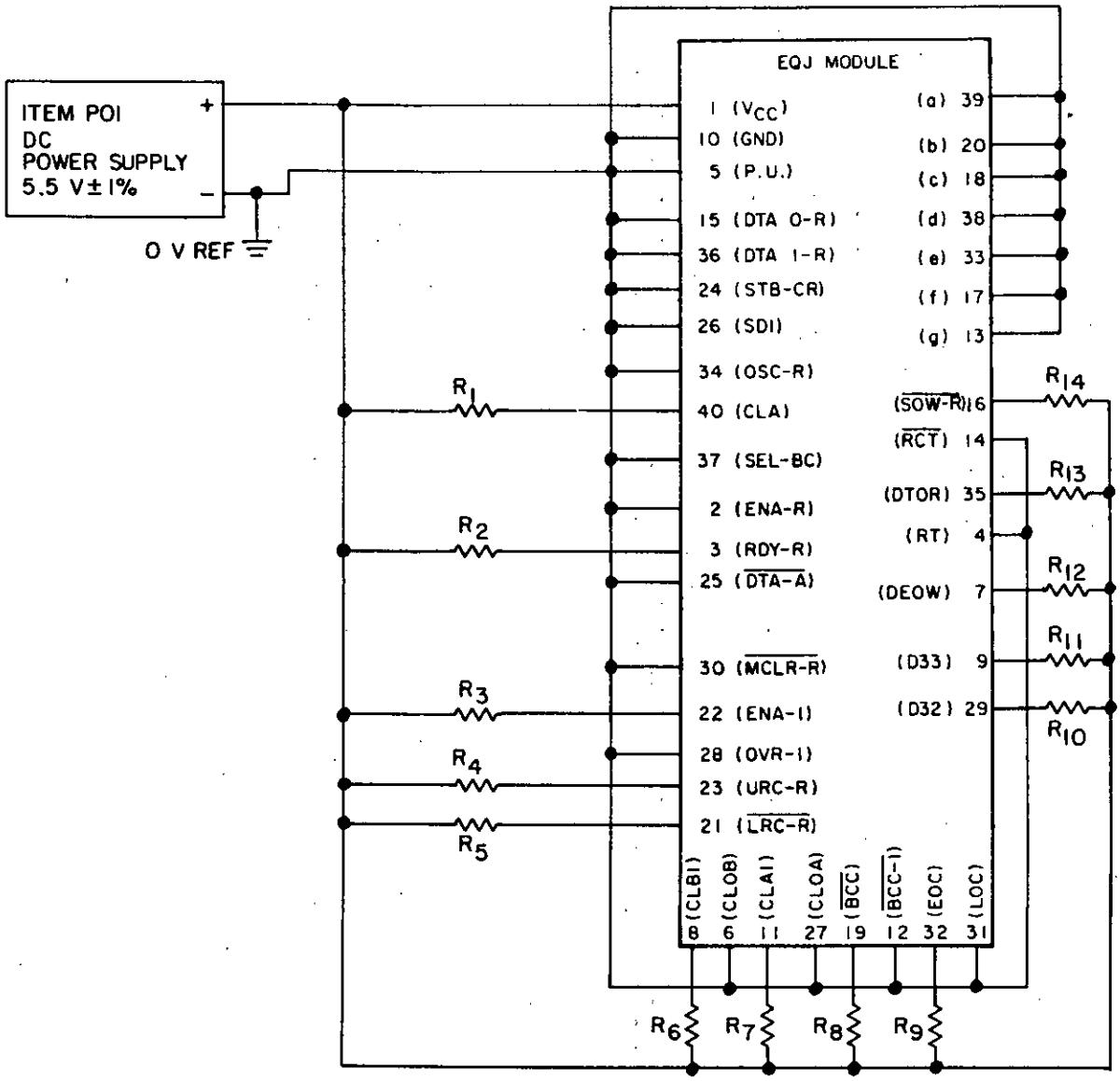
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NOTES:
1. Data input pulse parameters:
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
   $t_{TLH} = t_{THL} \leq 10$ ns.
2. Clock input pulse parameters:
   PWA = 70 ±5 ns.
   PRP = 140 ±10 ns.
   $t_{TLH} = t_{THL} \leq 10$ ns.
3. Verify that signals present on module pins 8 and 11 have a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V.
4. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 7. Operating temperature and life test circuit waveforms.

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NOTES:
1. R1 = R5 = R6 = R8 = R9 = 365Ω ±1%, 1/8 W.
2. R2 = R3 = R14 = 316Ω ±1%, 1/8 W.
3. R4 = 332Ω ±1%, 1/8 W.
4. R7 = 402Ω ±1%, 1/8 W.
5. R10 = R11 = 1.27 kΩ ±1%, 1/8 W.
6. R12 = 464Ω ±1%, 1/8 W.
7. R13 = 536Ω ±1%, 1/8 W.

FIGURE 8. Component temperature test circuit.

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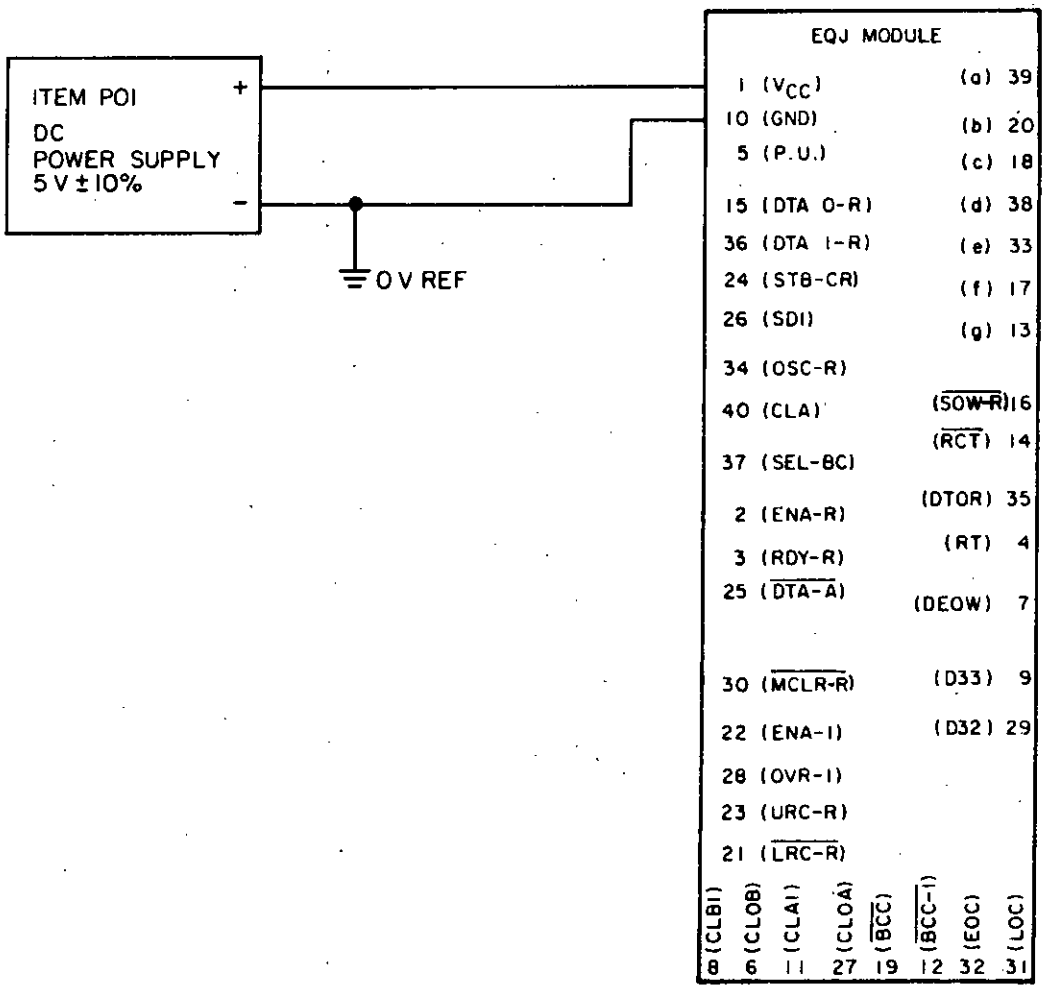


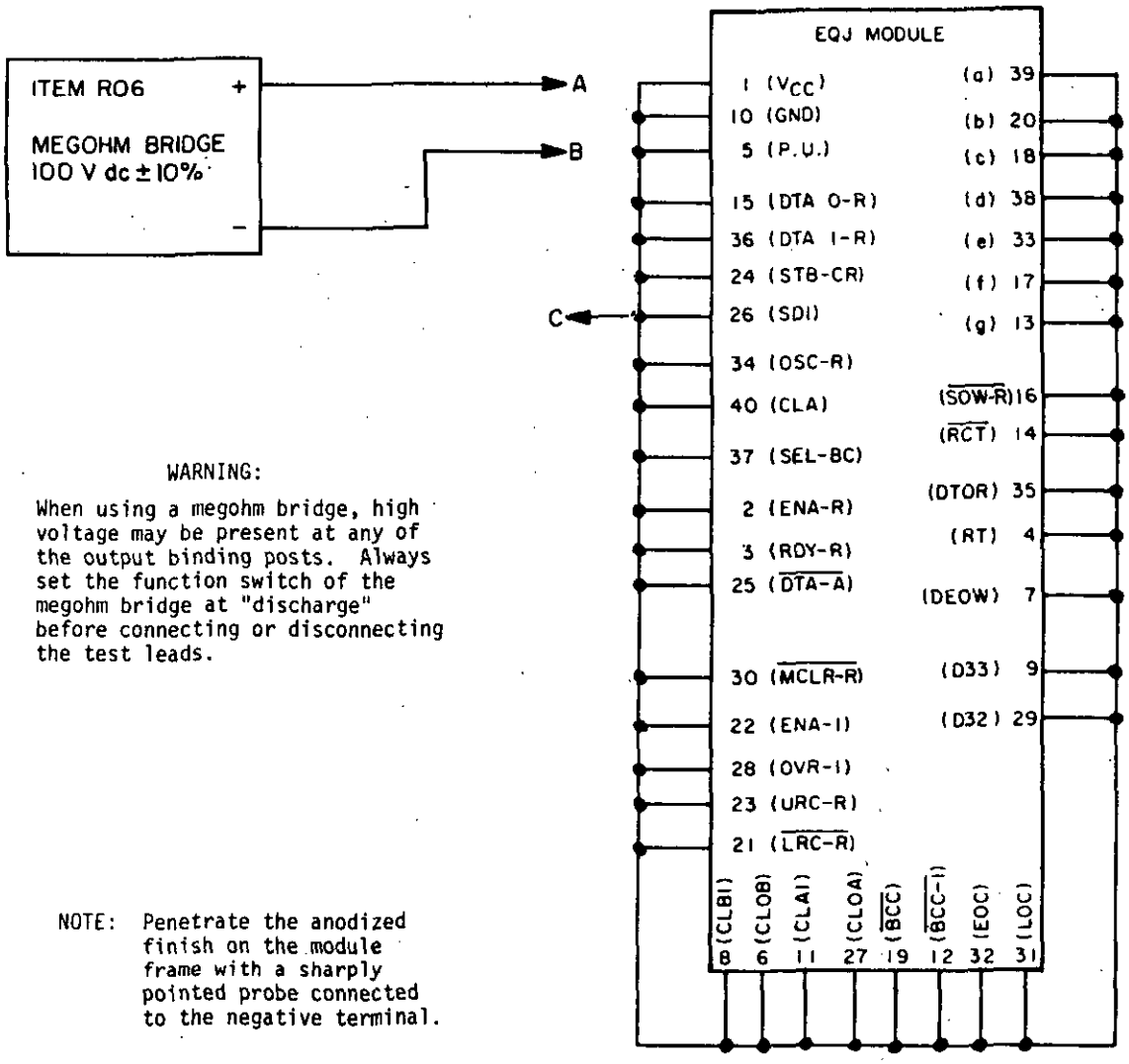
FIGURE 9. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	10

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

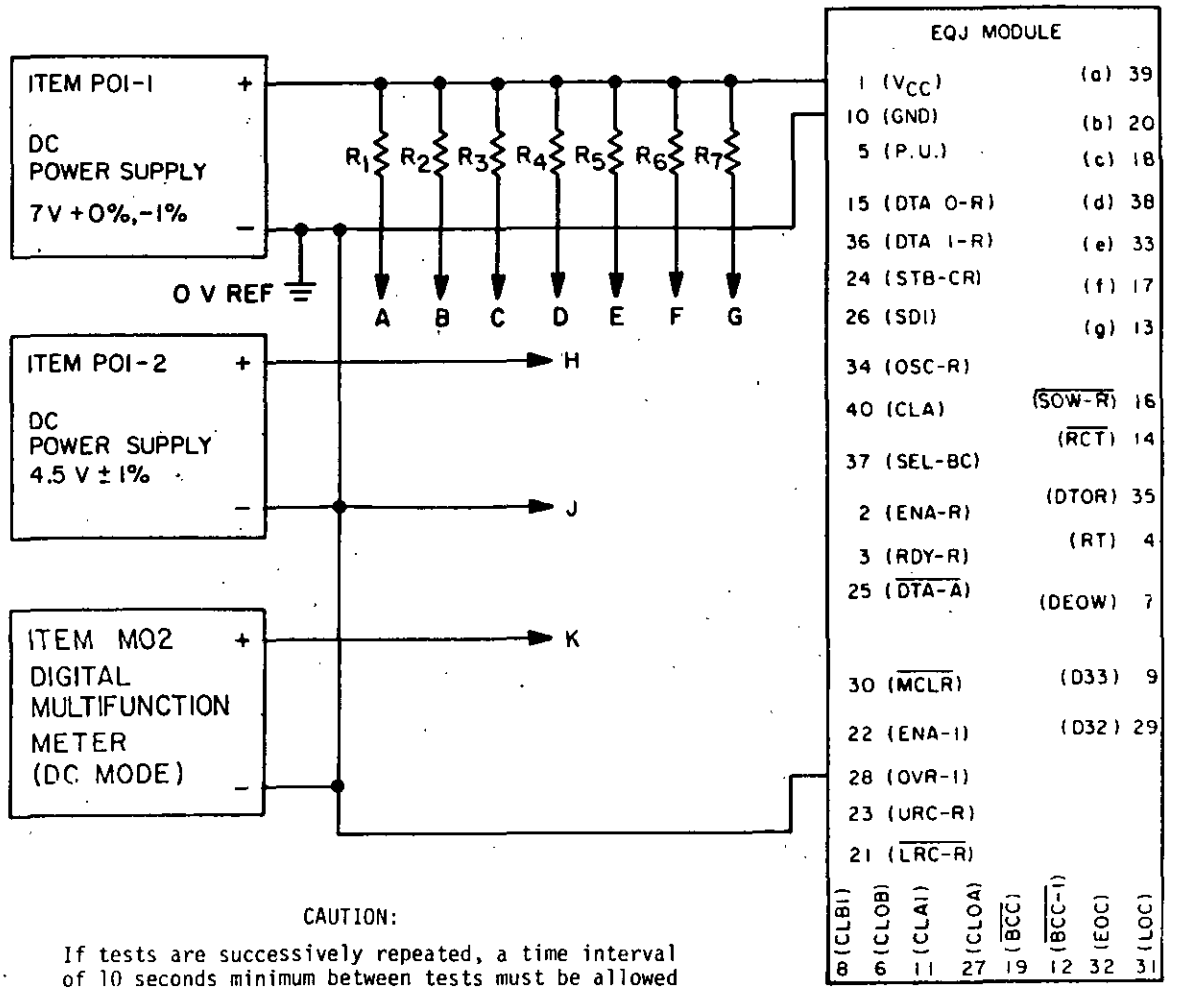
FIGURE 10. Isolation test circuit for table III.

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*TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect										Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	J to pin	K to pin	Min	Max		
B1	RDY-R V_{OUT} (low loaded)	3	Open	Open	Open	Open	Open	Open	Open	30	3	10	800	mV	11
B2	DEOW "	Open	"	Open	"	7	"	"	"	30,39	7	"	"	"	"
B3	CLBI "	"	"	8	"	Open	"	"	"	24,37	8	"	"	"	"
B4	D33 "	"	"	Open	"	"	"	9	"	5	9	"	"	"	"
B5	CLAI "	"	"	"	11	"	"	Open	37	15,36	11	"	"	"	"
B6	$\overline{SOW-R}$ "	16	"	"	Open	"	"	"	15,36	Open	16	"	"	"	"
B7	$\overline{BCC}$ "	Open	"	19	"	"	"	"	14,24	30,37	19	"	"	"	"
B8	$\overline{LRC-R}$ "	Open	"	21	"	"	"	"	13,17, 18,20, 24,33, 38,39	31,37	21	"	"	"	"
B9	ENA-I "	22	"	Open	"	"	"	"	Open	2	22	"	"	"	"
B10	URC-R "	Open	23	"	"	"	"	"	"	14	23	"	"	"	"
B11	D32 "	"	Open	"	"	"	"	29	"	5	29	"	"	"	"
B12	EOC "	"	"	32	"	"	"	Open	13,17, 18,20, 33,38, 39	31	32	"	"	"	"
B13	DTOR "	"	"	Open	"	"	35	"	Open	15,36	35	"	"	"	"
B14	CLA "	"	"	40	"	"	Open	"	Open	15,36	40	"	"	"	"

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CAUTION:

If tests are successively repeated, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating.

NOTES:
1. R1 = 316Ω ±1%, 1/8 W.
   R2 = 332Ω ±1%, 1/8 W.
   R3 = 365Ω ±1%, 1/8 W.
   R4 = 402Ω ±1%, 1/8 W.
   R5 = 464Ω ±1%, 1/8 W.
   R6 = 536Ω ±1%, 1/8 W.
   R7 = 1.27 kΩ ±1%, 1/8 W.
2. After point A, B, C, D, E, F, or G has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of five seconds before the measurement is recorded.

FIGURE 11. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

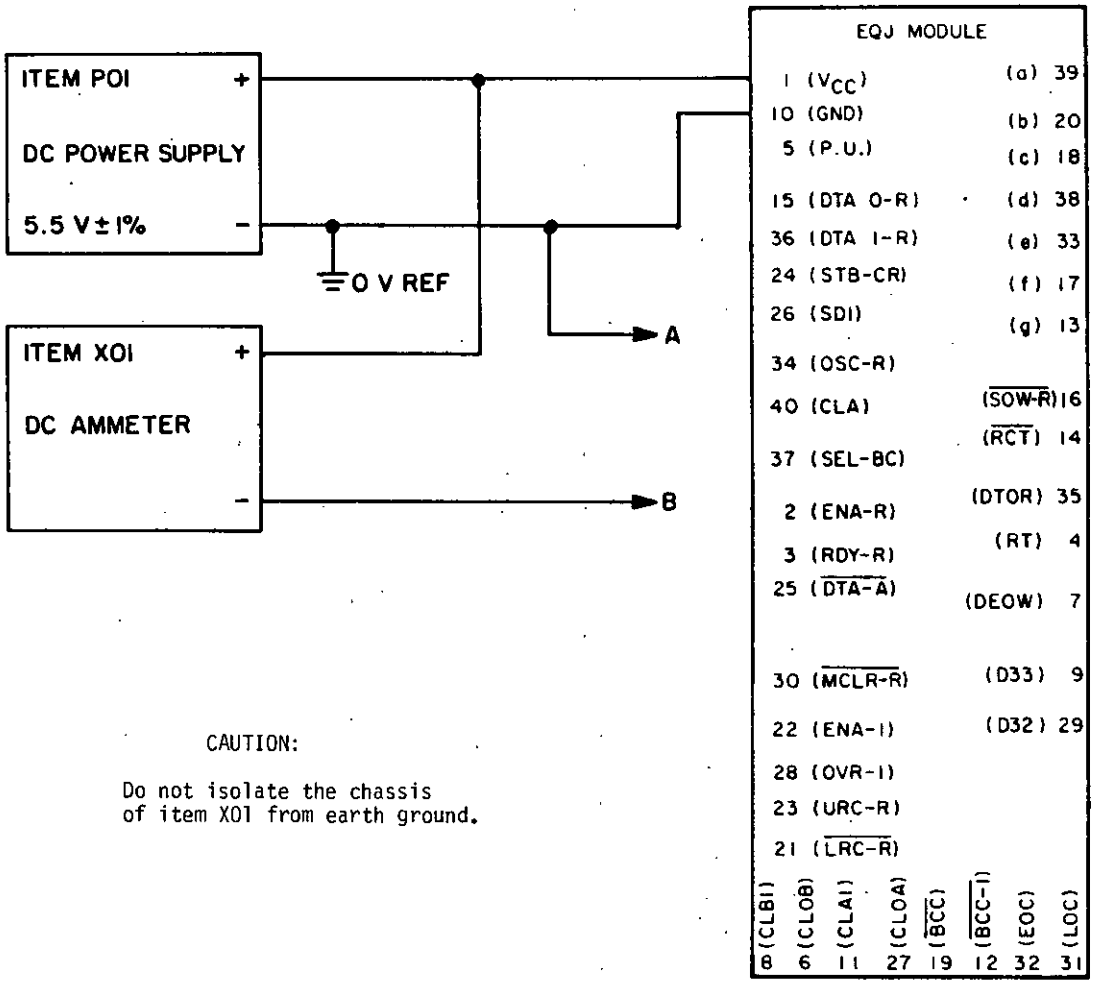
Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
C1	ENA-R	Input breakdown current	4,12,13,17,18,20,25,28 30,31,33,34,38,39,15,24, 26,27,36,14,37,6	2	0	1	mA	12
C2	RT	"	2,12,13,17,18,20,25,28 30,31,33,34,38,39,15,24, 26,27,36,14,37,6	4	"	"	"	"
C3	$\overline{\text{BCC-I}}$	"	2,4,13,17,18,20,25,28 30,31,33,34,38,39,15,24, 26,27,36,14,37,6	12	"	"	"	"
C4	Word length (g)	"	2,4,12,17,18,20,25,28 30,31,33,34,38,39,15,24, 26,27,36,14,37,6	13	"	"	"	"
C5	Word length (f)	"	2,4,12,13,18,20,25,28 30,31,33,34,38,39,15,24, 26,27,36,14,37,6	17	"	"	"	"
C6	Word length (c)	"	2,4,12,13,17,20,25,28 30,31,33,34,38,39,15,24, 26,27,36,14,37,6	18	"	"	"	"
C7	Word length (b)	"	2,4,12,13,17,18,25,28 30,31,33,34,38,39,15,24, 26,27,36,14,37,6	20	"	"	"	"
C8	$\overline{\text{DTA-A}}$	"	2,4,12,13,17,18,20,28 30,31,33,34,38,39,15,24, 26,27,36,14,37,6	25	"	"	"	"
C9	OVR-I	"	2,4,12,13,17,18,20,25 30,31,33,34,38,39,15,24, 26,27,36,14,37,6	28	"	2	"	"
C10	$\overline{\text{MCLR-R}}$	"	2,4,12,13,17,18,20,25 28,31,33,34,38,39,15,24, 26,27,36,14,37,6	30	"	1	"	"
C11	LOC	"	2,4,12,13,17,18,20,25 28,30,33,34,38,39,15,24, 26,27,36,14,37,6	31	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C12	Word length (e)	Input breakdown current	2,4,12,13,17,18,20,25 28,30,31,34,38,39,15,24, 26,27,36,14,37,6	33	0	1	mA	12
C13	OSC-R	"	2,4,12,13,17,18,20,25 28,30,31,33,38,39,15,24, 26,27,36,14,37,6	34	"	"	"	"
C14	Word length (d)	"	2,4,12,13,17,18,20,25 28,30,31,33,34,39,15,24, 26,27,36,14,37,6	38	"	"	"	"
C15	Word length (a)	"	2,4,12,13,17,18,20,25 28,30,31,33,34,38,15,24, 26,27,36,14,37,6	39	"	"	"	"
C16	DTA 0-R	"	2,4,12,13,17,18,20,25 28,30,31,33,34,38,39,24, 26,27,36,14,37,6	15	"	3	"	"
C17	STB-CR	"	2,4,12,13,17,18,20,25 28,30,31,33,34,38,39,15, 26,27,36,14,37,6	24	"	2	"	"
C18	SDI	"	2,4,12,13,17,18,20,25 28,30,31,33,34,38,39,15, 24,27,36,14,37,6	26	"	2	"	"
C19	CLOA	"	2,4,12,13,17,18,20,25 28,30,31,33,34,38,39,15, 24,26,36,14,37,6	27	"	1	"	"
C20	DTA 1-R	"	2,4,12,13,17,18,20,25 28,30,31,33,34,38,39,15, 24,26,27,14,37,6	36	"	3	"	"
C21	$\overline{RCT}$	"	2,4,12,13,17,18,20,25 28,30,31,33,34,38,39,15, 24,26,27,36,37,6	14	"	3	"	"
C22	SEL-BC	"	2,4,12,13,17,18,20,25 28,30,31,33,34,38,39,15, 24,26,27,36,14,6	37	"	4	"	"
C23	CLOB	"	2,4,12,13,17,18,20,25 28,30,31,33,34,38,39,15, 24,26,27,36,14,37	6	"	2	"	"

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CAUTION:

Do not isolate the chassis
of item X01 from earth ground.

FIGURE 12. Input breakdown current test circuit for table V.

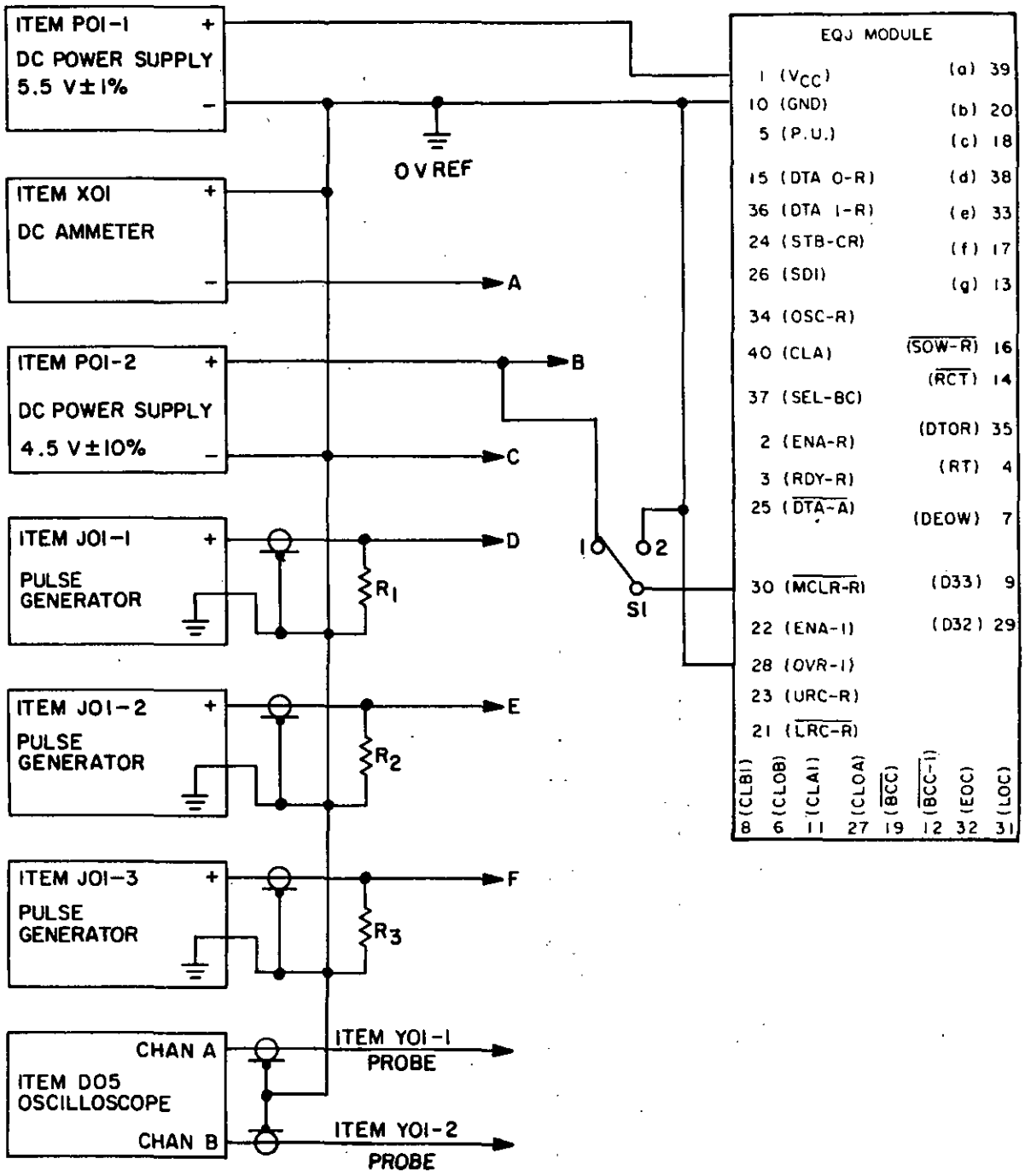
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TABLE VI. Output short circuit current test.

Test condition	Parameter		Connect 1/ 2/ A to pin	B to pin	C to pin	S1 pos	D to pin	E to pin	F to pin	Initial limits 25°C ±5°C ambient Min	Max	Unit	Test figure
D1	RDY-R	Output short circuit current	3	Open	5,24,37	1	Open	Open	Open	-20	-55	mA	13,14
D2	DEOW	"	7	13,17, 18,20, 33,38, 39	31	"	"	"	"	"	"	"	"
D3	CLBI	"	8	24	37	"	"	"	"	"	"	"	"
D4	CLAI	"	11	24	37	2	"	"	"	"	"	"	"
D5	$\overline{\text{SOW-R}}$	"	16	Open	36	1	"	"	"	"	"	"	"
D6	$\overline{\text{BCC}}$	"	19	"	14	1	"	"	"	"	"	"	"
D7	$\overline{\text{LCR-R}}$	"	21	"	Open	2	"	"	"	"	"	"	"
D8	ENA-I	"	22	2	Open	"	"	"	"	"	"	"	"
D9	URC-R	"	23	14,24	37	"	"	"	"	"	"	"	"
D10	EOC	"	32	Open	18,31	1	"	"	"	"	"	"	"
D11	DTOR	"	35	36	15	1	"	"	"	"	"	"	"
D12	CLA	"	40	36,4	15	**	34 (IP)	"	"	"	"	"	"
D13	D32	"	29	4,13, 17,18, 20,24, 26,33, 36,38, 39	15,31	**	"	14 (3P)	37 (1P)	-6	-130	"	"
D14	D33	"	9	4,13, 17,18, 20,24, 26,33, 36,38, 39	15,13	**	"	14 (3P)	37 (1P)	-6	-130	"	"

1/ Connections and operations A thru F and S1 will be performed in order.
2/ In columns D, E, and F, the number in parenthesis indicates the number of times the single pulse button on the indicated pulse generator will be depressed.

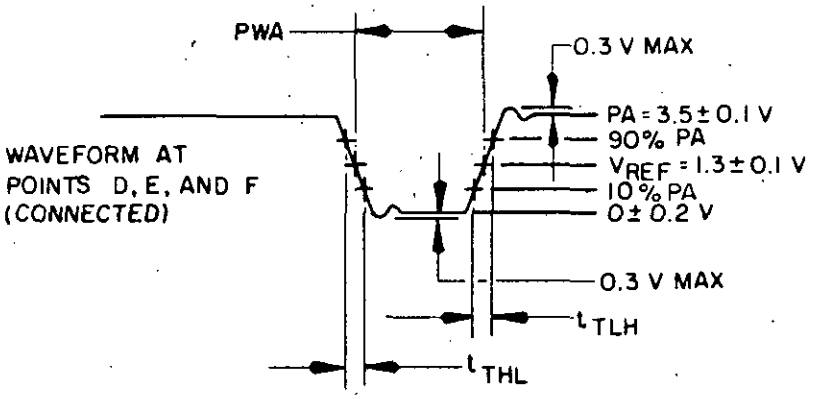
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:43Z
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NOTES:
1. R1 = R2 = R3 = 51.1Ω ±1%, 1/4 W.
2. S1 is a SPDT switch that will normally be in position 1. When indicated in the table by a double asterisk (**), the switch shall momentarily be placed in position 2, then returned to position 1.

* FIGURE 13. Output short circuit current test circuit for table VI.

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NOTES:
1. Data input pulse parameters:
   PWA = 200 ±20 ns.
   t_{THL}, t_{TLH} = 10 ±3 ns.
2. Items J01-1, J01-2, and J01-3 are operated in the single pulse mode. To obtain the proper output pulse characteristics, it is recommended the generator be setup as required at some frequency, then placed in the single pulse mode.

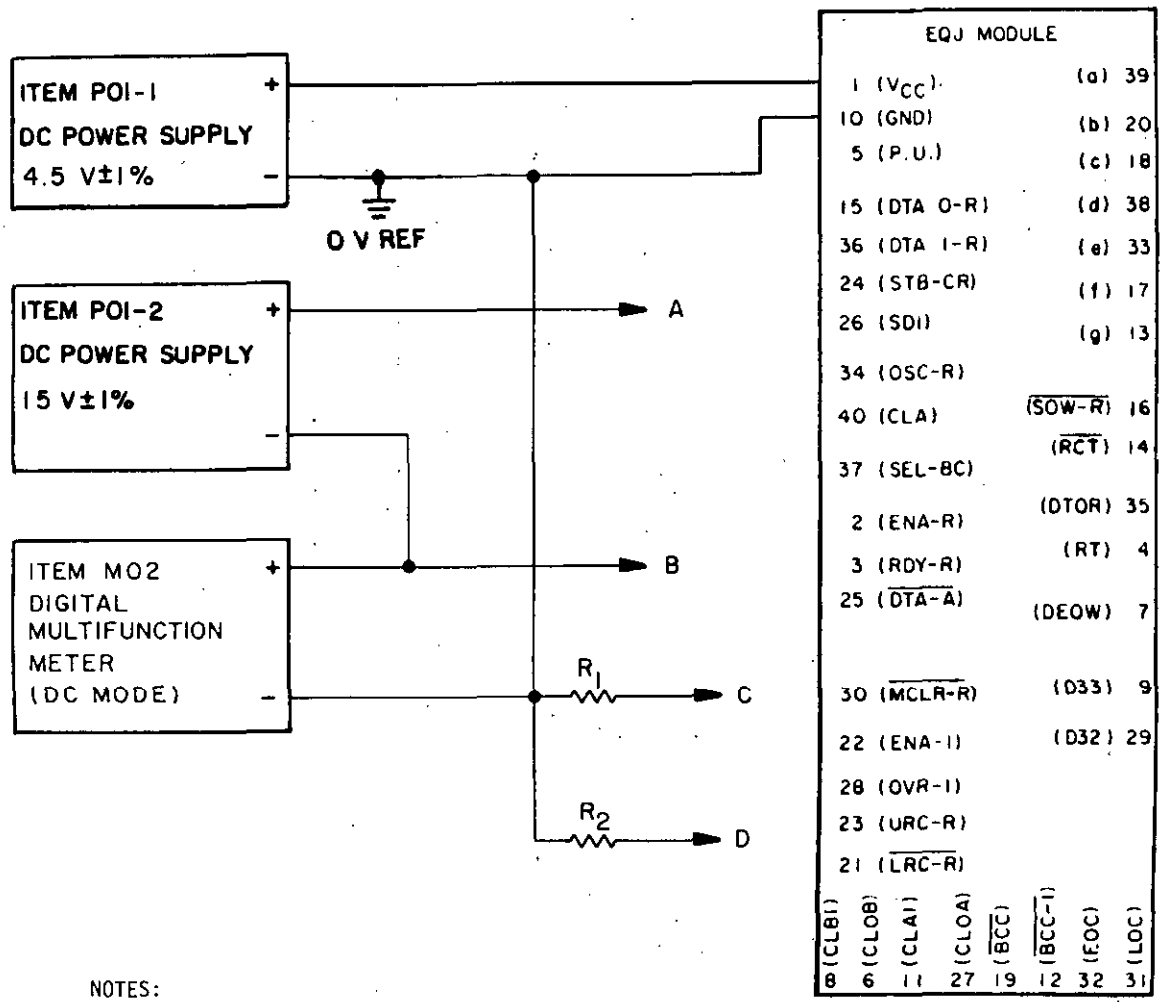
FIGURE 14. Output short circuit current test circuit waveform for table VI.

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TABLE VII. Input clamp diode voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	ENA-R Input clamp diode voltage	C	2	-0.4	-1.5	V	15
E2	RT "	"	4	"	"	"	"
E3	$\overline{\text{BCC-I}}$ "	"	12	"	"	"	"
E4	Word length (g) "	"	13	"	"	"	"
E5	Word length (f) "	"	17	"	"	"	"
E6	Word length (c) "	"	18	"	"	"	"
E7	Word length (b) "	"	20	"	"	"	"
E8	$\overline{\text{DTA-A}}$ "	"	25	"	"	"	"
E9	OVR-I "	"	28	"	"	"	"
E10	$\overline{\text{MCLR-R}}$ "	"	30	"	"	"	"
E11	LOC "	"	31	"	"	"	"
E12	Word length (e) "	"	33	"	"	"	"
E13	OSC-R "	"	34	"	"	"	"
E14	Word length (d) "	"	38	"	"	"	"
E15	Word length (a) "	"	39	"	"	"	"
E16	STB-CR "	"	24	"	"	"	"
E17	SDI "	D	26	"	"	"	"
E18	CLA "	C	27	-0.1	"	"	"
E19	DTA 7-R "	"	36	-0.4	"	"	"
E20	$\overline{\text{RCT}}$ "	"	14	"	"	"	"
E21	SEL-BC "	"	37	"	"	"	"
E22	CLOB "	"	6	-0.1	"	"	"
E23	DTA 0-R "	"	15	-0.4	"	"	"

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NOTES:
1. R1 = 1.1 kΩ ±1%, 1/4 W.
2. R2 = 750Ω ±1%, 1/4 W.

* FIGURE 15. Input clamp diode voltage test circuit for table VII.

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Check the source to verify that this is the current version before use.

TABLE VIII. Functional test. 1/ 2/

Test condition	Parameter	Connect																										Output																Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F0	Functional	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	16,17
	"	B	B	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	L	L	L	L	L	H	H	H	L	L	L	H	L	L	"
	"	A	A	A	B	B	B	A	B	B	B	B	B	A	B	B	B	A	B	B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	A	B	B	B	B	B	A	B	B	B	A	B	B	A	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	A	B	B	B	B	B	A	B	B	B	A	B	B	B	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	A	B	B	B	B	B	A	B	B	B	A	B	B	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	A	B	B	B	B	B	A	B	B	B	A	B	B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	A	B	B	B	B	B	A	B	B	B	A	B	B	B	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	A	B	B	B	B	B	A	B	B	B	A	B	B	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F1	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	B	A	L	H	L	L	L	H	H	H	H	L	L	L	H	H	"
F2	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	B	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
F3	"	A	A	A	B	A	A	A	B	A	A	B	B	A	B	B	B	A	B	A	A	A	B	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
F4	"	A	A	A	B	A	A	A	B	A	B	A	B	A	B	B	B	A	B	A	A	A	B	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
F5	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	B	B	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F6	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	L	H	L	L	L	H	H	H	H	L	L	L	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	B	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	B	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	B	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F7	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	B	A	A	B	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	B	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	B	A	A	B	A	B	B	B	A	B	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	B	A	A	B	A	B	B	B	A	A	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	B	A	A	B	A	B	B	B	A	A	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F8	"	A	A	A	B	A	A	A	B	B	A	A	B	A	B	B	B	A	A	A	A	A	B	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	B	A	A	B	A	B	B	B	A	B	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F9	"	A	A	A	B	A	B	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	B	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	B	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued. 1/ 2/

Test condition	Parameter	Connect																													Output															Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F10	Functional	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	L	H	H	16,17
	"	A	A	A	B	A	A	A	B	A	B	B	B	A	B	B	B	A	B	A	A	A	B	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F11	"	A	A	A	B	A	A	A	B	A	B	B	B	A	B	B	B	A	A	A	A	A	B	B	B	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	B	B	B	A	B	B	B	A	A	A	A	A	A	B	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F12	"	A	A	A	B	A	A	A	B	A	B	B	B	A	B	B	B	A	A	A	A	A	B	B	B	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	B	A	A	B	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F13	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	B	A	A	B	A	B	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	B	A	A	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	B	A	A	B	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	B	A	A	B	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	B	A	A	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F14	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	B	A	A	A	A	B	L	H	L	L	H	H	L	H	H	H	L	L	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F15	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	B	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F16	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	B	L	H	L	L	L	H	H	H	H	L	L	L	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F17	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	L	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F18	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F19	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F20	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F21	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
F22	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	

See footnotes at end of table.

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TABLE VIII. Functional test - Continued. 1/ 2/

Test condition	Parameter	Connect																										Output																Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F23	Functional	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	16,17
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F24	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F25	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F26	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F27	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F28	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F29	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F30	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F31	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F32	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	H	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F33	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	L	H	L	L	L	H	H	H	H	L	L	L	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F34	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F35	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F36	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued. 1/ 2/

Test condition	Parameter	Connect																									Output															Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F37	Functional	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	16,17
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F38	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F39	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F40	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F41	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F42	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F43	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F44	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F45	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F46	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F47	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F48	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	H	H	H	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F49	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	L	H	L	L	H	H	L	H	H	H	L	L	H	H	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	A	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	B	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	B	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued. 1/ 2/

Test condition	Parameter	Connect																										Output															Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F50	Functional	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	A	L	H	H	L	L	H	H	L	L	L	L	L	L	L	16,17
	"	A	A	A	A	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	A	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	B	A	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F51	"	A	A	A	A	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	A	L	H	H	L	L	H	H	L	L	L	L	L	L	L	"
	"	A	A	A	A	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	B	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	B	B	B	A	A	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	B	B	A	A	B	A	B	B	B	A	B	B	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F52	"	B	B	A	B	A	B	B	B	B	A	A	B	B	B	B	B	B	B	B	B	B	B	A	A	L	L	L	L	L	H	H	H	L	L	L	H	L	L	"
	"	B	B	A	B	A	B	B	B	B	A	A	B	B	B	A	B	B	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	B	A	A	A	B	B	B	B	A	A	B	B	B	B	B	B	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	B	A	B	A	B	B	B	B	A	A	B	B	B	B	B	B	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F53	"	B	A	A	B	A	B	A	A	B	A	A	A	A	A	B	A	A	B	B	B	A	B	A	A	L	L	H	L	H	L	L	H	L	H	L	H	H	L	"
	"	B	A	A	B	A	B	A	A	B	A	A	A	A	A	A	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	A	B	A	A	A	A	A	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	A	B	A	A	A	A	A	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F54	"	B	A	A	B	A	B	A	A	B	A	A	A	B	A	B	A	B	B	B	B	A	B	A	A	L	L	H	L	H	L	L	H	L	H	L	H	H	L	"
	"	B	A	A	B	A	B	A	A	B	A	A	A	B	A	A	A	B	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	A	B	A	A	A	B	A	B	A	B	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	A	B	A	A	A	B	A	B	A	B	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F55	"	A	A	A	B	A	B	A	A	B	A	A	B	B	A	B	B	A	B	B	B	A	A	A	A	L	L	L	L	L	L	H	H	H	L	L	H	H	L	"
	"	A	A	A	B	A	B	A	A	B	A	A	B	B	A	A	B	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	A	A	B	A	A	B	A	A	B	B	A	B	B	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	A	B	A	A	B	B	A	B	B	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F56	"	A	A	A	B	A	B	A	A	B	A	A	A	A	A	B	A	A	B	B	B	A	A	A	A	L	L	L	L	L	L	H	H	H	L	L	H	H	L	"
	"	A	A	A	B	A	B	A	A	B	A	A	A	A	A	A	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	A	A	B	A	A	B	A	A	A	A	A	B	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	A	B	A	A	A	A	A	B	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F57	"	B	A	A	B	A	B	A	B	B	A	A	A	B	A	B	A	A	B	B	B	A	A	A	A	L	L	L	L	L	H	H	H	L	L	L	H	H	L	"
	"	B	A	A	B	A	B	A	B	B	A	A	A	B	A	A	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	B	B	A	A	A	B	A	B	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	B	B	A	A	A	B	A	B	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued. 1/ 2/

Test condition	Parameter	Connect																										Output																Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F58	Functional	B	A	A	B	A	B	A	A	B	A	A	A	B	B	B	A	A	B	B	B	B	B	A	A	L	L	H	L	H	H	L	H	L	H	L	H	H	L	16,17
	"	B	A	A	B	A	B	A	A	B	A	A	A	B	B	A	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	A	B	A	A	A	B	B	B	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	A	B	A	A	A	B	B	B	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F59	"	B	A	A	B	A	B	A	B	B	A	A	B	B	B	A	B	A	A	B	B	B	B	A	A	L	L	L	L	L	H	H	H	L	L	L	H	L	L	"
	"	B	A	A	B	A	B	A	B	B	A	A	B	B	B	A	B	A	A	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F60	"	A	A	A	B	A	B	A	A	B	A	A	B	A	A	B	A	A	B	B	B	A	A	A	A	L	L	L	L	L	L	H	H	H	L	L	H	H	L	"
	"	A	A	A	B	A	B	A	A	B	A	A	B	A	A	A	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	A	A	B	A	A	B	A	A	B	A	A	B	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	A	B	A	A	B	A	A	B	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F61	"	B	B	A	B	A	B	A	B	B	A	A	A	B	B	B	A	B	B	B	A	A	A	A	A	L	L	L	L	L	H	H	H	L	L	L	H	H	L	"
	"	B	B	A	B	A	B	A	B	B	A	A	A	B	B	A	A	B	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	B	A	A	A	B	A	B	B	A	A	A	B	B	B	A	B	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	B	A	B	A	B	A	B	B	A	A	A	B	B	B	A	B	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F62	"	B	A	A	B	A	B	A	B	B	A	A	B	B	B	A	B	A	A	B	B	B	A	B	A	A	L	L	L	L	L	H	H	H	L	L	L	H	H	L	"
	"	B	A	A	B	A	B	A	B	B	A	A	B	B	B	A	A	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F63	"	B	A	A	B	A	B	A	B	B	A	A	B	B	B	A	B	A	A	B	B	B	B	B	A	A	L	L	L	L	L	H	H	H	L	L	L	H	H	L	"
	"	B	A	A	B	A	B	A	B	B	A	A	B	B	A	A	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F64	"	B	A	A	B	A	B	B	B	B	A	A	B	B	B	B	A	A	B	B	B	A	B	A	A	L	L	L	L	L	H	H	H	L	L	L	H	H	L	"
	"	B	A	A	B	A	B	B	B	B	A	A	B	B	B	A	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	B	B	B	A	A	B	B	B	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	B	B	B	A	A	B	B	B	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F65	"	A	B	A	B	A	B	B	B	B	A	A	A	A	A	B	B	A	B	B	B	B	B	A	A	L	L	H	L	H	H	H	H	H	L	L	H	L	L	"
	"	A	B	A	B	A	B	B	B	B	A	A	A	A	A	A	B	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	B	A	A	A	B	B	B	B	A	A	A	A	A	B	B	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	B	A	B	A	B	B	B	B	A	A	A	A	A	B	B	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F66	"	B	A	A	B	A	B	A	B	B	A	A	A	B	A	B	A	A	B	B	B	B	A	A	A	L	L	L	L	L	H	H	H	L	L	L	H	L	L	"
	"	B	A	A	B	A	B	A	B	B	A	A	A	B	A	A	A	A	B	B	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	B	B	A	A	A	B	A	B	A	A	B	B	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	B	B	A	A	A	B	A	B	A	A	B	B	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued. 1/ 2/

Test condition	Parameter	Connect																														Output																	Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F67	Functional	A	B	A	B	A	B	B	A	B	A	A	B	A	B	A	A	B	B	B	A	A	A	A	A	L	L	L	L	L	L	H	H	H	L	L	H	H	L	
	"	A	B	A	B	A	B	B	A	B	A	A	B	A	B	A	A	B	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	16,17
	"	A	B	A	A	A	B	B	A	B	A	A	B	A	B	B	A	B	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	B	A	B	A	B	B	A	B	A	A	B	A	B	B	A	B	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F68	"	B	A	A	B	A	B	A	A	B	A	A	B	A	A	B	B	B	B	B	B	A	A	A	A	L	L	L	L	L	L	H	H	L	L	L	H	H	L	"
	"	B	A	A	B	A	B	A	A	B	A	A	B	A	A	A	B	B	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	A	B	A	A	B	A	A	B	B	B	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	A	B	A	A	B	A	A	B	B	B	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F69	"	B	A	A	B	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	L	L	L	L	L	H	H	H	L	L	L	H	H	L	"
	"	B	A	A	B	A	B	A	B	B	A	A	B	B	A	A	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F70	"	B	A	A	B	A	B	B	B	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	L	L	L	L	L	H	H	H	L	L	L	H	H	L	"
	"	B	A	A	B	A	B	B	B	B	A	A	B	B	A	A	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	B	B	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	B	B	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F71	"	B	A	A	B	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	A	A	A	A	L	L	L	L	L	H	H	H	L	L	L	H	H	L	"
	"	B	A	A	B	A	B	A	B	B	A	A	B	B	A	A	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	B	B	A	A	B	B	A	B	A	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F72	"	A	A	A	B	A	B	A	A	B	A	A	B	A	B	B	A	A	B	B	A	A	A	A	A	L	L	L	L	L	L	H	H	H	L	L	H	H	L	"
	"	A	A	A	B	A	B	A	A	B	A	A	B	A	B	A	A	A	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	A	A	B	A	A	B	A	A	B	A	B	B	A	A	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	A	B	A	A	B	A	B	B	A	A	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F73	"	B	A	A	B	A	B	A	A	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	L	L	L	L	L	L	H	H	L	L	L	H	H	L	"
	"	B	A	A	B	A	B	A	A	B	A	A	B	B	A	A	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	A	A	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	A	B	A	A	B	B	A	B	A	A	B	B	B	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F74	"	B	A	A	B	A	B	B	A	B	A	A	A	B	A	B	A	A	B	B	B	B	B	A	A	L	L	H	L	H	H	H	H	L	L	L	H	H	L	"
	"	B	A	A	B	A	B	B	A	B	A	A	A	B	A	A	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	A	A	B	B	A	B	A	A	A	B	A	B	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"

See footnotes at end of tables.

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TABLE VIII. Functional test - Continued. 1/ 2/

Test condition	Parameter	Connect																											Output																Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F75	Functional	B	A	A	B	A	B	B	A	B	A	A	A	B	A	B	A	A	B	B	B	B	B	A	A	L	L	H	L	H	H	H	H	L	L	L	H	H	L	16,17
	"	B	A	A	B	A	B	A	A	B	A	A	A	B	A	B	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	B	A	A	B	A	B	A	A	B	A	A	A	B	A	A	A	A	B	B	B	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	B	A	B	A	B	A	B	B	A	A	B	A	A	A	B	B	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	B	B	A	A	B	A	A	A	B	A	B	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	B	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	A	A	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	B	A	B	A	A	A	B	A	A	A	B	A	A	A	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	A	A	A	B	A	A	A	B	A	A	A	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	A	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F76	"	A	A	B	B	B	A	A	B	A	A	A	B	A	A	A	B	A	B	A	A	A	A	A	A	L	H	H	L	L	H	H	L	L	L	L	H	H	L	"
F77	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	A	B	A	B	A	A	A	A	A	A	L	H	H	L	L	H	H	L	L	L	L	L	H	L	"
F78	"	A	B	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	H	H	L	L	L	H	H	H	L	L	L	L	H	L	"
F79	"	A	B	A	B	B	A	A	B	A	A	A	A	A	B	B	B	A	B	A	A	A	B	A	A	L	H	H	L	L	H	H	L	L	L	L	L	H	L	"
	"	A	B	B	B	B	A	A	B	A	A	A	A	A	B	B	B	A	B	A	A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F80	"	A	B	A	B	B	A	A	B	A	A	A	A	A	B	B	B	A	B	A	A	A	B	A	A	L	H	H	L	L	H	H	L	L	L	L	L	H	L	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F81	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	L	H	H	L	L	H	H	L	L	L	L	L	H	L	"
	"	A	B	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F82	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	L	H	H	L	L	H	H	L	L	L	L	L	H	L	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	B	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued. 1/ 2/

Test condition	Parameter	Connect																										Output																Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F83	Functional	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	L	H	H	L	L	H	H	L	L	L	L	L	H	L	16,17
	"	A	B	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F84	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	L	H	H	L	L	H	H	L	L	L	H	L	H	L	"
	"	A	B	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F85	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	L	H	H	H	L	H	H	L	L	L	H	L	H	L	"
	"	A	B	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	P	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F86	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	B	B	A	B	A	A	A	A	A	A	L	H	H	H	L	H	H	L	L	L	H	L	H	L	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F87	"	A	B	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	H	H	L	H	L	H	H	H	L	L	H	L	H	L	"
	"	A	B	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F88	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	L	H	H	H	L	H	H	L	L	L	H	L	H	L	"
	"	A	B	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued. 1/ 2/

Test condition	Parameter	Connect																								Output																Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F89	Functional	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	L	H	H	H	L	H	H	L	L	L	L	L	H	L	16,17
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F90	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	A	B	A	A	A	A	A	A	L	H	H	L	L	H	H	L	L	L	L	L	H	L	"
	"	A	A	A	B	B	A	B	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	A	A	A	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	B	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F91	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	L	L	H	L	L	H	H	H	H	L	L	H	H	L	"
F92	"	A	A	A	A	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	L	L	H	L	L	H	H	H	H	L	L	H	H	L	"
F93	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	L	L	H	L	L	H	H	H	H	L	L	H	H	L	"
	"	A	A	A	B	B	A	B	B	A	A	A	B	A	B	B	B	A	A	A	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F94	"	A	A	A	A	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	L	H	H	L	L	H	H	L	H	L	L	H	H	L	"
F95	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	H	L	H	L	L	H	H	H	H	L	L	H	H	L	"
F96	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	A	B	A	A	B	A	A	A	A	A	H	L	H	L	L	H	H	H	H	L	L	H	H	L	"
F97	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	H	L	H	L	L	H	H	H	L	L	L	H	H	L	"
	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	A	B	A	A	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F98	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	H	L	H	L	L	H	H	H	L	L	L	H	H	L	"
	"	A	A	A	B	B	B	B	B	B	A	A	B	B	B	B	B	A	A	B	A	A	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F99	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	A	A	A	L	L	H	L	L	H	H	H	H	L	L	H	H	L	"
	"	A	A	A	B	B	B	B	B	B	A	A	A	A	B	B	B	A	A	B	A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued. 1/ 2/

Test condition	Parameter	Connect																													Output																Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F100	Functional	A	A	A	B	B	A	B	B	A	A	A	A	A	B	B	B	A	A	A	A	A	B	A	A	L	H	H	L	L	H	H	L	H	L	L	L	H	L	16,17
	"	A	A	A	B	B	B	B	B	B	A	A	A	A	B	B	B	A	A	B	A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F101	"	A	A	A	B	B	B	B	B	B	A	A	A	A	B	B	B	B	A	B	A	A	B	A	A	L	L	H	L	H	H	H	H	H	L	L	H	H	L	"
	"	A	A	A	B	B	B	B	B	B	A	A	B	A	B	B	B	A	A	B	A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	B	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	A	B	A	B	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	B	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	A	B	A	B	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F102	"	A	A	A	B	B	A	B	B	A	A	A	B	A	B	B	B	A	A	A	A	A	B	A	A	H	H	L	L	L	H	H	H	H	L	L	L	H	L	"
F103	"	A	A	A	B	B	A	B	B	A	A	A	B	A	B	B	B	A	A	A	B	A	A	A	A	H	H	L	L	L	H	H	H	H	L	L	L	H	L	"
F104	"	A	A	A	B	B	A	B	B	A	A	A	B	A	B	B	B	A	A	A	A	B	A	A	A	H	H	L	L	L	H	H	H	H	L	L	L	L	L	"
		A	A	A	B	B	A	B	B	A	A	A	B	A	B	B	B	B	A	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
		A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	B	A	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F105	"	A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	B	A	B	A	B	A	A	A	L	L	L	L	L	H	H	H	H	L	L	H	L	L	"
		A	A	A	B	B	A	A	A	A	A	A	B	A	B	B	B	A	A	B	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
		A	A	A	B	B	A	A	A	A	A	A	B	A	B	B	B	A	A	B	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
		A	A	A	B	B	A	A	B	A	A	A	B	A	B	B	B	B	A	B	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
		A	A	A	B	B	A	A	A	A	A	A	B	A	B	B	B	A	A	B	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
		A	A	A	B	B	A	A	A	A	A	A	B	A	B	B	B	A	A	B	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F106	"	A	A	A	B	B	A	A	A	A	A	A	B	A	B	B	B	A	A	B	A	B	A	A	A	L	L	H	L	L	H	H	H	H	L	L	H	H	L	"
		A	A	A	B	B	A	A	A	A	A	A	B	A	B	B	B	A	B	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
		A	A	A	A	B	A	A	A	A	A	A	B	A	B	B	B	A	B	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
		A	A	A	B	B	A	A	A	A	A	A	B	A	B	B	B	A	B	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F107	"	A	A	A	A	B	A	A	A	A	A	A	B	A	B	B	B	A	B	A	A	B	B	A	A	H	H	L	L	L	H	H	H	H	L	L	L	H	L	"
		A	A	A	A	B	A	A	A	A	A	A	B	A	B	B	B	B	A	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F108	"	A	A	A	A	B	A	A	A	A	A	A	B	A	B	B	B	A	B	A	A	B	B	A	A	L	H	L	L	L	H	H	H	H	L	L	L	H	L	"
		A	A	A	A	B	A	A	A	A	A	A	B	A	B	B	B	A	B	B	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued. 1/ 2/

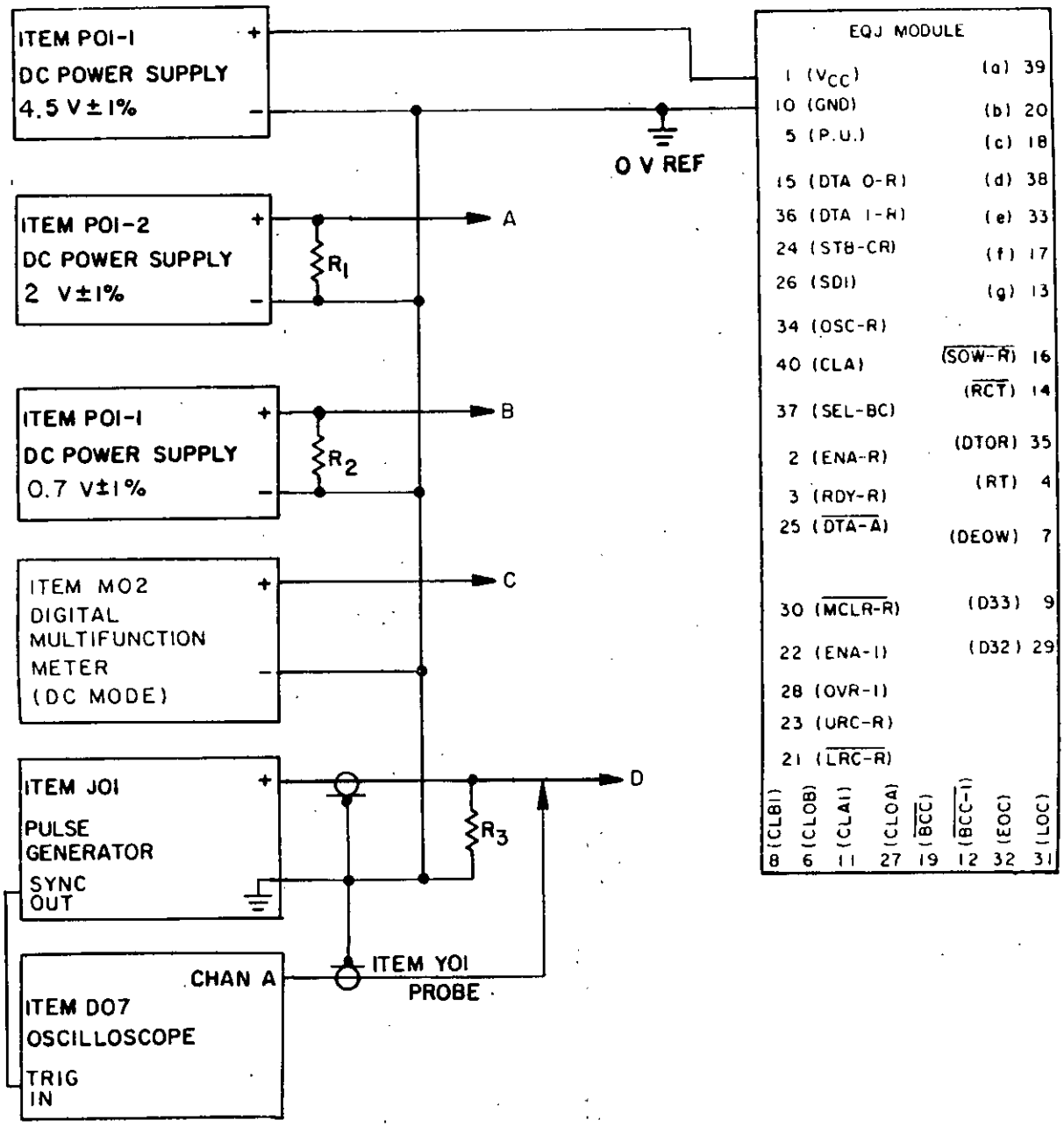
Test condition	Parameter	Connect																										Output																Test figure
		Pin 2 to	Pin 4 to	Pin 5 to	Pin 6 to	Pin 12 to	Pin 13 to	Pin 14 to	Pin 15 to	Pin 17 to	Pin 18 to	Pin 20 to	Pin 24 to	Pin 25 to	Pin 26 to	Pin 27 to	Pin 28 to	Pin 30 to	Pin 31 to	Pin 33 to	Pin 34 to	Pin 36 to	Pin 37 to	Pin 38 to	Pin 39 to	Pin 3	Pin 7	Pin 8	Pin 9	Pin 11	Pin 16	Pin 19	Pin 21	Pin 22	Pin 23	Pin 29	Pin 32	Pin 35	Pin 40	
F109	Functional	A	A	A	A	B	A	A	A	A	A	A	B	A	B	B	A	A	B	B	A	B	B	A	A	L	L	L	L	L	H	H	H	H	L	L	H	H	L	16,17
	"	A	A	A	A	B	A	A	A	A	A	A	B	A	B	B	B	A	B	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	A	B	A	B	B	B	A	B	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	A	B	A	A	A	A	A	A	B	A	B	B	B	A	B	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	A	B	A	B	B	B	A	B	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	A	B	A	B	A	B	A	B	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	A	B	A	B	A	B	A	B	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F110	"	A	A	A	B	B	A	A	A	A	A	B	A	B	A	B	A	B	B	A	A	B	B	A	A	L	H	L	L	L	H	H	H	H	L	L	L	H	L	"
	"	A	A	A	B	B	A	A	A	A	A	B	A	B	A	B	A	B	B	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	B	A	B	A	B	A	B	B	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	B	A	B	A	B	A	B	B	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	B	A	B	A	B	A	B	B	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	B	A	B	A	B	A	B	B	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	B	A	B	A	B	A	B	B	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	B	A	B	A	B	A	B	B	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F111	"	A	A	A	B	B	A	A	A	A	A	A	B	A	B	A	B	A	B	A	A	B	A	A	A	L	H	L	L	H	H	L	H	H	H	L	L	H	H	"
	"	A	A	A	B	B	A	A	A	A	A	A	B	A	A	A	B	A	B	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	B	A	A	A	A	B	A	A	A	B	A	B	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	A	B	A	A	A	B	A	B	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	B	A	A	A	B	A	A	A	B	A	B	A	B	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	A	B	A	A	A	B	A	B	A	A	B	A	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
	"	A	A	A	B	B	A	A	A	A	A	A	B	A	A	A	B	A	B	A	A	B	B	A	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	"
F112	"	A	A	A	B	B	A	A	A	A	A	A	B	A	A	A	B	A	B	A	A	B	B	A	A	L	H	L	L	L	H	H	H	H	L	H	L	H	L	"

1/ Initial and end-of-life limits at 25°C ± 5°C ambient and 0°C to 60°C operating temperature:

L = 0.0 V minimum, 0.4 V maximum
H = 2.4 V minimum, 4.5 V maximum
X = don't care.

2/ Verification of output logic levels L and H may be performed using a logic state indicator in place of item M02 of figure 16.

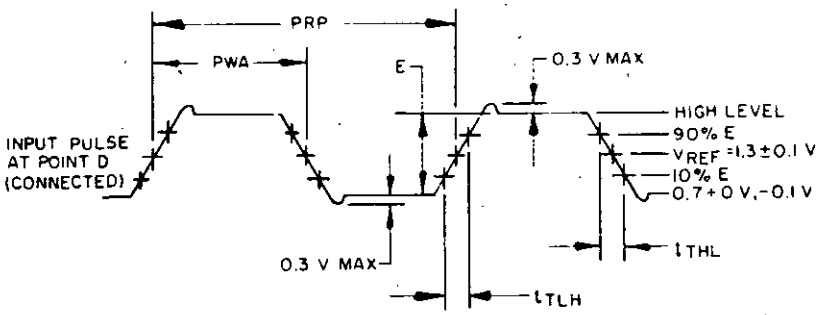
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 51.1Ω ±1%, 1/4 W.

* FIGURE 16. Functional test circuit for table VIII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 50 \pm 25$ ns.
   PWA = 500 ±100 ns.
   PRP = 1000 ±200 ns.
2. Clock threshold level setup: Make circuit connections per table VIII. Use item M02 and adjust dc offset of item J01 for a low level of 0.7 +0 V, -0.1 V (clock low level threshold). Adjust item J01 to obtain the required PWA and PRP. Use item D05 and adjust item J01 amplitude for a high level of 2.0 +0.1 V, -0.0 V (clock high level threshold). The high level amplitude can be adjusted to the required voltage by combining the dc offset as measured with item M02 with the pulse amplitude (PA) as measured with item D05. After making the required adjustments, item J01 should be placed in the single pulse mode.

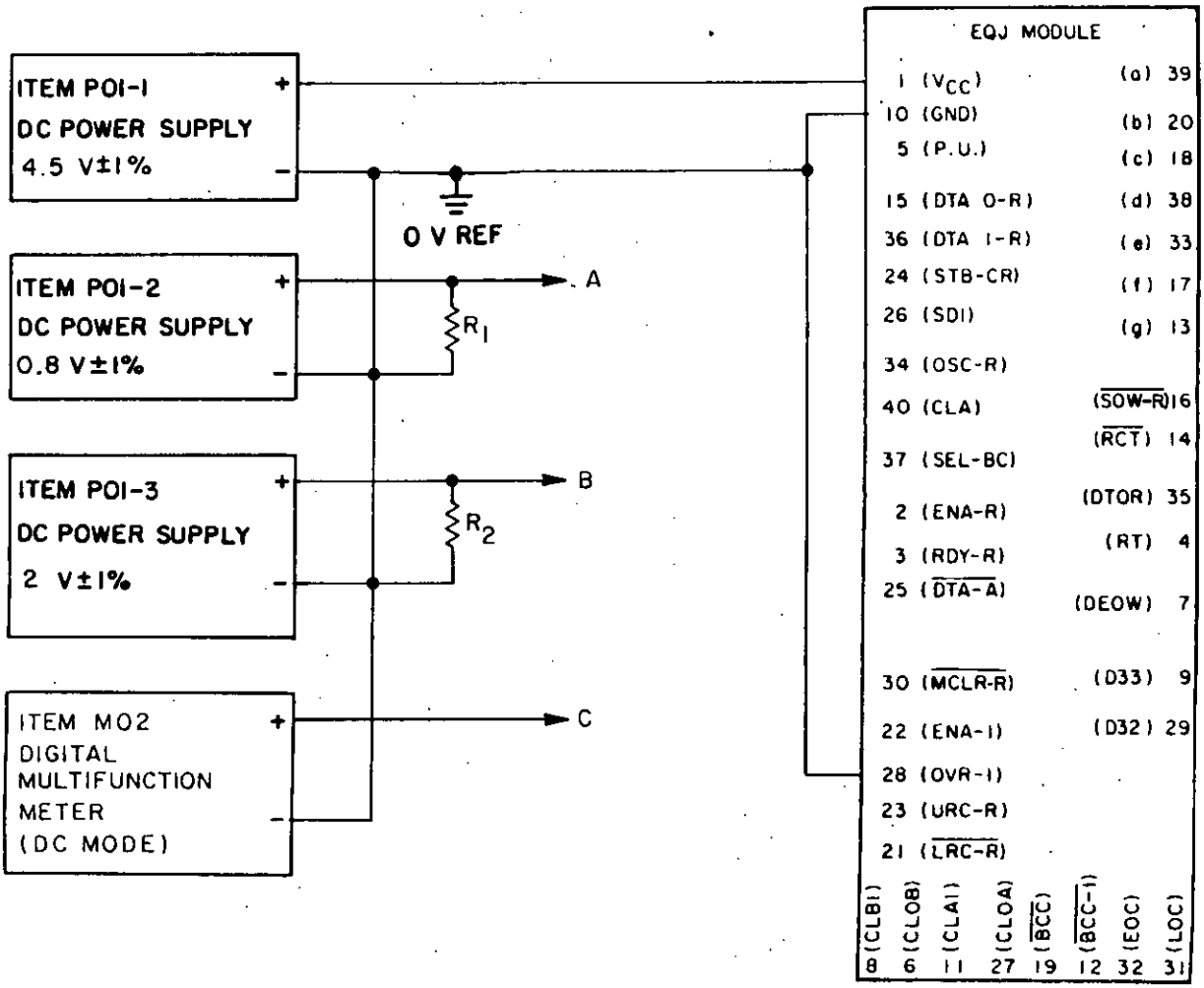
* FIGURE 17. Functional test circuit waveform for table VIII.

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*TABLE IX. Output voltage (low-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	RDY-R	Output voltage, low level	30	Open	3	0	400	mV	18
G2	DEOW	"	30,39	"	7	"	"	"	"
G3	CLBI	"	24,37	"	8	"	"	"	"
G4	D33	"	5	"	9	"	"	"	"
G5	CLAI	"	15,36	37	11	"	"	"	"
G6	$\overline{\text{SOW-R}}$	"	Open	15,36	16	"	"	"	"
G7	$\overline{\text{BCC}}$	"	30,37	14,24	19	"	"	"	"
G8	$\overline{\text{LRC-R}}$	"	31,37	13,17,18, 20,24,33, 38,39	21	"	"	"	"
G9	ENA-I	"	2	Open	22	"	"	"	"
G10	URC-R	"	14	"	23	"	"	"	"
G11	D32	"	5	"	29	"	"	"	"
G12	EOC	"	31	13,17,18, 20,33,38, 39	32	"	"	"	"
G13	DTOR	"	15,36	Open	35	"	"	"	"
G14	CLA	"	15,36	Open	40	"	"	"	"

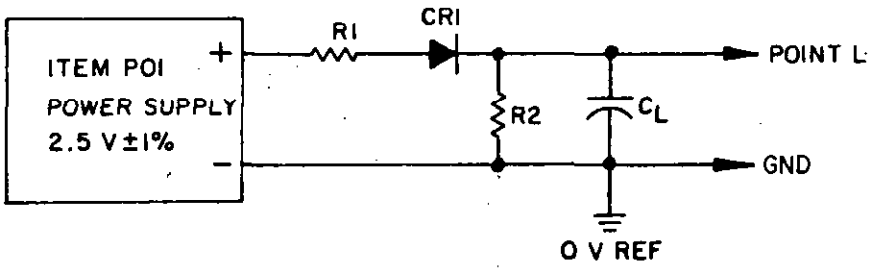
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load the outputs simultaneously with the load as specified on figure 19.

* FIGURE 18. Output voltage (low-level) test circuit for table IX.

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NOTES:
1. CR1 = 1N3600, 1N4150 or equivalent.
2. For output pins 3, 16, 22
   R1 = Select value for a current flow of 16.0 mA ±5% into point L with point L held at 0.4 V.

   For output pin 23
   R1 = Select value for a current flow of 15.5 mA ±5% into point L with point L held at 0.4 V.

   For output pins 8, 19, 40
   R1 = Select value for a current flow of 14.4 mA ±5% into point L with point L held at 0.4 V.

   For output pin 21
   R1 = Select value for a current flow of 14.0 mA ±5% into point L with point L held at 0.4 V.

   For output pin 11
   R1 = Select value for a current flow of 12.8 mA ±5% into point L with point L held at 0.4 V.

   For output pins 7, 32
   R1 = Select value for a current flow of 11.2 mA ±5% into point L with point L held at 0.4 V.

   For output pin 35
   R1 = Select value for a current flow of 9.6 mA ±5% into point L with point L held at 0.4 V.

   For output pins 9, 29
   R1 = Select value for a current flow of 4.0 mA ±5% into point L with point L held at 0.4 V.
3. For output pins 9, 29
   R2 = 6.19 kΩ ±1%, 1/8 W.

   For output pins 3, 22
   R2 = 3.01 kΩ ±1%, 1/8 W.

   For output pins 8, 19
   R2 = 6.65 kΩ ±1%, 1/8 W.

   For output pin 40
   R2 = 3.16 kΩ ±1%, 1/8 W.

   For output pin 23
   R2 = 3.09 kΩ ±1%, 1/8 W.

   For output pin 11
   R2 = 7.5 kΩ ±1%, 1/8 W.

   For output pin 21
   R2 = 7.15 kΩ ±1%, 1/8 W.

   For output pins 7, 32
   R2 = 8.66 kΩ ±1%, 1/8 W.

   For output pin 35
   R2 = 10.0 kΩ ±1%, 1/8 W.

   For output pin 16
   R2 = 6.04 kΩ ±1%, 1/8 W.
4. For test conditions M1-M14, N1-N14, P1, Q1, R1, S1, T1, U1, V1, V2, W1-W6, X1-X3, Y1, Y2, Z1-Z4, AA1-AA14, and BB1-BB14:
   C_L = 50 ±5 pF (including probe and fixture capacitance).

FIGURE 19. EQJ logic load circuit.

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NOTES: Continued
5. For test conditions M15-M28, N15-N28, P2, Q2, R2, S2, T2, U2, V3, V4, W7-W12, and X4-X6:
   C_L = 150 ±5 pF (including probe and fixture capacitance).
6. Capacitor C_L is not required for test conditions G1-G14 and H1-H14.
7. Lead length from load circuit to module output pin under test shall be less than one inch (25.4 mm).

FIGURE 19. EQJ logic load circuit - Continued.

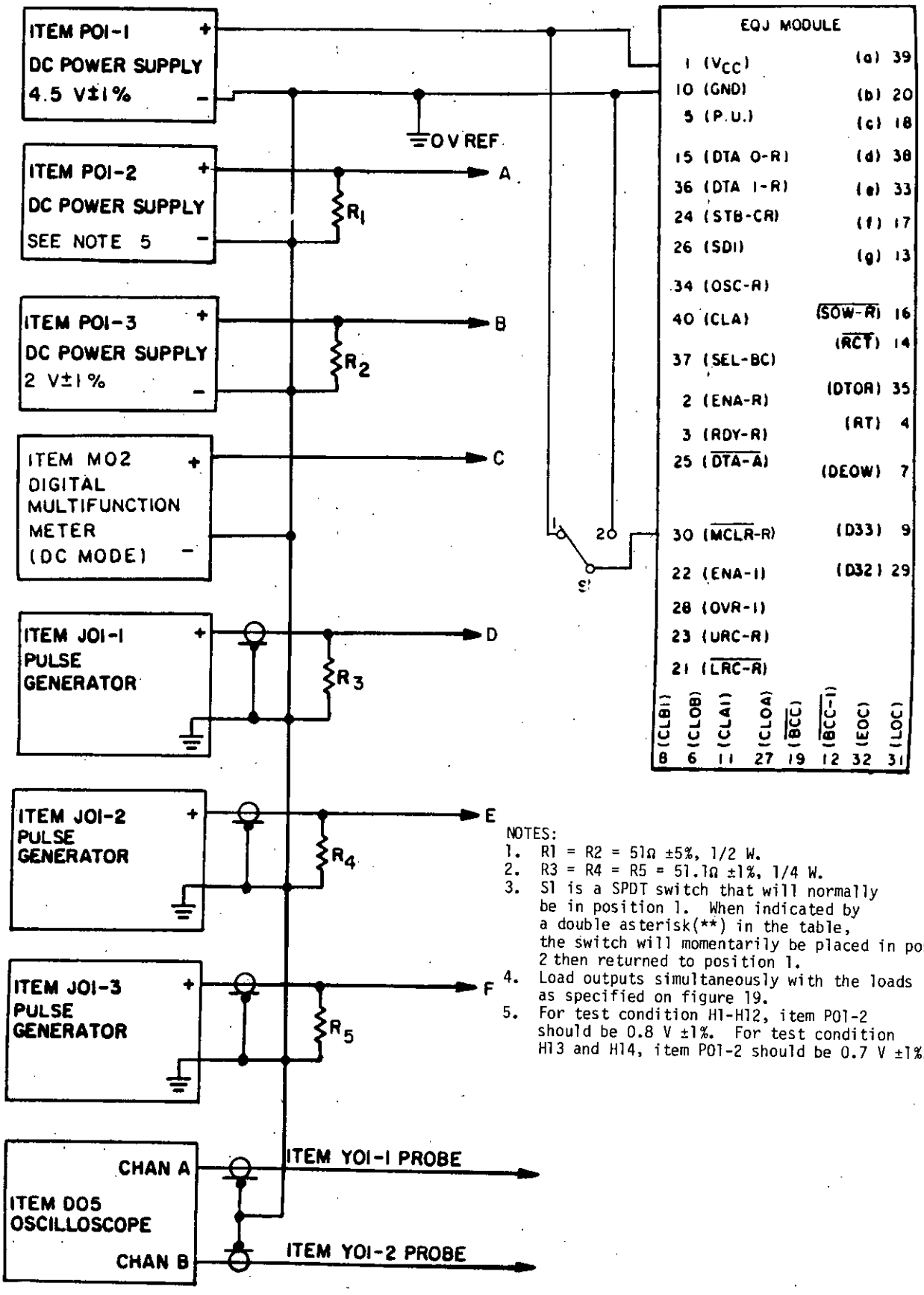
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*TABLE X. Output voltage (high-level) test.

Test condition	Parameter		Connect 1/ 2/ A to pin	B to pin	C to pin	S1 pos	D to pin	E to pin	F to pin	Initial limits 25°C ±5°C ambient Min	Max	Unit	Test figure
H1	RDY-R	Output voltage, high level	5,24,37	Open	3	1	Open	Open	Open	2.4	4.5	V	20,21
H2	DEOW	"	31	13,17, 18,20, 33,38, 39	7	"	"	"	"	"	"	"	"
H3	CLBI	"	37	24	8	"	"	"	"	"	"	"	"
H4	CLAI	"	37	24	11	2	"	"	"	"	"	"	"
H5	$\overline{\text{SOW-R}}$	"	36	Open	16	1	"	"	"	"	"	"	"
H6	$\overline{\text{BCC}}$	"	14	"	19	1	"	"	"	"	"	"	"
H7	$\overline{\text{LRC-R}}$	"	Open	"	21	2	"	"	"	"	"	"	"
H8	ENA-I	"	Open	2	22	"	"	"	"	"	"	"	"
H9	URC-R	"	37	14,24	23	"	"	"	"	"	"	"	"
H10	EOC	"	18,31	Open	32	1	"	"	"	"	"	"	"
H11	DTOR	"	15	36	35	1	"	"	"	"	"	"	"
H12	CLA	"	15	36,4	40	**	34 (1P)	"	"	"	"	"	"
H13	D32	"	15,31	4,13, 17,18, 20,24, 26,33, 36,38, 39	29	**	"	14 (1P)	37 (1P)	2.5	"	"	"
H14	D33	"	15,31	4,13, 17,18, 20,24, 26,33, 36,38, 39	9	**	"	14 (3P)	37 (1P)	2.5	"	"	"

1/ Connections and operations A thru F and S1 will be performed in order.
2/ In columns D, E, and F, the number in parenthesis indicates the number of times the single pulse button on the indicated pulse generator will be depressed.

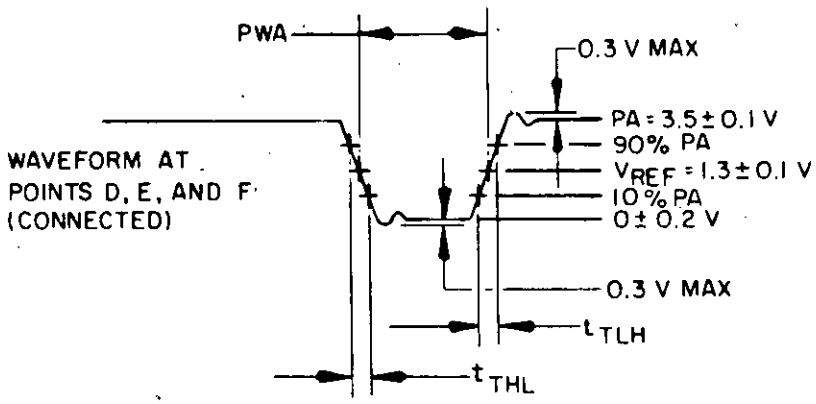
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = R4 = R5 = 51.1Ω ±1%, 1/4 W.
3. S1 is a SPDT switch that will normally be in position 1. When indicated by a double asterisk(**) in the table, the switch will momentarily be placed in position 2 then returned to position 1.
4. Load outputs simultaneously with the loads as specified on figure 19.
5. For test condition H1-H12, item POI-2 should be 0.8 V ±1%. For test condition H13 and H14, item POI-2 should be 0.7 V ±1%.

* FIGURE 20. Output voltage (high-level) test circuit for table X.

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NOTES:
1. Data input pulse parameters:
   PWA = 200 ±20 ns.
   $t_{THL} = t_{TLH}$ = 10 ±3 ns.
2. Items J01-1, J01-2, and J01-3 are operated in the single pulse mode. To obtain the proper output pulse characteristics, it is recommended the generator be setup as required at some frequency then placed in the single pulse mode.

FIGURE 21. Output voltage (high-level) test circuit waveform for table X.

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TABLE XI. Input current (high-level) test.

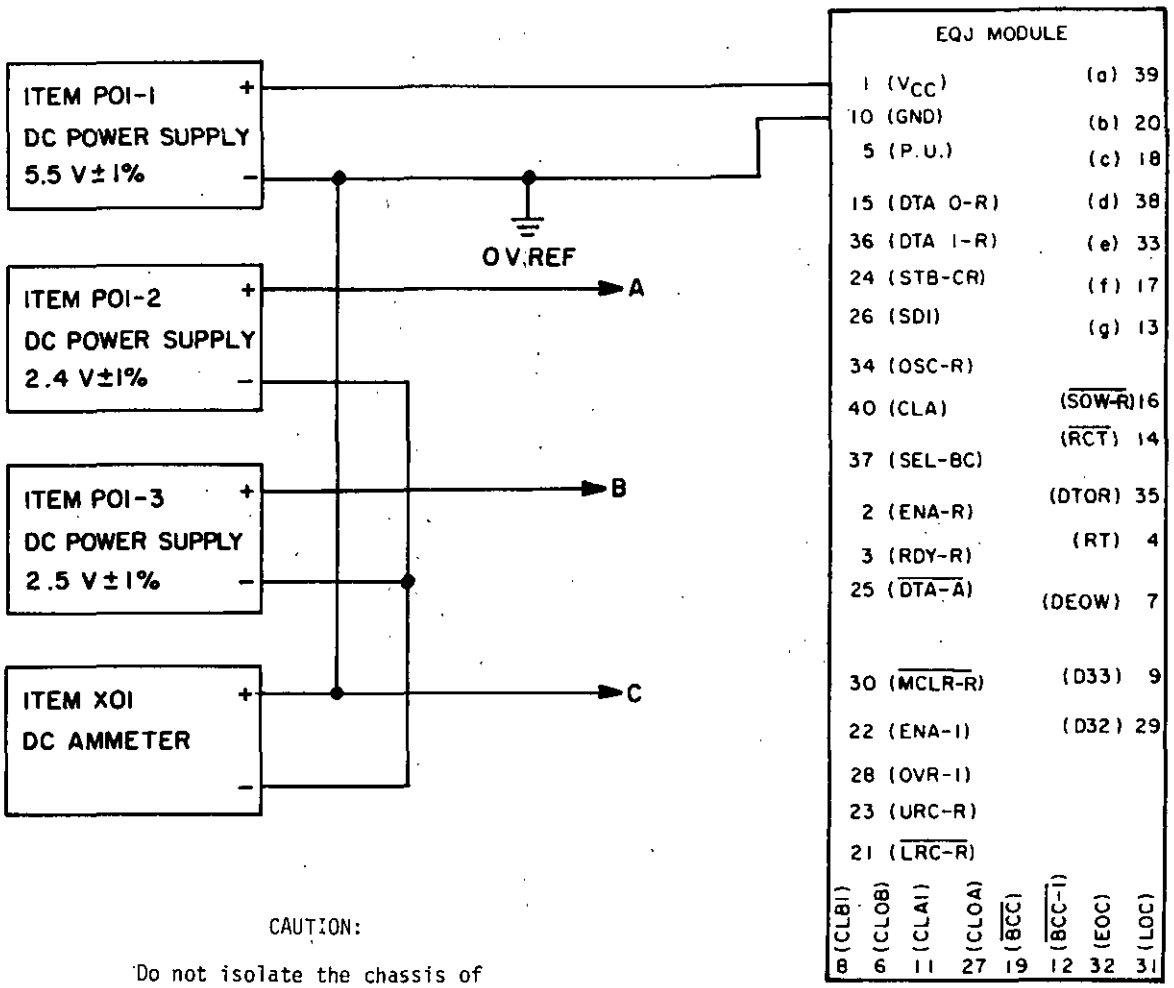
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 5 to pin	Min	Max		
J1	ENA-R Input current, high level	2	Open	30	Open	0	40	µA	22
J2	RT "	4	"	30	"	"	"	"	"
J3	$\overline{\text{BCC-I}}$ "	12	"	Open	"	"	"	"	"
J4	Word length (g) "	13	"	"	"	"	"	"	"
J5	Word length (f) "	17	"	"	"	"	"	"	"
J6	Word length (c) "	18	"	"	"	"	"	"	"
J7	Word length (b) "	20	"	"	"	"	"	"	"
J8	$\overline{\text{DTA-A}}$ "	25	"	30	"	"	"	"	"
J9	$\overline{\text{MCLR-R}}$ "	30	"	Open	28	"	"	"	"
J10	LOC "	31	"	14	Open	"	"	"	"
J11	Word length (e) "	33	"	Open	"	"	"	"	"
J12	Word length (d) "	38	"	"	"	"	"	"	"
J13	Word length (a) "	39	"	"	"	"	"	"	"
J14	STB-CR "	24	"	28,31	30,37,39, 20,18,38, 33,17,13	"	80	"	"
J15	CLOA "	27	"	Open	Open	-2250	"	"	"
J16	OVR-I "	28	"	Open	"	0	"	"	"
J17	$\overline{\text{RCT}}$ "	14	"	30,37,24	"	"	120	"	"
J18	DTA 0-R "	15	"	36	"	"	"	"	"
J19	DTA 7-R "	36	"	15	"	"	"	"	"
J20	CLOB "	6	"	Open	"	-4500	160	"	"

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Pin 5 to pin	Min	Max		
J21	OSC-R	Input current, high level	34	Open	Open	Open	0	160	µA	22
J22	SEL-BC	"	37	Open	30	"	"	160	"	"
J23	SDI	"	Open	26	Open	"	"	40	"	"

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CAUTION:

Do not isolate the chassis of item XOI from earth ground.

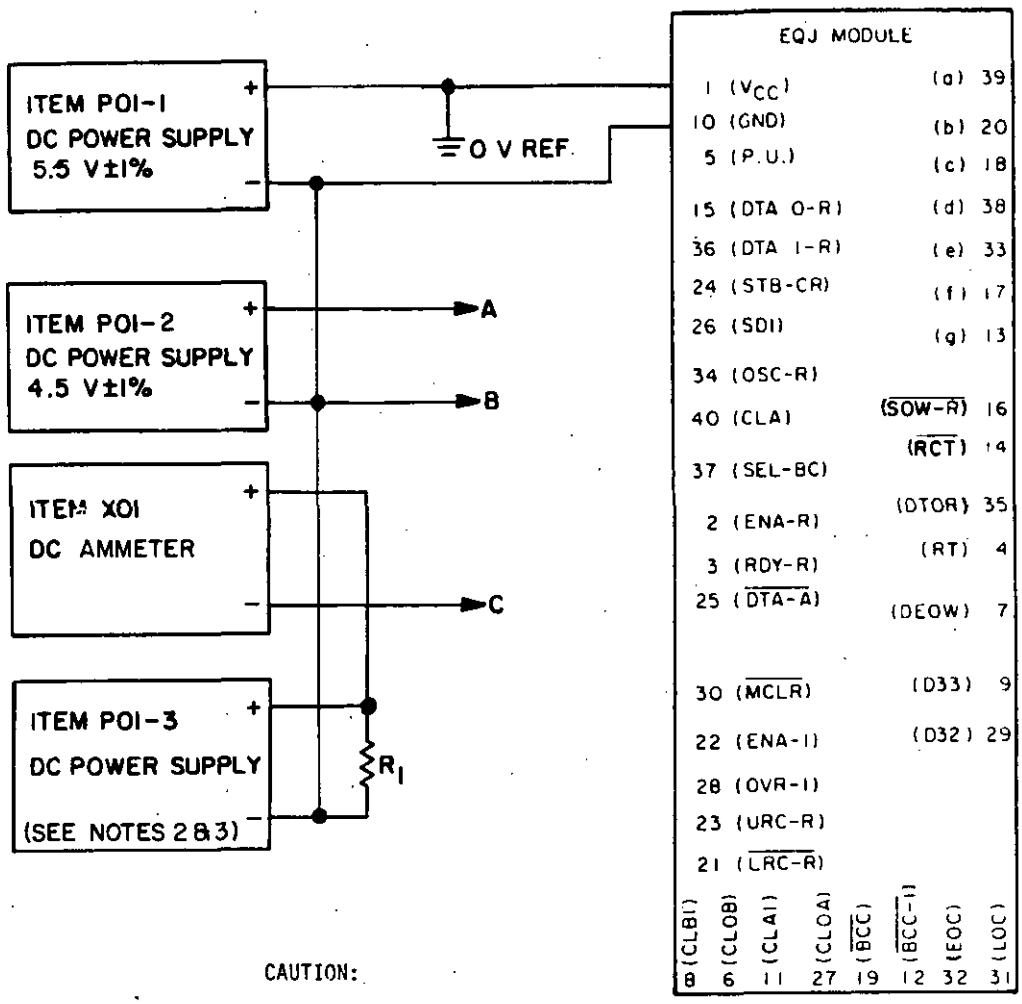
FIGURE 22. Input current (high-level) test circuit for table XI.

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TABLE XII. Input current (low-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K1	ENA-R	Input current, low level	Open	30	2	-0.1	-1.73	mA	23
K2	RT	"	30	28	4	"	"	"	"
K3	$\overline{\text{BCC-I}}$	"	Open	Open	12	"	"	"	"
K4	Word length (g)	"	"	31	13	"	"	"	"
K5	Word length (f)	"	"	"	17	"	"	"	"
K6	Word length (c)	"	"	"	18	"	"	"	"
K7	Word length (b)	"	"	"	20	"	"	"	"
K8	$\overline{\text{DTA-A}}$	"	30	28	25	"	"	"	"
K9	$\overline{\text{MCLR-R}}$	"	Open	"	30	"	"	"	"
K10	LOC	"	14,30	"	31	"	"	"	"
K11	Word length (e)	"	Open	31	33	"	"	"	"
K12	Word length (d)	"	"	"	38	"	"	"	"
K13	Word length (a)	"	"	"	39	"	"	"	"
K14	DTA 0-R	"	36	Open	15	-0.2	-3.46	"	"
K15	STB-CR	"	Open	30,37	24	"	-3.46	"	"
K16	CLOA	"	"	Open	27	"	-3.2	"	"
K17	OVR-I	"	"	"	28	"	-3.46	"	"
K18	OSC-R	"	"	"	34	"	"	"	"
K19	DTA 1-R	"	15	"	36	"	"	"	"
K20	$\overline{\text{RCT}}$	"	30,34,36,37	4,5,15,28	14	-0.3	-5.2	"	"
K21	SEL-BC	"	34,36	4,5,15	37	-0.3	-5.2	"	"
K22	CLOB	"	Open	Open	6	-0.4	-6.4	"	"
K23	SDI	"	Open	Open	26	-0.1	-0.78	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. R1 = 10Ω ±5%, 1/4 W.
2. For test conditions K1-K15 and K17-K21 and K23:
   V = 0.0 ±10 mV.
3. For test conditions K16 and K22:
   V = 0.4 V ±10%.

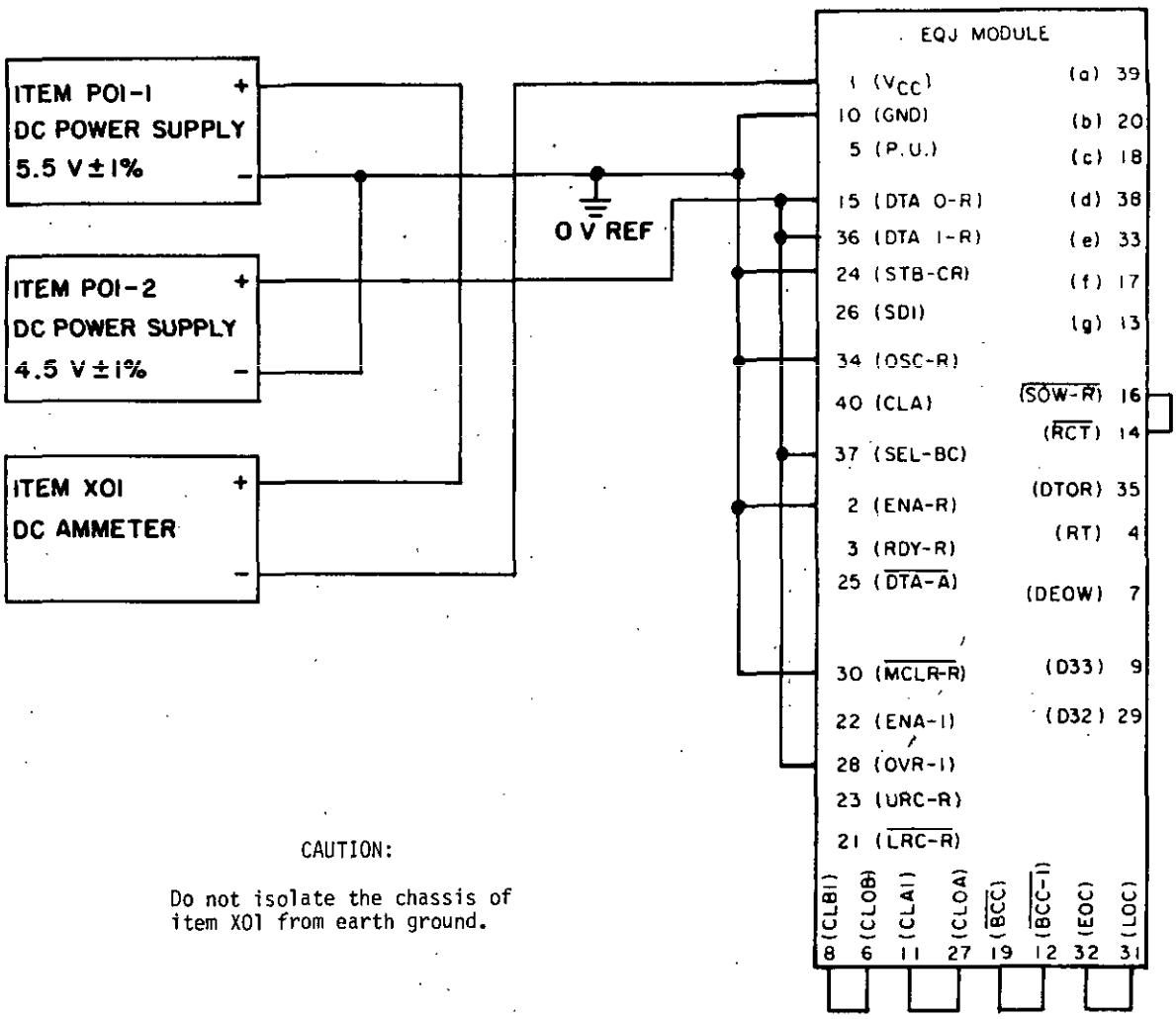
FIGURE 23. Input current (low-level) test circuit for table XII.

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TABLE XIII. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current	135	520	mA	24

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

FIGURE 24. Power supply current test circuit for table XIII.

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TABLE XIV. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M1	RDY-R	Transition time (t_{TLH}), C_L = 50 pF	13,17, 18,20, 33,38, 39,25, 30	31,37	Open	24,6	3	4	30	ns	25,26
M2	DEOW	"	17,18, 20,33, 38,39,	30	"	13	7	"	"	"	"
M3	CLBI	"	Open	37	"	24	8	"	"	"	"
M4	CLAI	"	Open	30,37	"	24	11	"	"	"	"
M5	$\overline{\text{SOW-R}}$	"	36	Open	"	15	16	"	"	"	"
M6	$\overline{\text{BCC}}$	"	14	30,37	"	24	19	"	"	"	"
M7	$\overline{\text{LRC-R}}$	"	13,17, 18,20, 33,38, 39,30	31,37	"	24	21	"	"	"	"
M8	ENA-I	"	Open	30	"	2	22	"	"	"	"
M9	URC-R	"	14	30,37	"	24	23	"	"	"	"
M10	EOC	"	17,18, 20,33, 38,39	30	"	13	32	"	"	"	"
M11	DTOR	"	Open	15	"	36	35	"	"	"	"
M12	CLA	"	35	4,5	"	34	40	"	"	"	"
M13	D32	"	13,17, 18,20, 33,38, 39,14	31,37, 30	26	24	29	5	25	"	"
M14	D33	"	13,17, 18,20, 33,38, 39,14	31,37, 30	26	24	9	5	25	"	"

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TABLE XIV. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M15	RDY-R	Transition time (t_{TLH}), C_L = 150 pF	13,17, 18,20, 33,38, 39,25, 30	31,37	Open	24,6	3	4	50	ns	25,26
M16	DEOW	"	17,18, 20,33, 38,39,	30	"	13	7	"	"	"	"
M17	CLBI	"	Open	37	"	24	8	"	"	"	"
M18	CLAI	"	Open	30,37	"	24	11	"	"	"	"
M19	$\overline{SOW-R}$	"	36	Open	"	15	16	"	"	"	"
M20	$\overline{BCC}$	"	14	30,37	"	24	19	"	"	"	"
M21	$\overline{LRC-R}$	"	13,17, 18,20, 33,38, 39,30	31,37	"	24	21	"	"	"	"
M22	ENA-I	"	Open	30	"	2	22	"	"	"	"
M23	URC-R	"	14	30,37	"	24	23	"	"	"	"
M24	EOC	"	17,18, 20,33, 38,39	30	"	13	32	"	"	"	"
M25	DTOR	"	Open	15	"	36	35	"	"	"	"
M26	CLA	"	35	4,5	"	34	40	"	"	"	"
M27	D32	"	13,17, 18,20, 33,38, 39,14	31,37, 30	26	24	29	6	35	"	"
M28	D33	"	13,17, 18,20, 33,38, 39,14	31,37, 30	26	24	9	6	35	"	"

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TABLE XV. Transition time (t_{THL}) test.

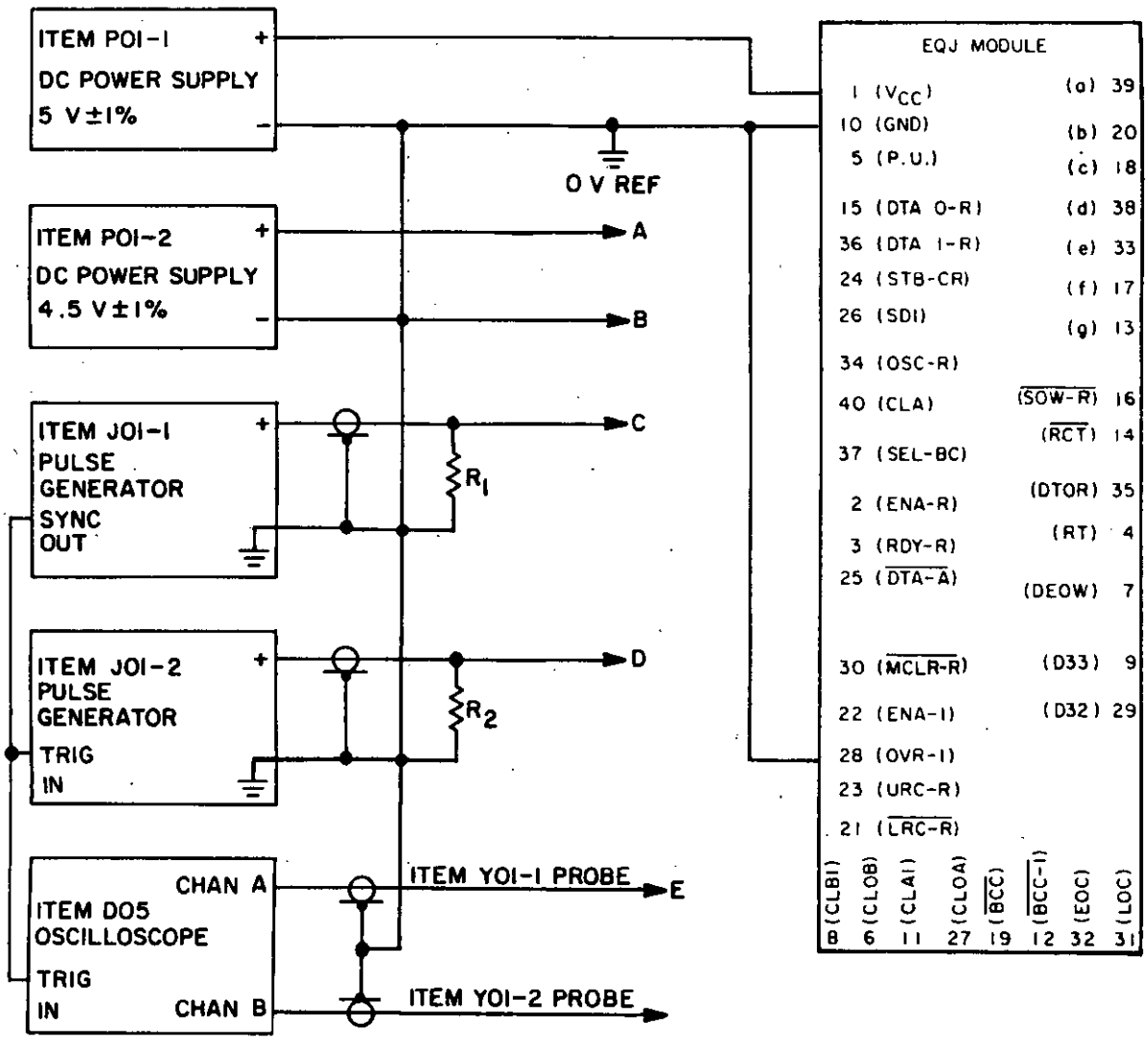
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N1	RDY-R	Transition time (t_{THL}), C_L = 50 pF	13,17, 18,20, 33,38, 39,25, 30	31,37	Open	24,6	3	4	13	ns	25,26
N2	DEOW	"	17,18, 20,33, 38,39,	30	"	13	7	"	"	"	"
N3	CLBI	"	Open	37	"	24	8	"	"	"	"
N4	CLAI	"	Open	30,37	"	24	11	"	"	"	"
N5	$\overline{\text{SOW-R}}$	"	36	Open	"	15	16	"	"	"	"
N6	$\overline{\text{BCC}}$	"	14	30,37	"	24	19	"	"	"	"
N7	$\overline{\text{LRC-R}}$	"	13,17, 18,20, 33,38, 39,30	31,37	"	24	21	"	"	"	"
N8	ENA-I	"	Open	30	"	2	22	"	"	"	"
N9	URC-R	"	14	30,37	"	24	23	"	"	"	"
N10	EOC	"	17,18, 20,33, 38,39	30	"	13	32	"	"	"	"
N11	DTOR	"	Open	15	"	36	35	"	"	"	"
N12	CLA	"	35	4,5	"	34	40	"	"	"	"
N13	D32	"	13,17, 18,20, 33,38, 39,14	31,37, 30	26	24	29	1	10	"	"
N14	D33	"	13,17, 18,20, 33,38, 39,14	31,37, 30	26	24	9	1	10	"	"

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TABLE XV. Transition time (t_{THL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N15	RDY-R	Transition time (t_{THL}), C_L = 150 pF	13,17, 18,20, 33,38, 39,25, 30	31,37	Open	24,6	3	4	20	ns	25,26
N16	DEOW	"	17,18, 20,33, 38,39	30	"	13	7	"	"	"	"
N17	CLBI	"	Open	37	"	24	8	"	"	"	"
N18	CLAI	"	Open	30,37	"	24	11	"	"	"	"
N19	$\overline{\text{SOW-R}}$	"	36	Open	"	15	16	"	"	"	"
N20	$\overline{\text{BCC}}$	"	14	30,37	"	24	19	"	"	"	"
N21	$\overline{\text{LRC-R}}$	"	13,17, 18,20, 33,38, 39,30	31,37	"	24	21	"	"	"	"
N22	ENA-I	"	Open	30	"	2	22	"	"	"	"
N23	URC-R	"	14	30,37	"	24	23	"	"	"	"
N24	EOC	"	17,18, 20,33, 38,39	30	"	13	32	"	"	"	"
N25	DTOR	"	Open	15	"	36	35	"	"	"	"
N26	CLA	"	35	4,5	"	34	40	"	"	"	"
N27	D32	"	13,17, 18,20, 33,38, 39,14	31,37, 30	26	24	29	2	15	"	"
N28	D33	"	13,17, 18,20, 33,38, 39,14	31,37, 30	26	24	9	2	15	"	"

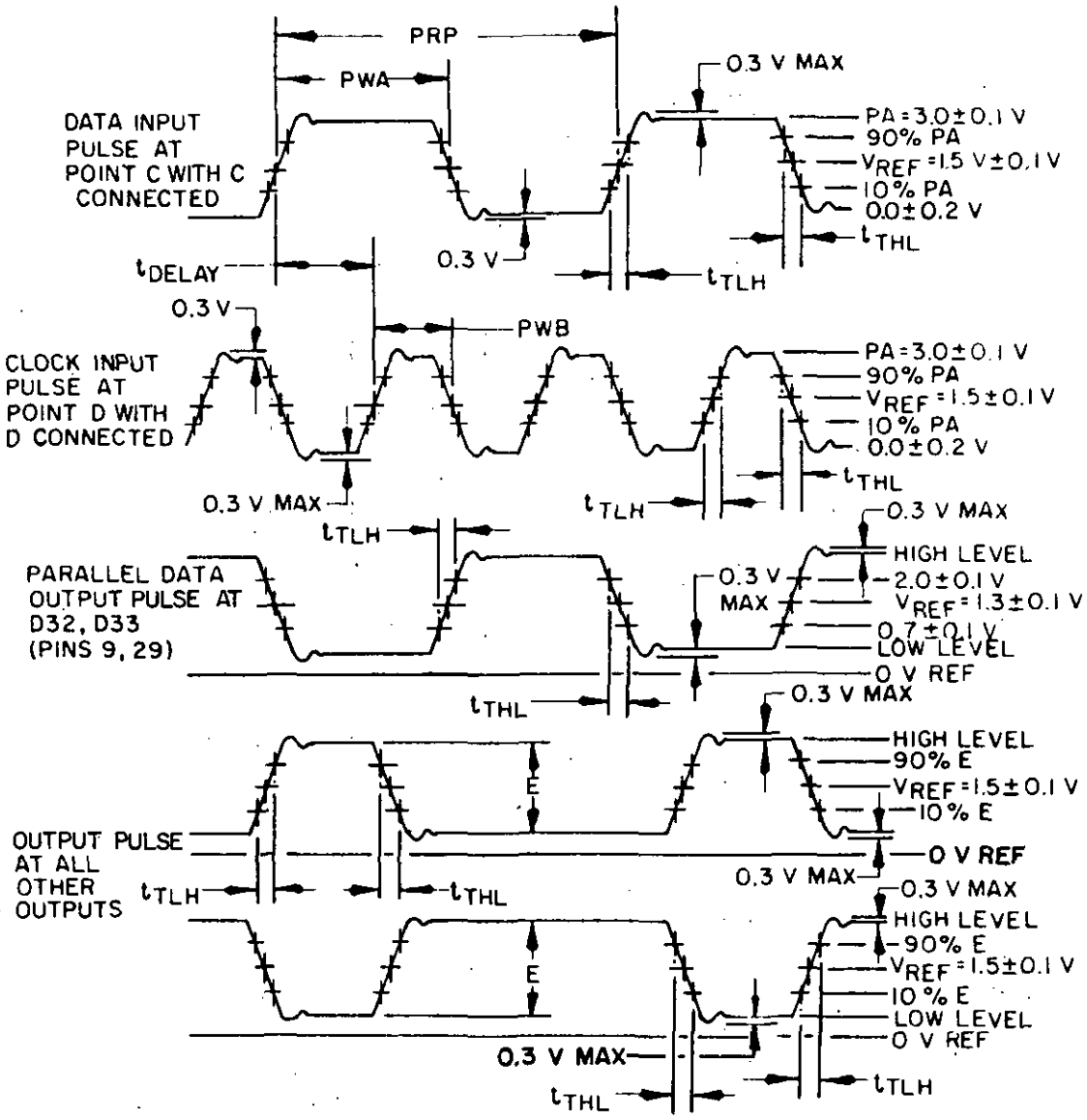
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/4 W.
2. Load pins 16 and 35 with the load as specified on figure 16.

* FIGURE 25. Transition time (t_{TLH} and t_{THL}) test circuit for tables XIV and XV.

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NOTES:
1. Data input pulse parameters:
   PWA = 1000 ±50 ns.
   PRP = 2000 ±100 ns.
   t_{TLH} and t_{THL} = 10 ±2 ns.
2. Clock input pulse parameters:
   PWB = 400 ±40 ns.
   t_{DELAY} = 300 ±50 ns.
   t_{TLH} and t_{THL} = 10 ±2 ns.
3. Item J01-2 is operated in the double pulse mode, positive output. Using item D05, adjust the appropriate rate, delay, pulse width and amplitude controls to obtain the required waveforms and logic levels.
4. Before making measurements, verify all outputs have a high level ≥ 2.4 V and low level ≤ 0.4.
5. All voltages are referenced to 0 V reference unless otherwise specified.

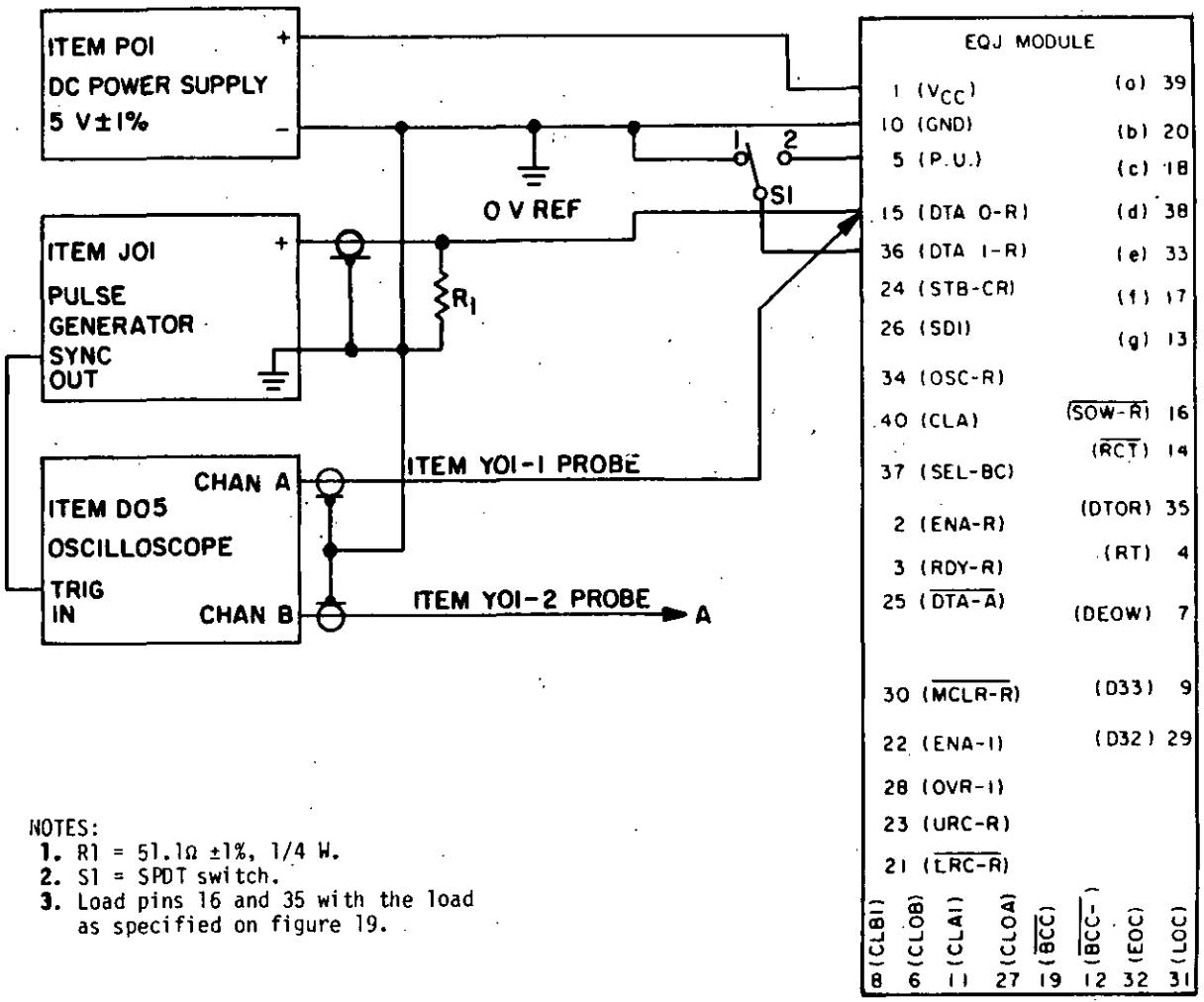
* FIGURE 26. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XIV and XV.

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	S1	Min	Max		
P1	Propagation delay time, from DTA 0-R (t_{PLH}), $\overline{SOW\text{-}R}$ (C_L = 50 pF)	16	2	4	26	ns	27,28
P2	Propagation delay time, from DTA 0-R (t_{PLH}), $\overline{SOW\text{-}R}$ (C_L = 150 pF)	"	"	"	32	"	"
Q1	Propagation delay time, from DTA 0-R (t_{PHL}), $\overline{SOW\text{-}R}$ (C_L = 50 pF)	"	"	"	19	"	"
Q2	Propagation delay time, from DTA 0-R (t_{PHL}), $\overline{SOW\text{-}R}$ (C_L = 150 pF)	"	"	"	25	"	"
R1	Propagation delay time, from DTA 0-R (t_{PLH}), DTOR (C_L = 50 pF)	35	1	8	45	"	"
R2	Propagation delay time, from DTA 0-R (t_{PLH}), DTOR (C_L = 150 pF)	"	"	"	57	"	"
S1	Propagation delay time, from DTA 0-R (t_{PHL}), DTOR (C_L = 50 pF)	"	"	"	45	"	"
S2	Propagation delay time, from DTA 0-R (t_{PHL}), DTOR (C_L = 150 pF)	"	"	"	57	"	"

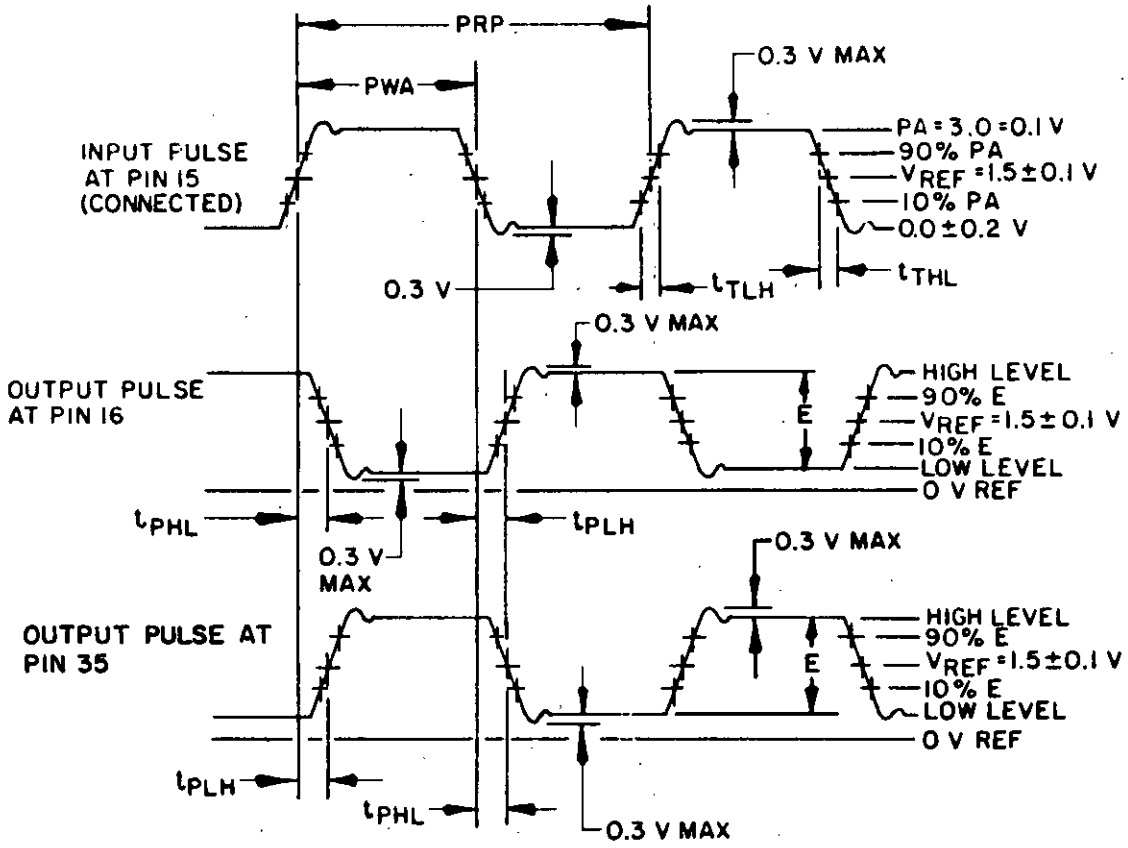
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:43Z
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. S1 = SPDT switch.
3. Load pins 16 and 35 with the load as specified on figure 19.

* FIGURE 27. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
   t_{TLH} and t_{THL} = 6 ±2 ns.
2. Before making measurements, verify both outputs have a high level > 2.4 V and a low level < 0.4 V.
3. All voltages are referenced to 0 V unless otherwise specified.

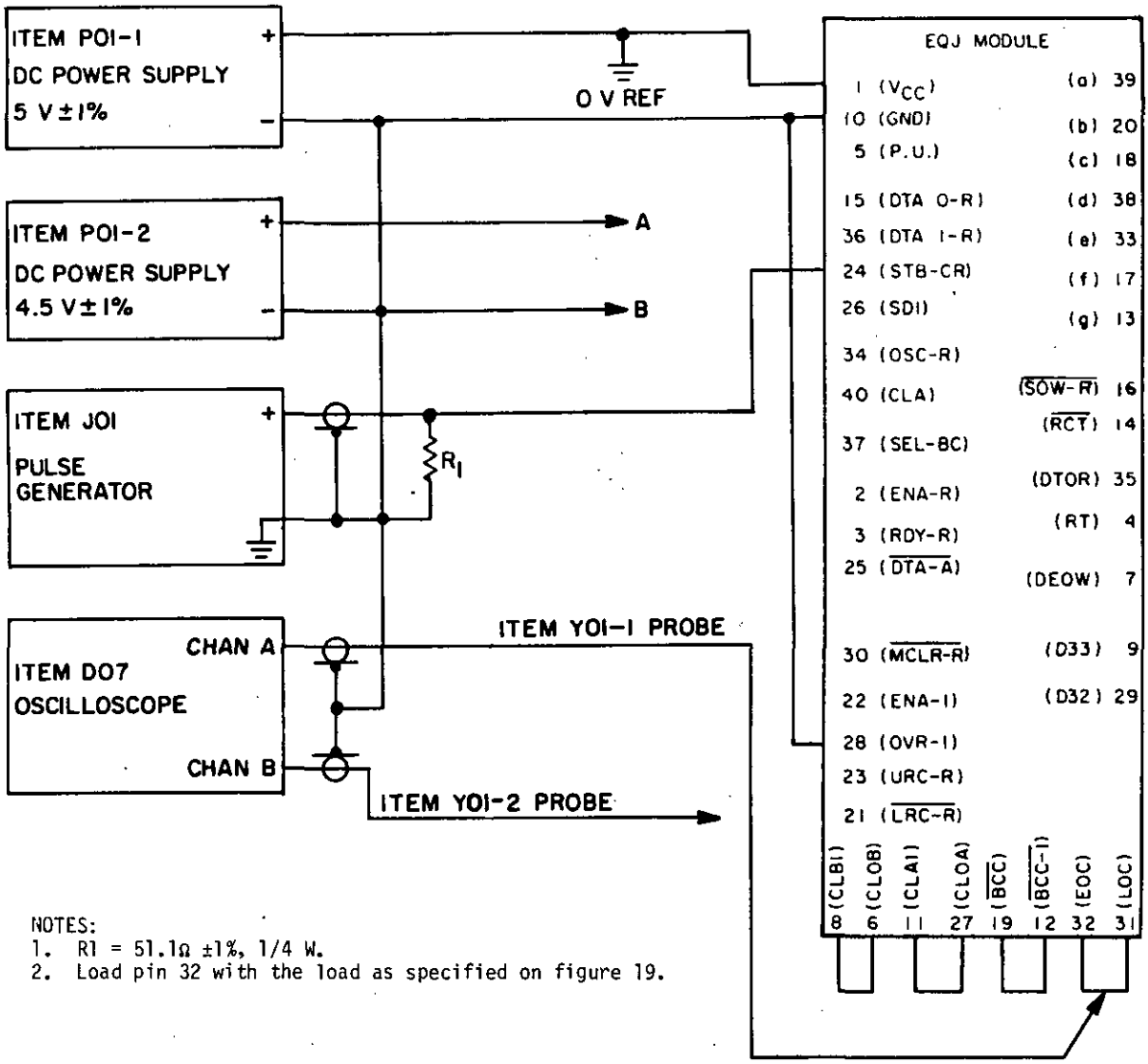
* FIGURE 28. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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TABLE XVII. Propagation delay time (end-of-count) and EOC spike test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
T1	EOC	Propagation delay time, end-of-count (measuring output pulse width) (C_L = 50 pF)	13,18,20, 33,38,39	17,37	35	135	ns	29,30
T2	EOC	Propagation delay time, end-of-count (measuring output pulse width) (C_L = 150 pF)	"	"	40	142	"	"
T3	EOC	Spike	"	"	0	3	"	"

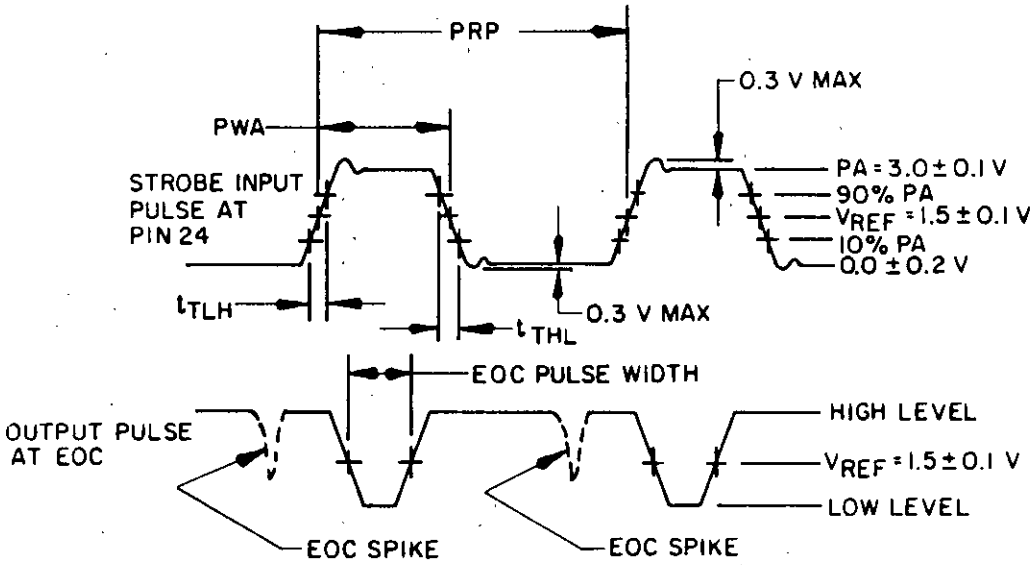
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load pin 32 with the load as specified on figure 19.

* FIGURE 29. Propagation delay time (end-of-count) and EOC spike test circuit for table XVII.

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NOTES:
1. Strobe input pulse parameters:
   PWA = 200 ±10 ns.
   PRP = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
2. Before making measurements, verify EOC output has a high level ≥ 2.4 V and a low level ≤ 0.4 V.
3. EOC spike width should not exceed 3 ns pulse width at V_{REF} (1.5 V).
4. All voltages are referenced to 0 V unless otherwise specified.

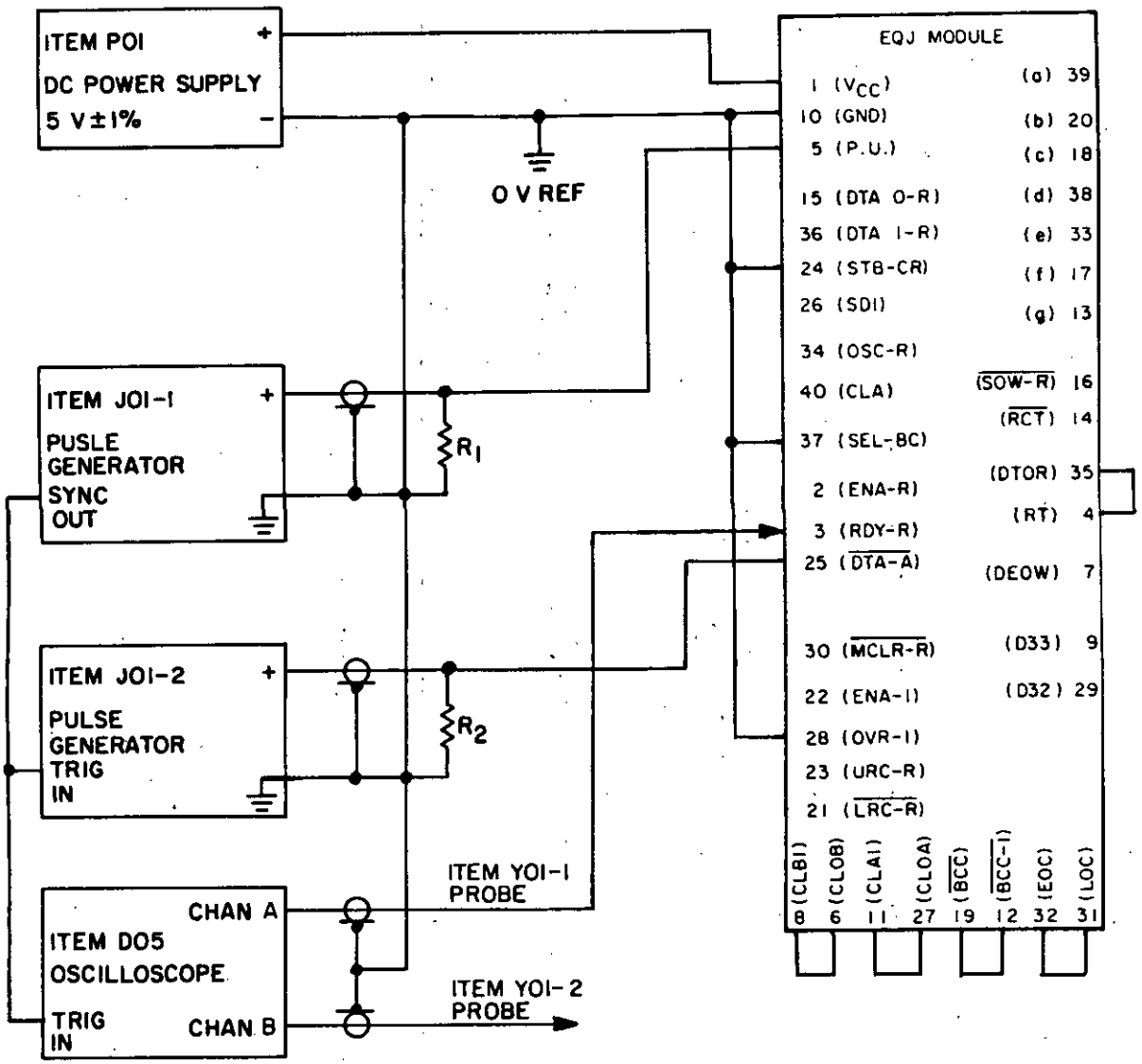
FIGURE 30. Propagation delay time (end-of-count) and EOC spike test circuit waveforms for table XVII.

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TABLE XVIII. Propagation delay time (t_{PHL}), minimum pulse width test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
U1	Propagation delay time (t_{PHL}), $\overline{DTA-A}$ to RDY-R, minimum pulse width, $\overline{DTA-A}$ (C_L = 50 pF)	20	85	ns	31,32
U2	Propagation delay time (t_{PHL}), $\overline{DTA-A}$ to RDY-R, minimum pulse width, $\overline{DTA-A}$ (C_L = 150 pF)	25	92	ns	31,32

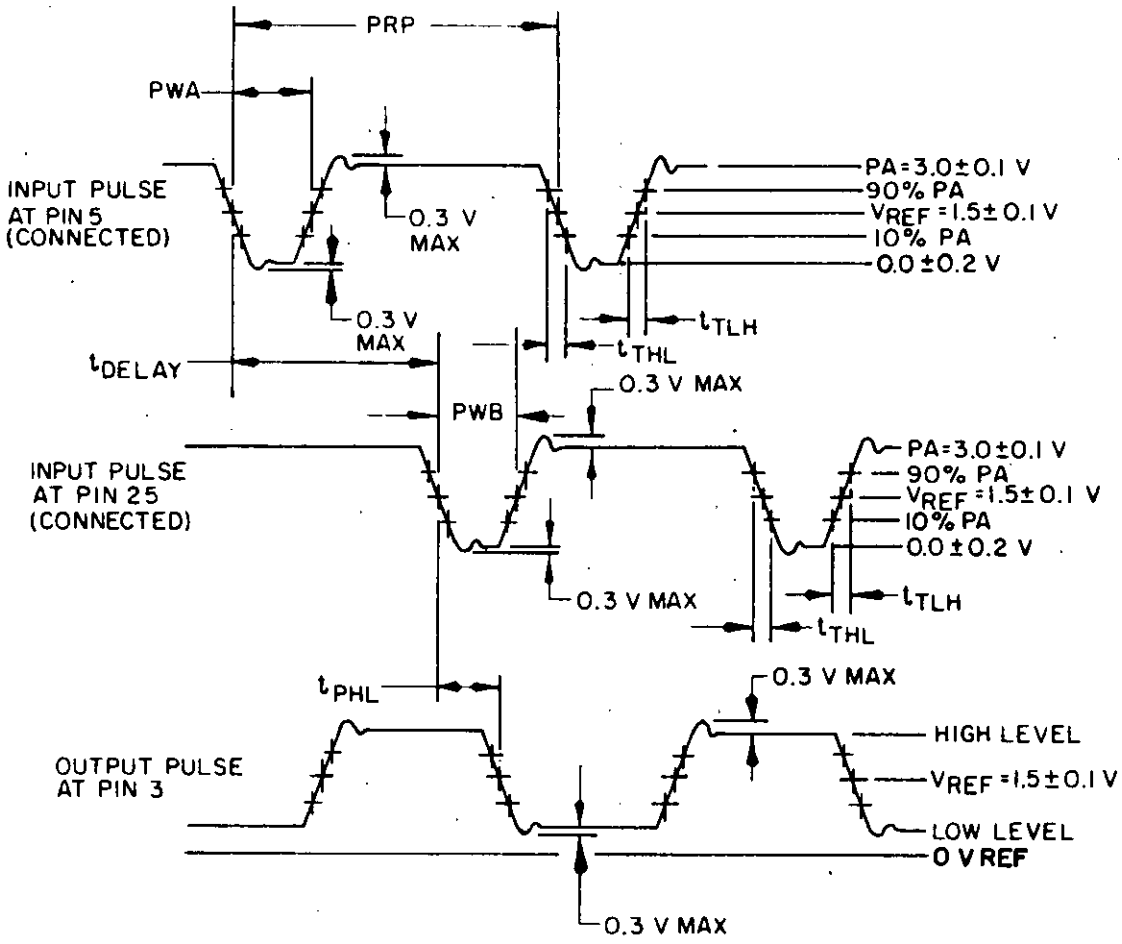
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load pin 3 with the load as specified on figure 19.

* FIGURE 31. Propagation delay time (t_{PHL}), minimum pulse width test circuit for table XVIII.

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NOTES:
1. Item J01-1 input pulse parameters (pin 5):
   PWA = 150 ±15 ns.
   PRP = 1000 ±100 ns.
   $t_{THL} = t_{TLH} = 6 \pm 2$ ns.
2. Item J01-2 input pulse parameters (pin 25):
   PWB = 60 +5, -0 ns.
   $t_{THL} = t_{TLH} = 6 \pm 2$ ns.
3. Item J01-2 is externally triggered by the sync output of item J01-1. The delay control of item J01-1 should be adjusted so that $t_{DELAY} = 350 \pm 35$ ns.
4. Before making measurement, verify RDY-R output pulse has a high level $\geq$ 2.4 V and low level $\leq$ 0.4 V.
5. All voltages are referenced to 0 V unless otherwise specified.

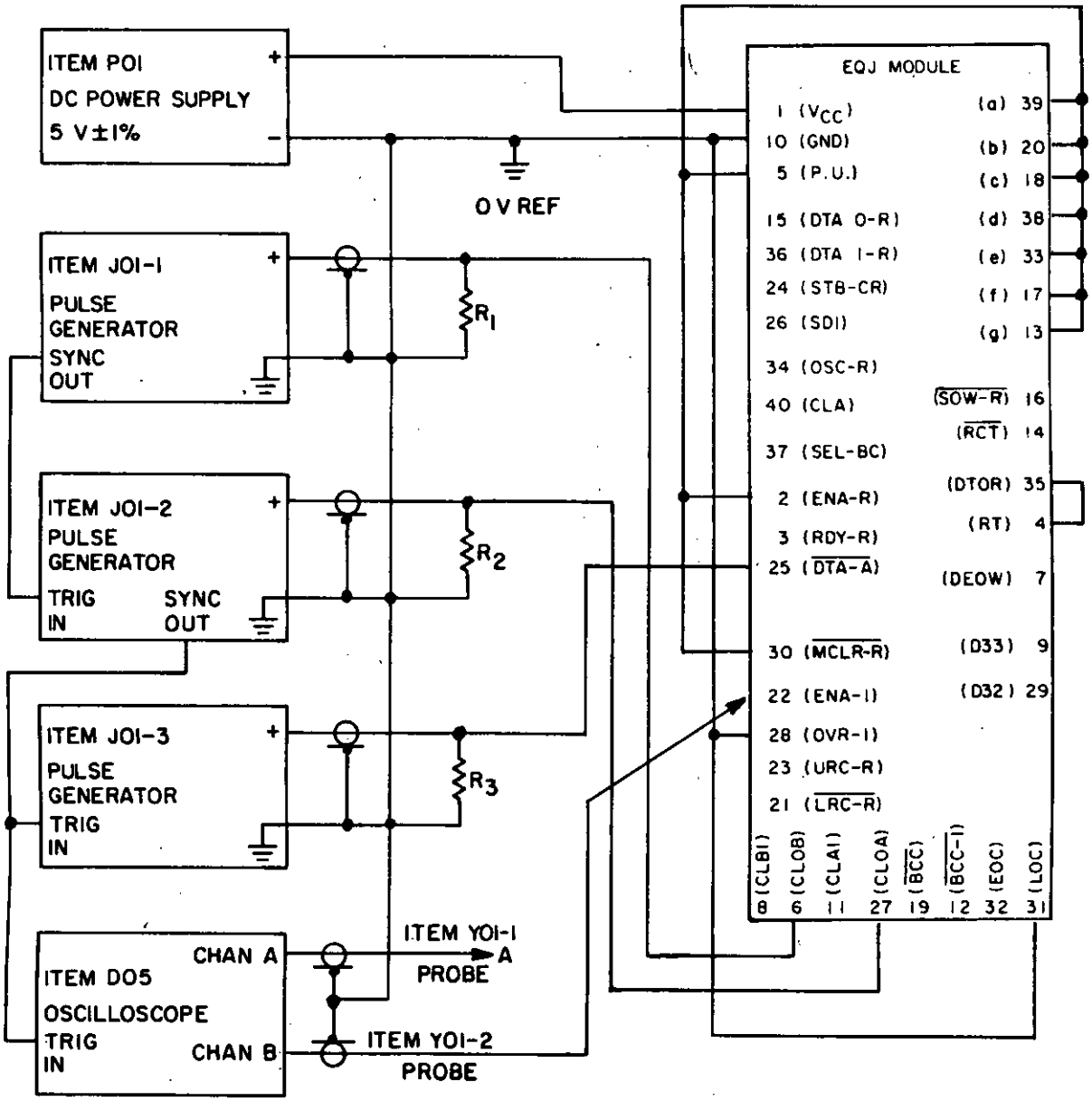
FIGURE 32. Propagation delay time (t_{PHL}), minimum pulse width test circuit waveforms for table XVIII.

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*TABLE XIX. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
V1	Propagation delay time, $\overline{DTA\text{-}A}$ to ENA-I (t_{PLH}) (C_L = 50 pF)	25	20	130	ns	33,34
V2	Propagation delay time, CLOA to ENA-I (t_{PHL}) (C_L = 50 pF)	27	15	45	"	"
V3	Propagation delay time, $\overline{DTA\text{-}A}$ to ENA-I (t_{PLH}) (C_L = 150 pF)	25	25	137	"	"
V4	Propagation delay time, CLOA to ENA-I (t_{PHL}) (C_L = 150 pF)	27	20	52	"	"

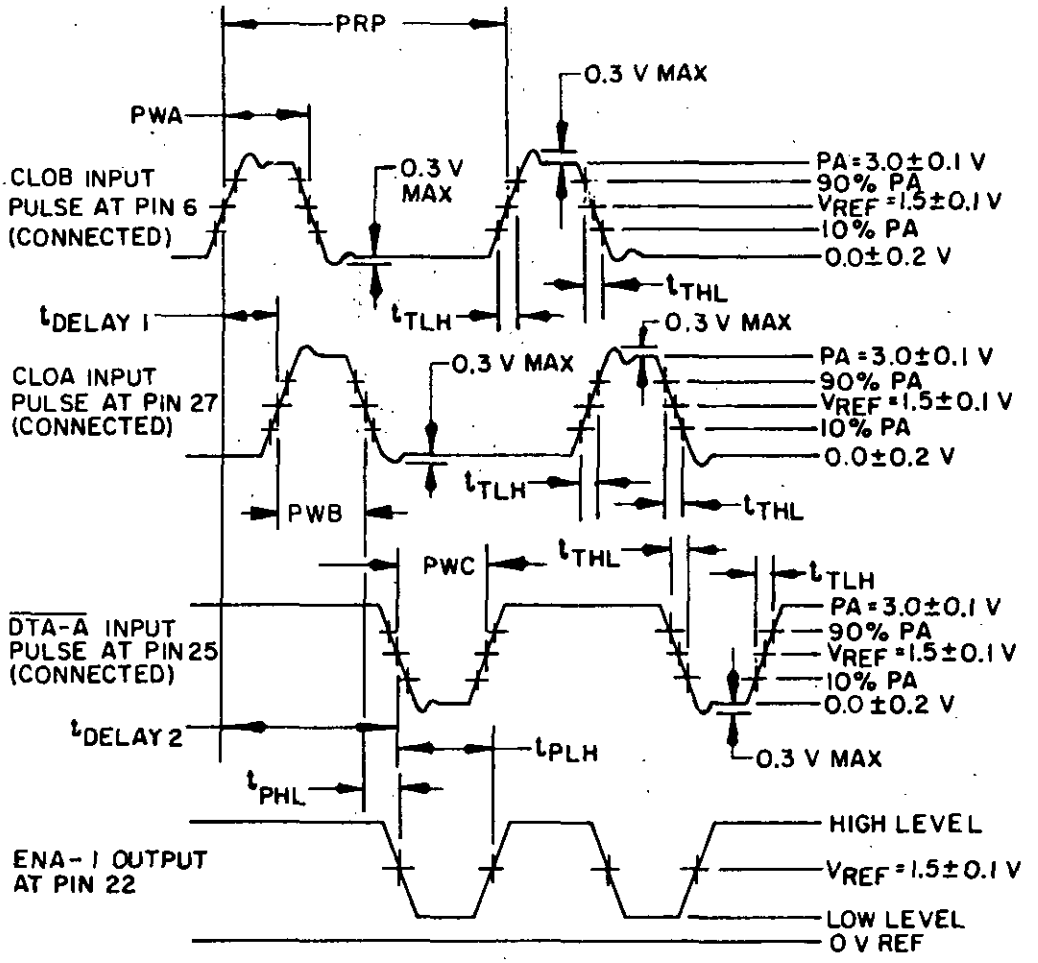
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NOTES:
1. R1 = R2 = R3 = 51.1Ω ±1%, 1/4 W.
2. Load pin 22 with the load as specified on figure 19.

* FIGURE 33. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIX.

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Check the source to verify that this is the current version before use.



NOTES:
1. CLOB input pulse parameters:
   PWA = 100 ±5 ns.
   PRP = 1000 ±50 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
2. CLOA input pulse parameters:
   PWB = 100 ±5 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
   $t_{DELAY\ 1}$ = 200 ±10 ns.
3. $\overline{DTA-A}$ input pulse parameters:
   PWC = 60 +5 ns, -0 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
   $t_{DELAY\ 2}$ = 600 ±30 ns.
4. Before making measurement, verify that outputs have a high level $\geq$ 2.4 V and low level $\leq$ 0.4 V.
5. All voltages are referenced to 0 V unless otherwise specified.

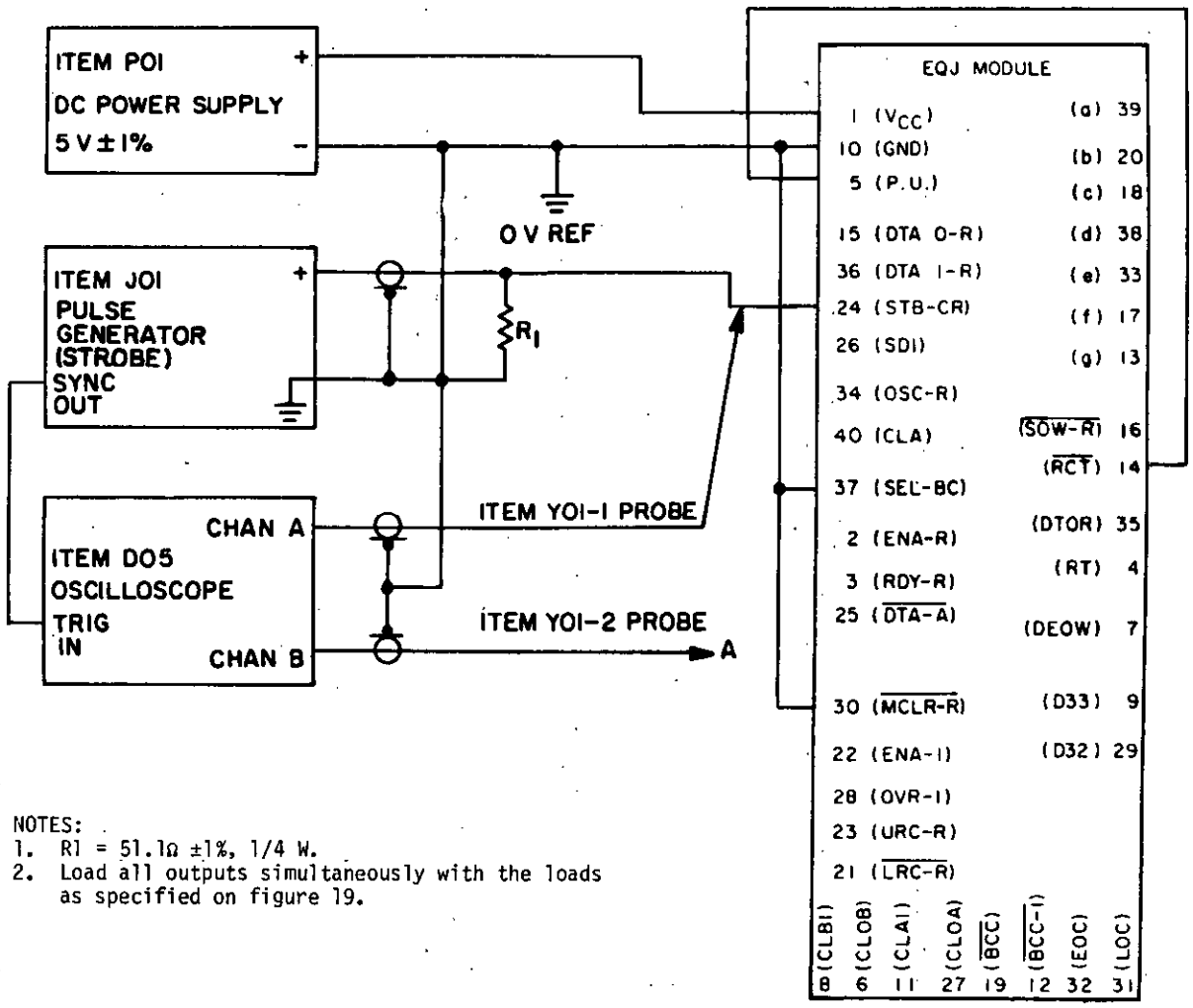
FIGURE 34. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIX.

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Check the source to verify that this is the current version before use.

TABLE XX. Propagation delay time (t_{PLH} and t_{PHL}) and minimum pulse width test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
W1	Propagation delay time, strobe to CLBI (t_{PLH}) (C_L = 50 pF)	8	6	42	ns	35,36
W2	Minimum pulse width, CLBI (C_L = 50 pF)	8	90	110	"	"
W3	Propagation delay time, strobe to URC-R (t_{PLH}) (C_L = 50 pF)	23	14	91	"	"
W4	Minimum pulse width, URC-R (C_L = 50 pF)	23	70	110	"	"
W5	Propagation delay time, strobe to $\overline{BCC}$ (t_{PHL}) (C_L = 50 pF)	19	10	79	"	"
W6	Minimum pulse width, $\overline{BCC}$ (C_L = 50 pF)	19	82	135	"	"
W7	Propagation delay time, strobe to CLBI (t_{PLH}) (C_L = 150 pF)	8	11	49	"	"
W8	Minimum pulse width, CLBI (C_L = 150 pF)	8	95	117	"	"
W9	Propagation delay time, strobe to URC-R (t_{PLH}) (C_L = 150 pF)	23	19	98	"	"
W10	Minimum pulse width, URC-R (C_L = 150 pF)	23	77	117	"	"
W11	Propagation delay time, strobe to $\overline{BCC}$ (t_{PHL}) (C_L = 150 pF)	19	15	86	"	"
W12	Minimum pulse width, $\overline{BCC}$ (C_L = 150 pF)	19	87	142	"	"

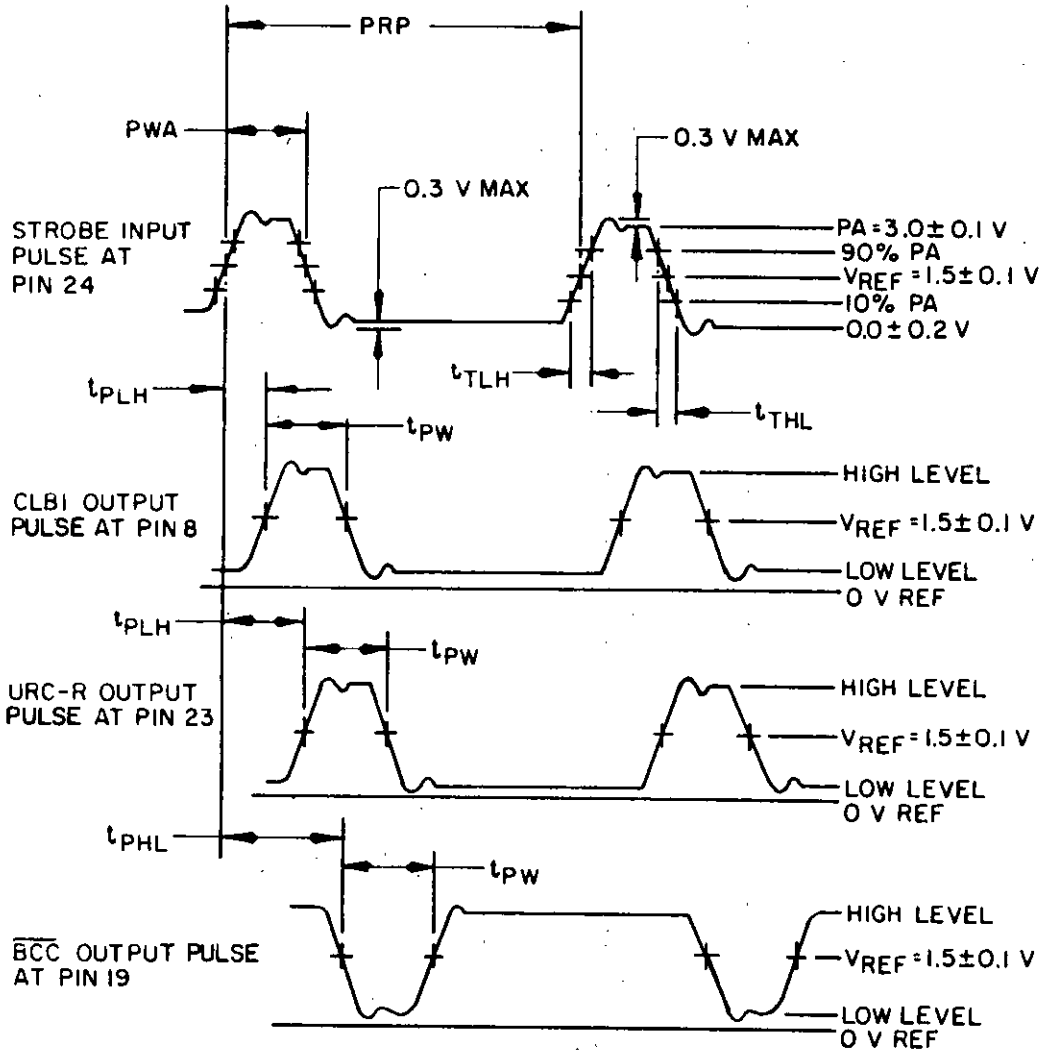
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs simultaneously with the loads as specified on figure 19.

* FIGURE 35. Propagation delay time (t_{PLH} and t_{PHL}) and minimum pulse width test circuit for table XX.

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NOTES:
1. Strobe input pulse parameters:
   PWA = 100 ±4 ns.
   PRP = 1000 ±40 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
2. Before making measurement, verify all outputs have a high level ≥ 2.4 V and low level ≤ 0.4 V.
3. All voltages are referenced to 0 V unless otherwise specified.

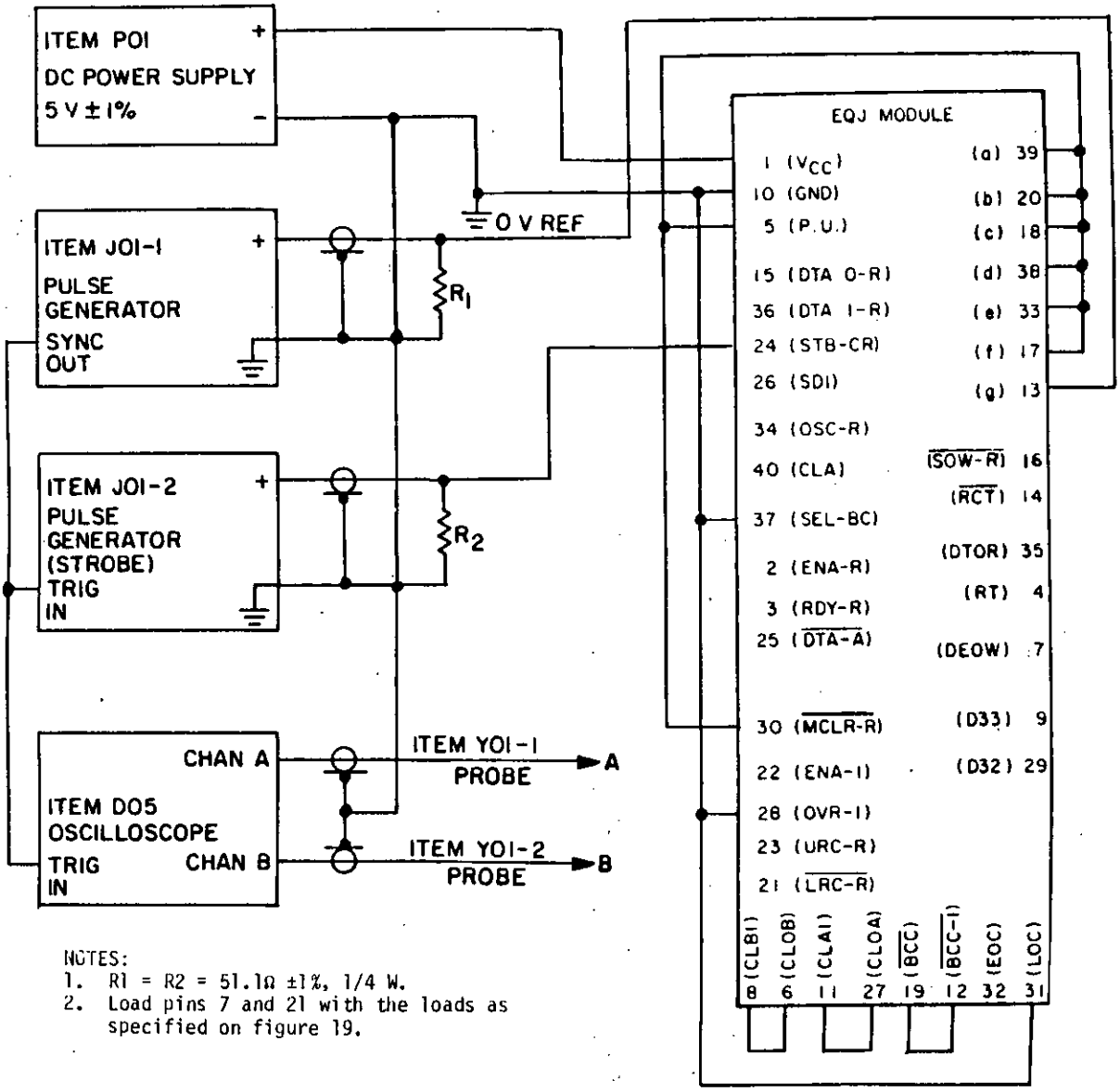
FIGURE 36. Propagation delay time (t_{PLH} and t_{PHL}) and minimum pulse width test circuit for table XX.

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*TABLE XXI. Propagation delay time (t_{PLH} and t_{PHL}) and pulse width test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	Propagation delay time, word length (g) to DEOW (t_{PLH}), C_L = 50 pF	13	7	25	110	ns	37,38
X2	Propagation delay time, STROBE to DEOW (t_{PHL}), C_L = 50 pF	24	7	16	77	"	"
X3	$\overline{\text{LRC-R}}$ pulse width (C_L = 50 pF)	21	Open	65	100	"	"
X4	Propagation delay time, word length (g) to DEOW (t_{PLH}), C_L = 150 pF	13	7	30	117	"	"
X5	Propagation delay time, STROBE to DEOW (t_{PHL}), C_L = 150 pF	24	7	20	84	"	"
X6	$\overline{\text{LRC-R}}$ pulse width (C_L = 150 pF)	21	Open	60	107	"	"

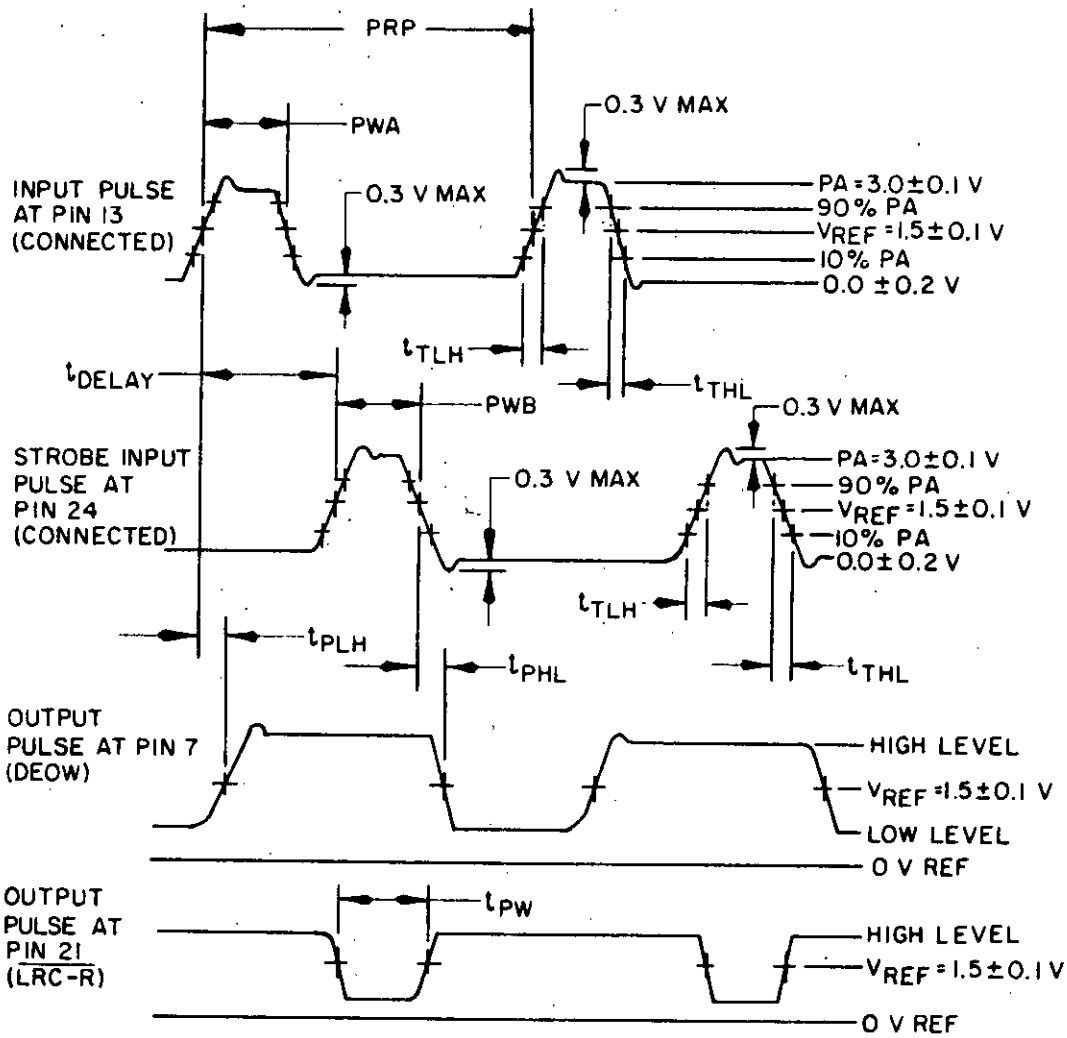
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load pins 7 and 21 with the loads as specified on figure 19.

* FIGURE 37. Propagation delay time (t_{PLH} and t_{PHL}) and pulse width test circuit for table XXI.

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NOTES:
1. Item J01-1 input pulse parameters (pin 13):
   PWA = 80 ±5 ns.
   PRP = 1000 ±50 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
2. Item J01-2 input pulse parameters (pin 24):
   PWB = 80 ±5 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
3. Item J01-2 is externally triggered by the SYNC output of item J01-1. The delay control of item J01-1 should be adjusted so the t_{DELAY} = 320 ±20 ns.
4. Before making measurement, verify DEOW and LRC-R output pulses have a high level ≥ 2.4 V and low level ≤ 0.4 V.
5. All voltages are referenced to 0 V unless otherwise specified.

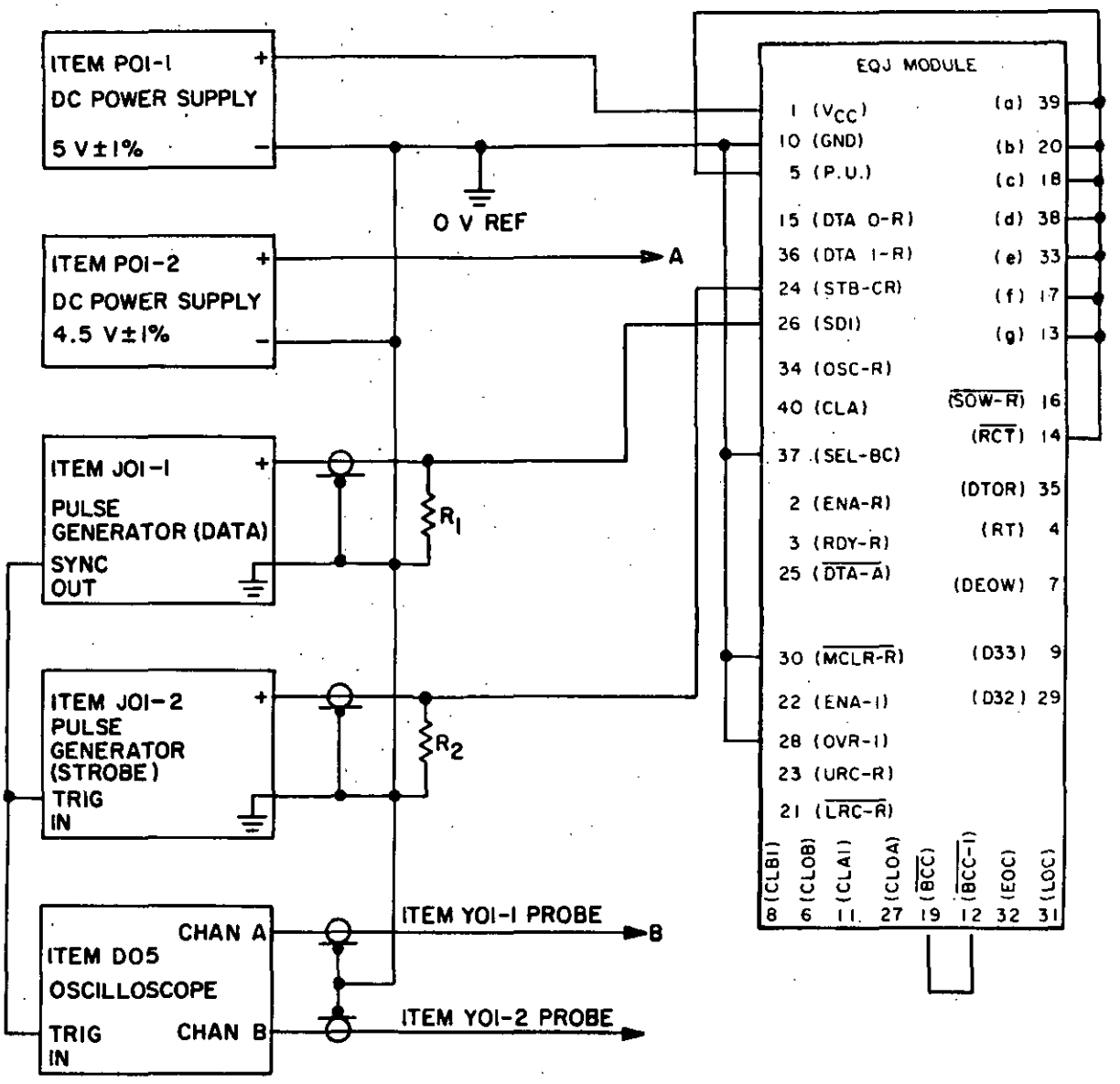
* FIGURE 38. Propagation delay time (t_{PLH} and t_{PHL}) and pulse width test circuit waveforms for table XXI.

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TABLE XXII. Maximum operating frequency check test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y1	D32 Maximum operating frequency check, measuring output period (t_{PD}), C_L = 50 pF	Open	29	850	1150	ns	39,40
Y2	D33 Maximum operating frequency check, measuring output period (t_{PD}), C_L = 50 pF	Open	9	850	1150	ns	39,40

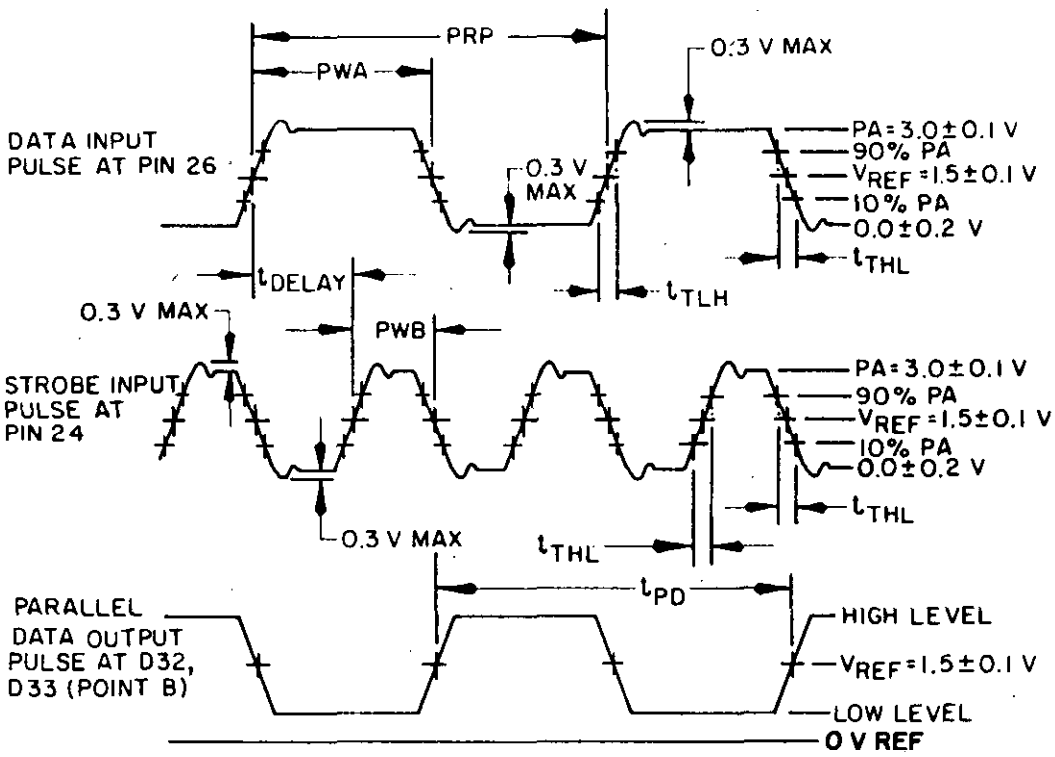
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load pins 9 and 29 with the load as specified on figure 19.

* FIGURE 39. Maximum operating frequency check test circuit for table XXII.

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NOTES:
1. Data input pulse parameters:
   PWA = 500 ±20 ns.
   PRP = 1000 ±40 ns.
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns.
2. Strobe input pulse parameters:
   PWB = 100 ±5 ns.
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns.
   $t_{DELAY} = 400 \pm 10$ ns.
3. Item J01-2 is operated in the double pulse mode.
4. Before making measurement, verify that parallel data outputs have a high level $\geq$ 2.4 V and a low level $\leq$ 0.4 V.
5. All voltages are referenced to 0 V unless otherwise specified.

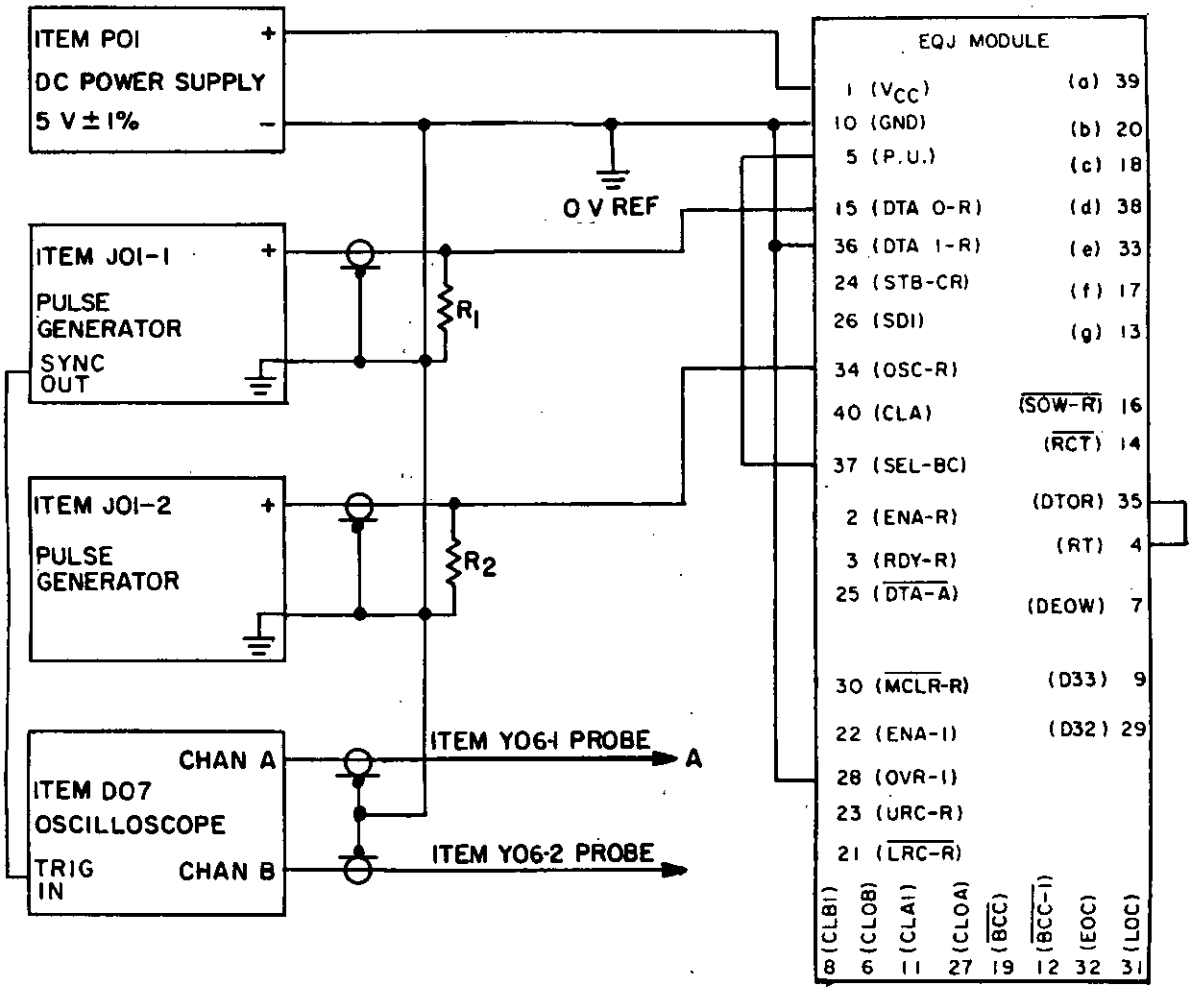
FIGURE 40. Maximum operating frequency check test circuit waveforms for table XXII.

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TABLE XXXIII. Maximum operating frequency check and pulse width test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
Z1	CLAI	Maximum operating frequency check (C_L = 50 pF)	11	850	1150	ns	41,42
Z2	CLAI	Pulse width (C_L = 50 pF)	11	51	107	"	"
Z3	CLBI	Maximum operating frequency check (C_L = 50 pF)	8	850	1150	"	"
Z4	CLBI	Pulse width A (C_L = 50 pF)	"	220	390	"	"
Z5	CLBI	Pulse width B (C_L = 50 pF)	"	220	390	"	"

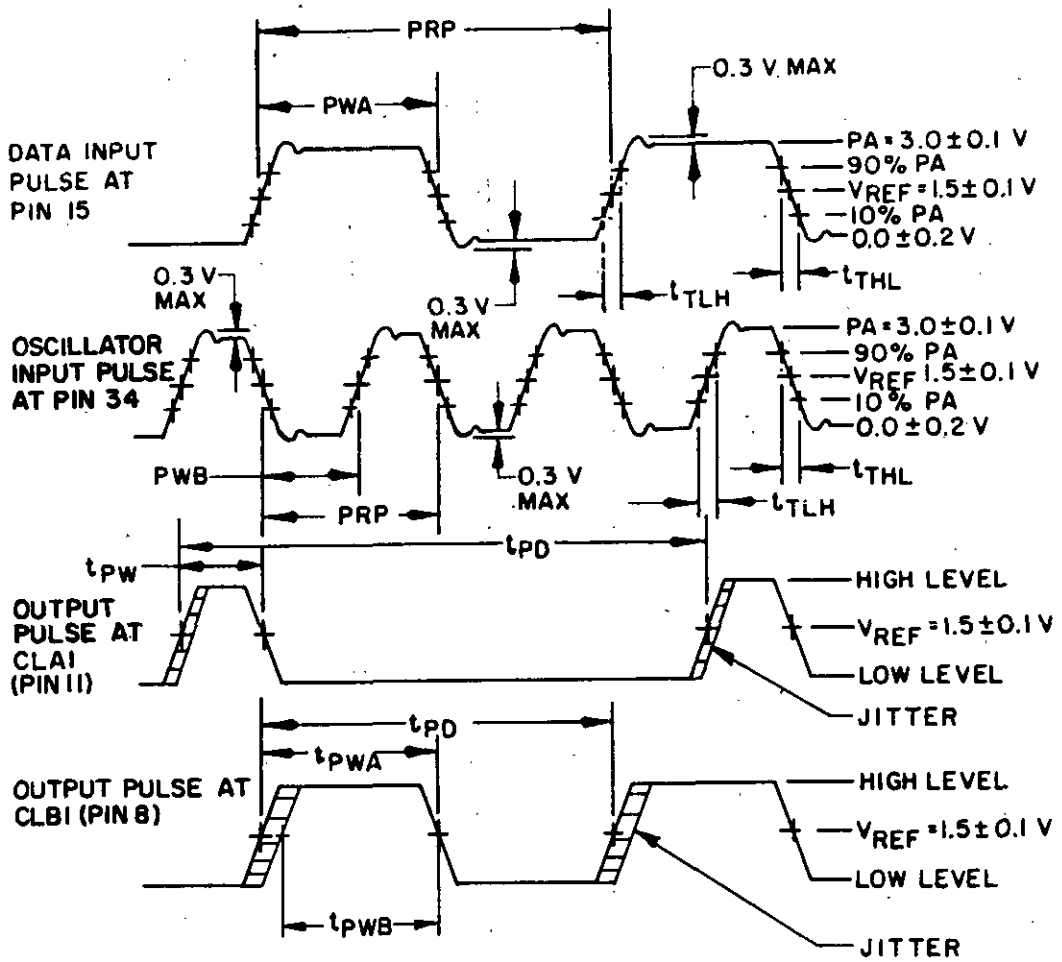
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load pins 8 and 11 with the load as specified on figure 19.
3. Set item D07 to internal trigger for test condition Z2.

* FIGURE 41. Maximum operating frequency check and pulse width test circuit for table XXIII.

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NOTES:
1. Data input pulse parameters:
   PWA = 500 ±20 ns.
   PRP = 1,000 ±50 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
2. Oscillator input pulse parameters:
   PWB = 44 ±4 ns.
   PRP = 131 ±8 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
3. It should be noted that due to the asynchronous relationship between data and oscillator inputs, jitter will be present on both the CLAI and CLBI outputs. The measurement should be made from the first leading edge trace to the first trailing edge trace.
4. Before making measurements, verify CLAI and CLBI outputs have a high level ≥ 2.4 V and a low level ≤ 0.4 V.
5. All voltages are referenced to 0 V unless otherwise specified.
6. For test condition Z5, the measurement should be made from the last leading edge to the first trailing edge.
7. When performing test condition Z2 using internal trigger, jitter may not be present on the output waveform.

* FIGURE 42. Maximum operating frequency check and pulse width test circuit waveforms for table XXIII.

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TABLE XXIV. Output crosstalk (low-level) test.

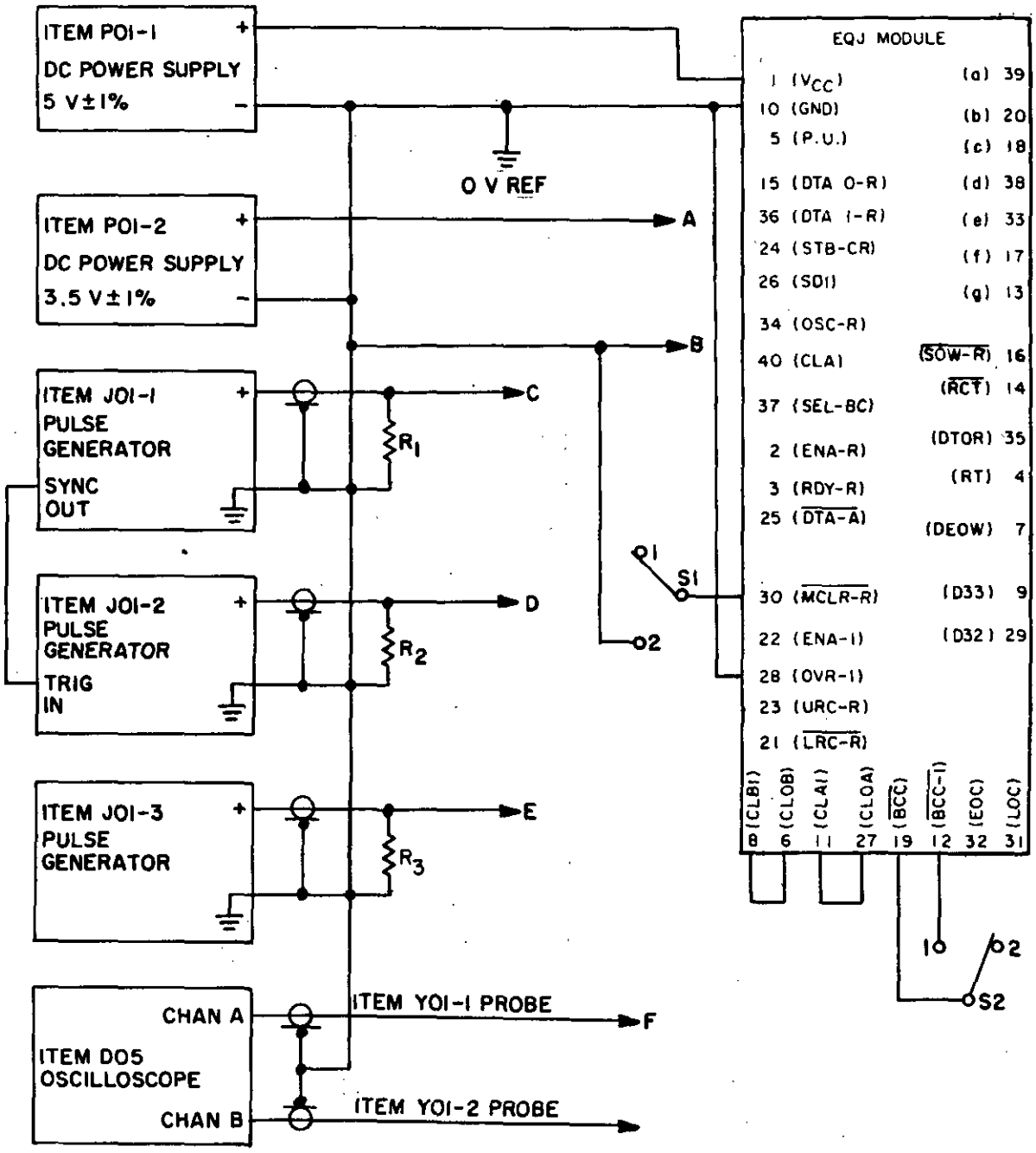
Test condition	Parameter		Connect								Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	S1 pos	S2 pos	C to pin	D to pin	E to pin	F to pin	Min	Max		
AA1	RDY-R	Output crosstalk, low level, C_L = 50 pF	14,37	25,36	1	1	15	Open	34	3	0	400	mV	43,44
AA2	DEOW	"	14,37	17,36	**	2	15	Open	34	7	"	"	"	"
AA3	CLBI	"	14	24,37	1	1	30	5	Open	8	"	"	"	"
AA4	D33	"	14,37	20,26, 36	**	"	15	Open	34	9	"	"	"	"
AA5	CLAI	"	14,37	15,36	1	"	30	5	Open	11	"	"	"	"
AA6	$\overline{\text{SOW-R}}$	"	15,36	20,37	"	"	14,24	Open	"	16	"	"	"	"
AA7	$\overline{\text{BBC}}$	"	14,24	15,20, 36,37	"	2	30	5	"	19	"	"	"	"
AA8	$\overline{\text{LRC-R}}$	"	13,17, 18,20, 24,33, 36,38, 39	31,37	"	1	15	Open	34	21	"	"	"	"
AA9	ENA-I	"	14,37	2,36	"	"	"	"	"	22	"	"	"	"
AA10	URC-R	"	37	14,20, 36	**	"	"	"	"	23	"	"	"	"
AA11	D32	"	14,37	20,26, 36	**	"	"	"	"	29	"	"	"	"
AA12	EOC	"	13,14, 17,18, 20,33, 37,38, 39	31,36,	1	"	"	"	"	32	"	"	"	"
AA13	DTOR	"	14	15,20, 36,37	"	"	24	"	Open	35	"	"	"	"
AA14	CLA	"	14	15,20, 36,37	"	"	24	"	Open	40	"	"	"	"

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TABLE XXV. Output crosstalk (high-level) test.

Test condition	Parameter		Connect								Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	S1 pos	S2 pos	C to pin	D to pin	E to pin	F to pin	Min	Max		
BB1	RDY-R	Output crosstalk, high level, C_L = 50 pF	14	5,31, 37	1	1	24	Open	Open	3	0	500	mV	43,44
BB2	DEOW	"	13,14 17,18, 20,33, 37,38, 39	31,36	"	"	15	Open	34	7	"	"	"	"
BB3	CLBI	"	14,24	37	"	"	30	5	Open	8	"	"	"	"
BB4	D33	"	14,26, 37	20,36	**	"	15	Open	34	9	"	"	"	"
BB5	CLAI	"	14,24	30,36, 37	1	"	"	"	"	11	"	"	"	"
BB6	$\overline{\text{SOW-R}}$	"	14,37	36	"	"	"	"	"	16	"	"	"	"
BB7	$\overline{\text{BBC}}$	"	37	14,36	"	2	"	"	"	19	"	"	"	"
BB8	$\overline{\text{LRC-R}}$	"	14	13,30, 37	"	1	24	"	Open	21	"	"	"	"
BB9	ENA-I	"	2,14	30,37	"	"	24	"	Open	22	"	"	"	"
BB10	URC-R	"	14,24	30,36, 37	"	"	15	"	34	23	"	"	"	"
BB11	D32	"	14,26, 37	20,36	**	"	"	"	"	29	"	"	"	"
BB12	EOC	"	14,37	31,33, 36	1	2	"	"	"	32	"	"	"	"
BB13	DTOR	"	36	15,37	"	1	24	"	Open	35	"	"	"	"
BB14	CLA	"	15,34 36	5,30, 37	"	1	24	"	Open	40	"	"	"	"

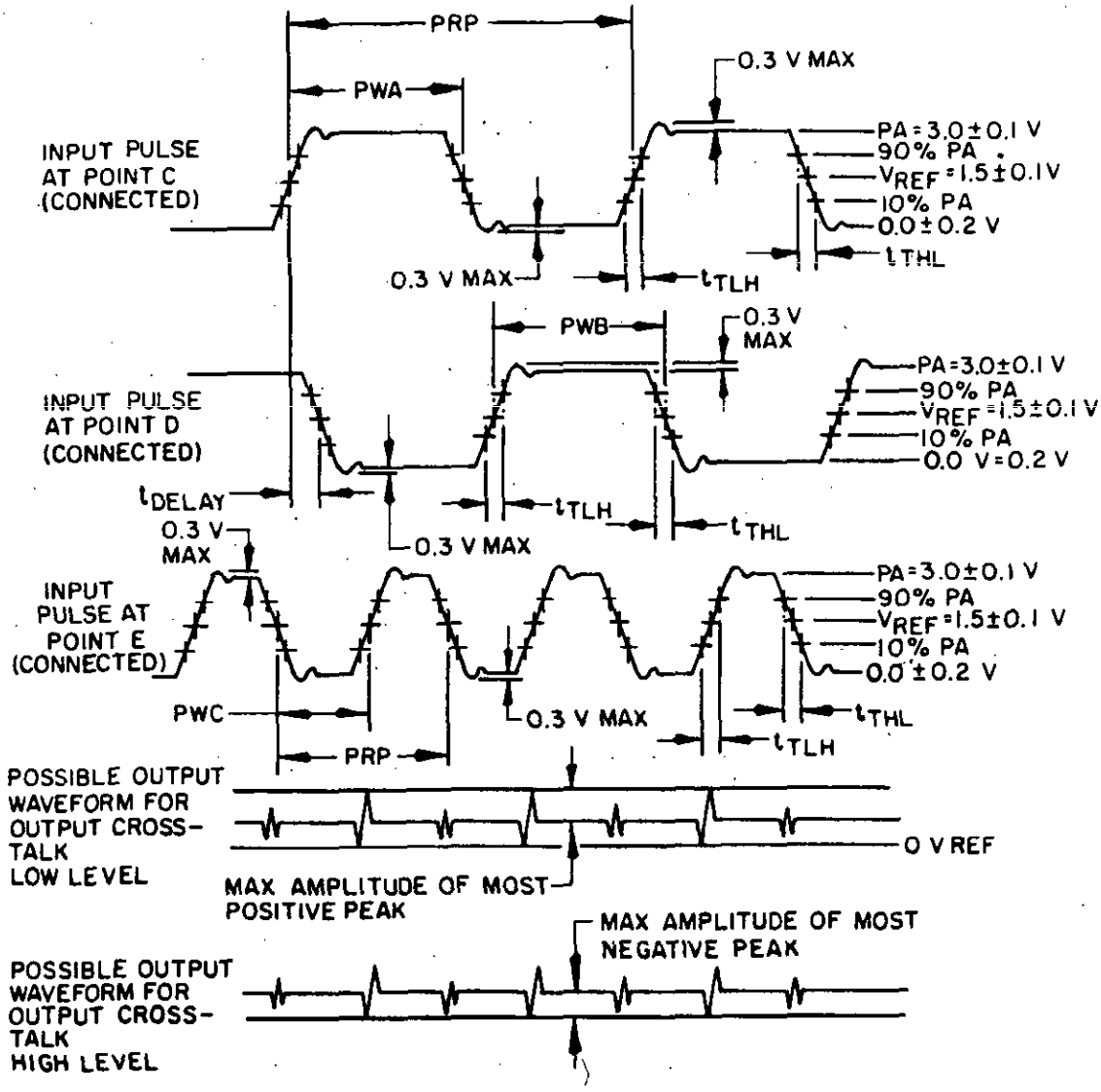
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NOTES:
1. R1 = R2 = R3 = 51.5Ω ±1%, 1/4 W.
2. S1 = S2 = SPDT switch.
3. Load all outputs simultaneously with the loads as specified on figure 19 except pins 8, 11, 16, 19, and 32.
4. S1 should normally be in position 1. When indicated by a double asterisk(**) in table XXIV and XXV, S1 should be placed momentarily in position 2, then returned to position 1. S2 should be placed momentarily in position 2, then returned to position 1. S2 should be placed in the indicated position.

* FIGURE 43. Output crosstalk (low-level and high-level) test circuit for tables XXIV and XXV.

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NOTES:
1. Item J01-1 input pulse parameters (point C):
   PWA = 500 ±25 ns.
   PRP = 1,000 ±50 ns.
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns.
2. Item J01-2 input pulse parameters (point D):
   PWB = 500 ±25 ns.
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns.
   $t_{DELAY} = 125 \pm 10$ ns.
3. Item J01-3 input pulse parameters (point E):
   PWC = 50 ±5 ns.
   PRP = 130 ±5 ns.
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns.
4. Item J01-1 sync output is used to drive the trigger input of item J01-2. The delay control of item J01-1 should be used to adjust t_{DELAY} as required.

FIGURE 44. Output crosstalk (low-level and high-level) test circuit waveforms for tables XXIV and XXV.

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TABLE XXVI. Power supply transient amplitude test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
CC	Power supply transient amplitude	13,17,18, 20,33,37, 38,39	36	1	0	400	mV	45,46

TABLE XXVII. Power supply filter capacitance test.

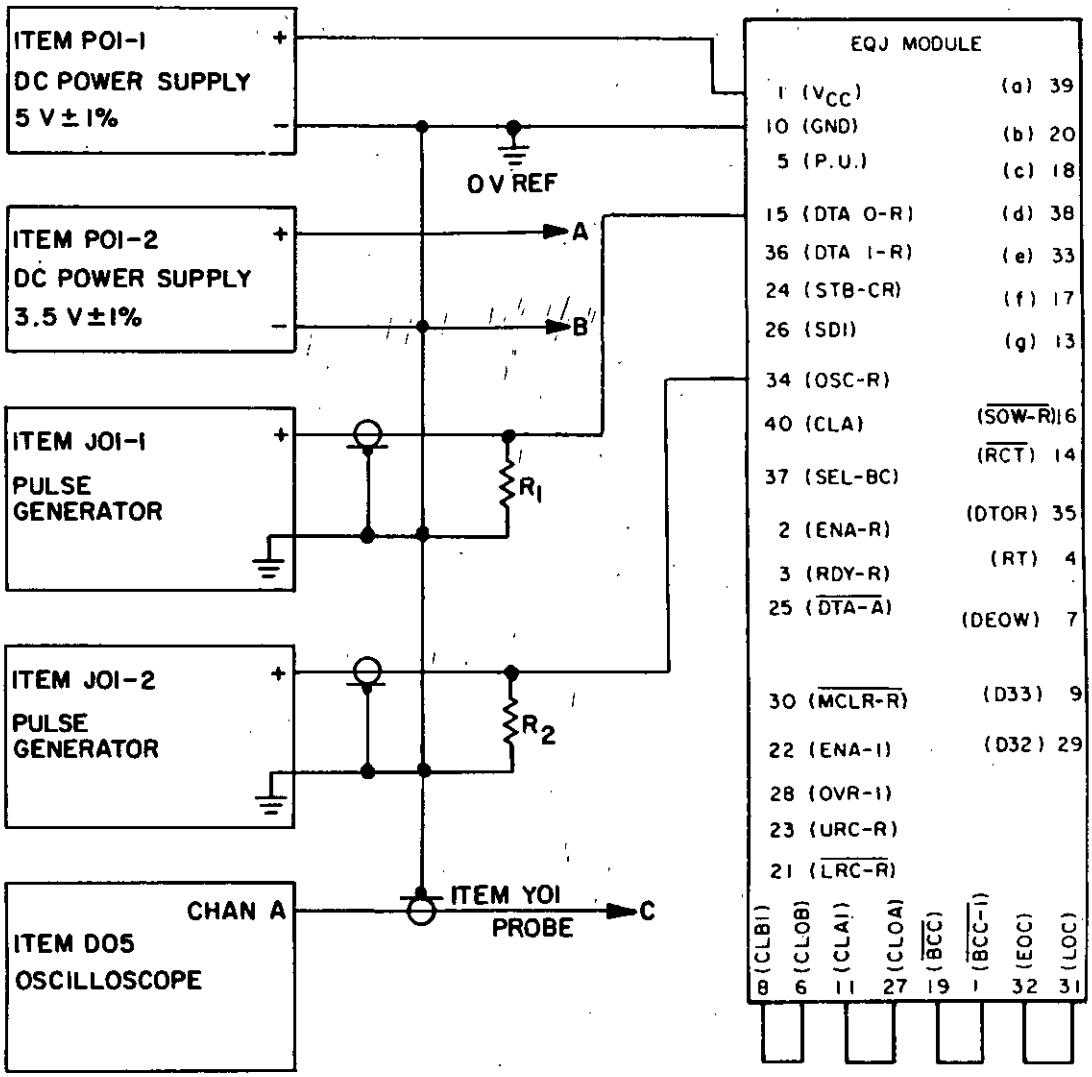
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
DD	Power supply filter capacitance	Open	1	0.07	0.13	µF	47

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TABLE XXVIII. Input capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
EE1	ENA-R	Input capacitance	2	Open	2	15	pF	47
EE2	RT	"	4	"	"	"	"	"
EE3	$\overline{BCC}$	"	12	"	"	"	"	"
EE4	Word length (g)	"	13	"	"	"	"	"
EE5	Word length (f)	"	17	"	"	"	"	"
EE6	Word length (c)	"	18	"	"	"	"	"
EE7	Word length (b)	"	20	"	"	"	"	"
EE8	$\overline{DTA-A}$	"	25	"	"	"	"	"
EE9	$\overline{MCLR-R}$	"	30	"	"	"	"	"
EE10	LOC	"	31	"	"	"	"	"
EE11	Word length (e)	"	33	"	"	"	"	"
EE12	Word length (d)	"	38	"	"	"	"	"
EE13	Word length (a)	"	39	"	"	"	"	"
EE14	STB-CR	"	24	"	"	25	"	"
EE15	CLOA	"	27	"	"	"	"	"
EE16	OVR-I	"	28	"	"	"	"	"
EE17	SDI	"	26	"	"	30	"	"
EE18	$\overline{RCT}$	"	14	"	"	35	"	"
EE19	DTA 0-R	"	15	"	3	"	"	"
EE20	DTA 1-R	"	36	"	2	"	"	"
EE21	CLOB	"	6	"	"	45	"	"
EE22	OSC-R	"	34	"	"	"	"	"
EE23	SEL-BC	"	37	"	"	"	"	"

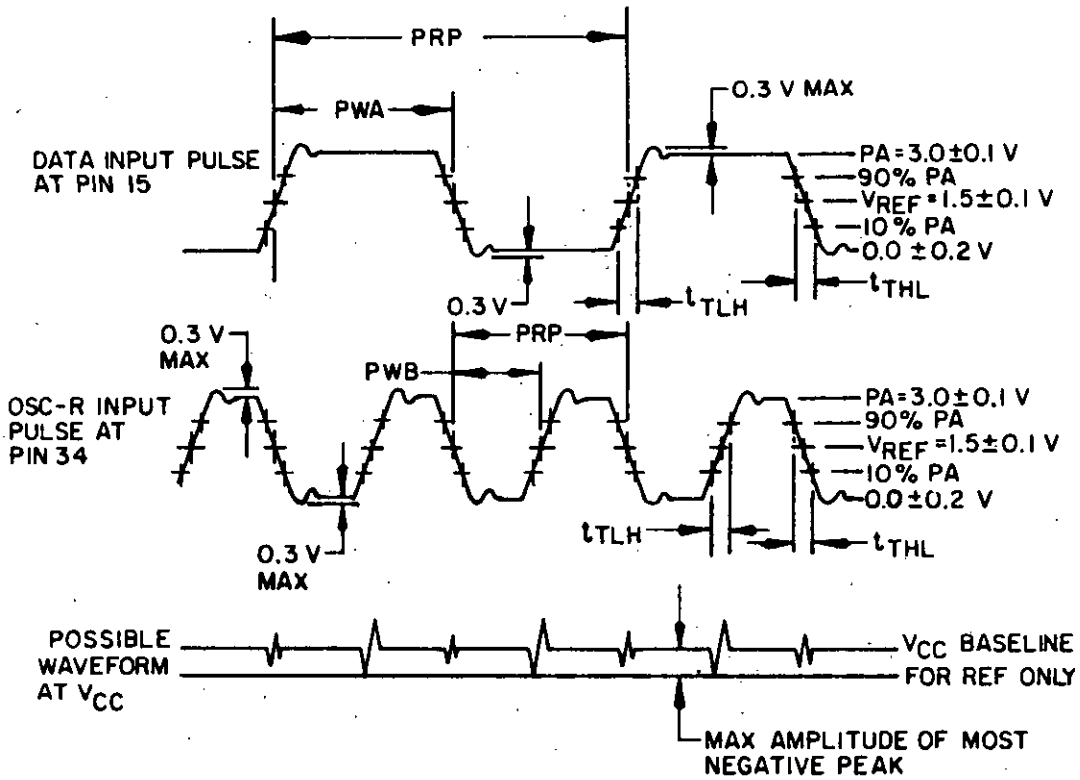
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NOTE: R1 = R2 = 51.1Ω ±1%, 1/4 W.

* FIGURE 45. Power supply transient amplitude test circuit for table XXVI.

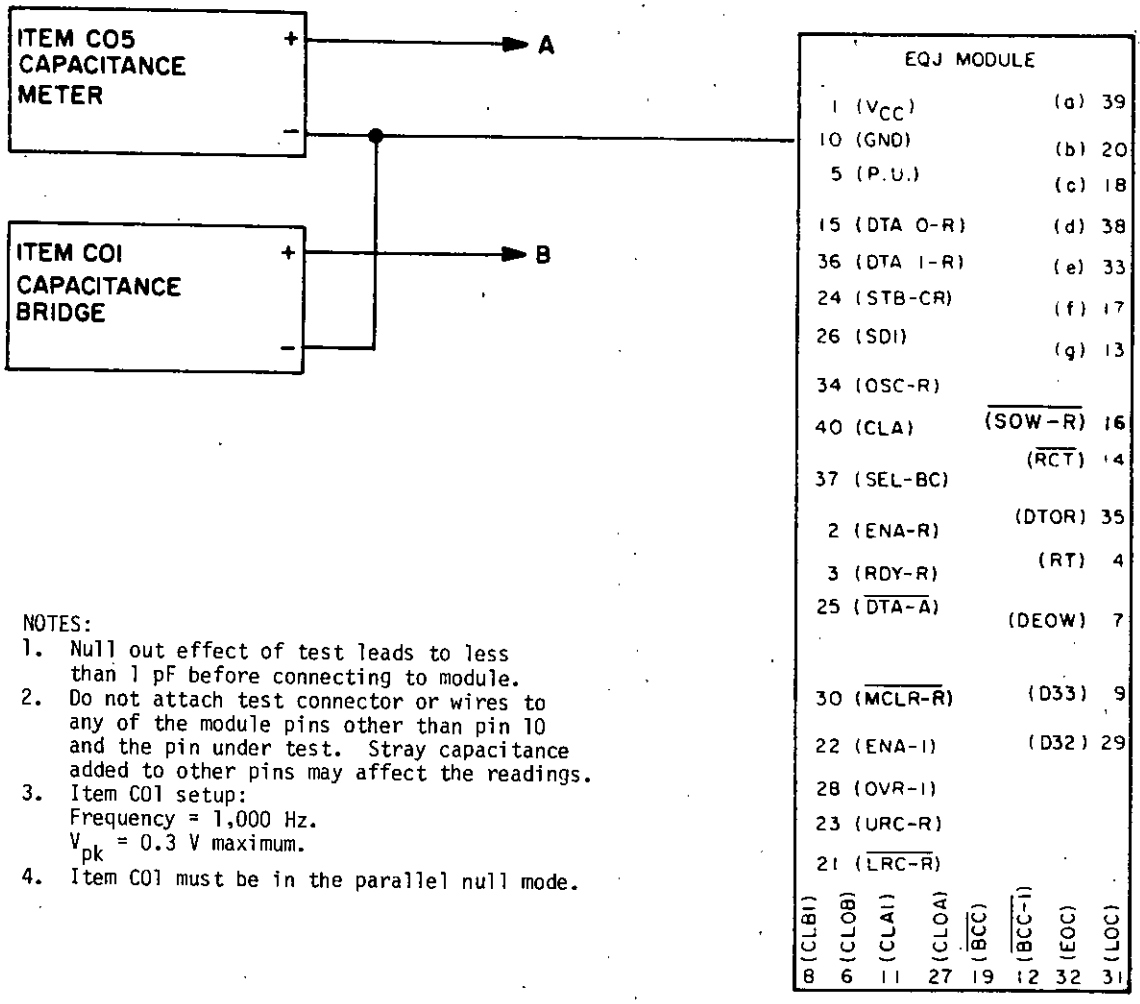
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NOTES:
1. Data input pulse parameters:
   PWA = 500 ±20 ns.
   PRP = 1,000 ±40 ns.
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns.
2. OSC-R input pulse parameters:
   PWB = 50 ±5 ns.
   PRP = 130 ±5 ns.
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns.
3. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 46. Power supply transient amplitude test circuit waveforms for table XXVI.

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NOTES:
1. Null out effect of test leads to less than 1 pF before connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item COl setup:
   Frequency = 1,000 Hz.
   V_{pk} = 0.3 V maximum.
4. Item COl must be in the parallel null mode.

FIGURE 47. Power supply filter and input capacitance test circuit for tables XXVII and XXVIII.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/215A(EC)	Modules, Standard Electronic, Standard Serial Input, Digital, Key Code EQJ

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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